

Near-Lossless Knowledge Distillation: Definitions, Error Decompositions, Capacity Lower Bounds and Objective Design

A theoretical treatment

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Abstract

We give a mathematically explicit account of distilling a teacher $p_T(y | x)$ into a smaller autoregressive student $q_\theta(y | x)$. We (i) formalise “lossless” through divergences between sequence-level laws under a deployment input distribution and prove the relations among forward KL, reverse KL, total variation, JSD_β , Hellinger and task-level excess risk, including exact chain-rule identities that pair each KL direction with a specific prefix distribution; (ii) decompose the student’s gap into approximation, estimation, optimisation, prefix-shift (exposure bias) and covariate-shift terms and bound each, recovering the $O(\varepsilon T)$ versus $O(\varepsilon T^2)$ compounding dichotomy and showing exactly when on-policy training removes it; (iii) prove information-theoretic (rate–distortion) lower bounds on the achievable divergence as a function of the student’s description length, together with a softmax-bottleneck rank bound, and derive the power-law forms they predict; (iv) analyse the logit gradients of the standard objectives, the role of temperature, and the Fisher-information content of soft labels, and propose a sequence-level, prefix-adaptive JSD_β objective with proven exactness, boundedness, agnostic total-variation competitiveness, consistency, and an unbiased Rao–Blackwellised gradient estimator; (v) analyse intermediate-representation matching, proving when it is redundant, when it helps (noisy supervision, conditioning, compositional hardness) and when it over-constrains, and propose a Fisher-aligned variant with no approximation cost in a linear-Gaussian model; (vi) assemble an end-to-end bound, separate what is proved from what is conjectured, and specify experiments; (vii) give sharp bounds on the probability that the greedy outputs of teacher and student differ, with extensions to beam search and speculative decoding: exact flip thresholds for every f -divergence, an exact knapsack characterisation of the worst case in terms of the teacher’s margins along its greedy paths, and a path-weighted exchange rate which shows that behavioural and decoding-level losslessness are logically independent. We close with open problems. Every inequality marked *numerically checked* is verified by the scripts in `verification/`.

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Setting, notation and standing assumptions

Context. Distillation in its modern form goes back to [Buciluă et al. \(2006\)](#), [Ba and Caruana \(2014\)](#) and [Hinton et al. \(2015\)](#). The sequence-level variant for autoregressive models is due to [Kim and Rush \(2016\)](#). [Xu et al. \(2024\)](#) survey methods for large language models. Here we care about the theory: what “lossless” should mean, what limits it, and which objectives provably approach it.

Objects. $\mathcal{X}$ is a countable input space and $\mathcal{V}$ a finite vocabulary, $|\mathcal{V}| = V$. Outputs are sequences $y = (y_1, \dots, y_T) \in \mathcal{Y} := \mathcal{V}^T$. Variable-length outputs are handled by an absorbing end-of-sequence token: after it, every model emits it again with probability one. This does not change any divergence between sequence laws, so a fixed horizon T loses no generality. To avoid a clash with the horizon T , we write the *teacher* as

$$p(y | x) := p_T(y | x) = \prod_{t=1}^T p(y_t | x, y_{<t}), \quad q_\theta(y | x) = \prod_{t=1}^T q_\theta(y_t | x, y_{<t}), \quad \theta \in \Theta_S.$$

The *state* (context) at step t is $s_t = (x, y_{<t}) \in \mathcal{S}_t$. We abbreviate the next-token conditionals as $p_t(\cdot | s) := p(\cdot | s)$ and $q_t(\cdot | s) := q_\theta(\cdot | s)$. The sequence laws are $P^x := p(\cdot | x)$ and $Q_\theta^x := q_\theta(\cdot | x)$ on $\mathcal{Y}$. Inputs follow a *training* distribution $\mathcal{D}$. Behaviour is evaluated under a *deployment* distribution μ , which may differ from $\mathcal{D}$.

Prefix (state) distributions. For a policy $\pi \in \{p, q_\theta\}$, d_t^π is the law of $s_t = (x, y_{<t})$ when $x \sim \mathcal{D}$ and $y \sim \pi(\cdot | x)$. We write d_t^p for the teacher and d_t^θ for the student. Distillation on teacher-generated prefixes (“teacher forcing”) trains on d_t^p . *On-policy* distillation trains on d_t^θ .

Per-token errors. For a divergence D between next-token distributions, define the off- and on-policy average per-token errors

$$\varepsilon_{\text{off}}^D := \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim d_t^p} D(p_t(\cdot | s) \| q_t(\cdot | s)), \quad \varepsilon_{\text{on}}^D := \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim d_t^\theta} D(p_t(\cdot | s) \| q_t(\cdot | s)). \quad (1)$$

For reverse KL, “ $D(p \| q)$ ” in (1) means $\text{KL}(q \| p)$. We write e_t^{off} and e_t^{on} for the individual summands, so that $\varepsilon_{\text{off}} = \frac{1}{T} \sum_t e_t^{\text{off}}$.

Logits. The student’s next-token distribution is $q_t(\cdot | s) = \text{softmax}(z_\theta(s))$ with logits $z_\theta(s) \in \mathbb{R}^V$. The teacher’s logits are $v(s)$. $C := I - \frac{1}{V} \mathbf{1}\mathbf{1}^\top$ is the centring projector and $\tilde{z} := Cz$ denotes centred logits.

Conventions. $\log$ is natural. Rates R are in bits unless stated otherwise, so a class of 2^R students has $\log |\Theta_S| = R \ln 2$ nats. $a \lesssim b$ hides absolute constants. $h(\beta) = -\beta \ln \beta - (1 - \beta) \ln(1 - \beta)$ is the binary entropy in nats.

Standing assumptions. Each result states exactly which of the following it uses.

Assumption A1 (Bounded student log-probabilities). There is $B < \infty$ with $\log q_\theta(a | s) \geq -B$ for all θ, s and all a with $p(a | s) > 0$. This can be enforced, for example, by mixing with the uniform distribution at every state before end-of-sequence: $q = (1 - \eta)q' + \eta/V$ gives $B \leq \log(V/\eta)$. Restricting to the teacher’s support keeps the absorbing end-of-sequence convention intact, and it is all that Theorem 2.6 uses.

Assumption A2 (Bounded covariate shift). $\mu \ll \mathcal{D}$ and $\rho := \|\text{d}\mu/\text{d}\mathcal{D}\|_\infty < \infty$.

Assumption A3 (Realisability). There is $\theta^\circ \in \Theta_S$ with $q_{\theta^\circ}(\cdot | x) = p(\cdot | x)$ for $\mathcal{D}$ -a.e. x .

Assumption A4 (Finite description length). $|\Theta_S| \leq 2^R$, for example P parameters stored with b bits each, so that $R = Pb$. Alternatively Θ_S has ε -covering number $\mathcal{N}(\varepsilon)$ in the metric $\|\log q_\theta - \log q_{\theta'}\|_\infty$.

Assumption A5 (Linear read-out). $z_\theta(s) = Wh_\theta(s) + b$ with $W \in \mathbb{R}^{V \times d_S}$, $h_\theta(s) \in \mathbb{R}^{d_S}$. The same form holds for the teacher, with width d_T .

Assumption A6 (Stochastic optimisation). The training objective F is L -smooth and bounded below. The optimiser sees unbiased stochastic gradients with variance at most σ^2 .

Status labels. Each result carries one of these tags: [proved] means proved here with no assumption beyond the definitions; [proved under ...] means proved under the listed assumptions; [known result; proof sketch] means a result from the literature with a pointer to its key lemma; [proved within the stated model] means an exact calculation inside a stylised model, whose relevance to real networks is itself an assumption. Conjectures and heuristics are labelled as such. Several inequalities are additionally [numerically checked] by the scripts in `verification/`.

1 Formalising “lossless”

1.1 Divergences, distillation risk and task-level excess risk

Definition 1.1 (f -divergences used throughout). For convex $f : (0, \infty) \rightarrow \mathbb{R}$ with $f(1) = 0$, set $D_f(P\|Q) = \sum_y Q(y) f(P(y)/Q(y))$. We use the conventions $0f(0/0) = 0$ and $Qf(P/0) = Pf^*(0)$, where $f^*(0) := \lim_{u \rightarrow \infty} f(u)/u$. The instances we use are:

divergence	$f(u)$	$f''(1)$	bounded?
forward KL, $\text{KL}(P\ Q)$	$u \log u$	1	no
reverse KL, $\text{KL}(Q\ P)$	$-\log u$	1	no
total variation $\text{TV} = \frac{1}{2}\ P - Q\ _1$	$\frac{1}{2} u - 1 $	—	≤ 1
Hellinger ² , $H^2 = \frac{1}{2} \sum (\sqrt{P} - \sqrt{Q})^2$	$\frac{1}{2}(\sqrt{u} - 1)^2$	$\frac{1}{4}$	≤ 1
$\chi^2(P\ Q)$	$(u - 1)^2$	2	no
$\text{JSD}_\beta(P\ Q)$	$\beta u \log u - m \log m$, $m = \beta u + 1 - \beta$	$\beta(1 - \beta)$	$\leq h(\beta)$

Here $\text{JSD}_\beta(P\|Q) := \beta \text{KL}(P\|M_\beta) + (1 - \beta) \text{KL}(Q\|M_\beta)$ with $M_\beta = \beta P + (1 - \beta)Q$. This follows the convention of [Agarwal et al. \(2024\)](#); $\beta = \frac{1}{2}$ is the classical Jensen–Shannon divergence ([Lin, 1991](#)). Conventions differ between papers (some swap β and $1 - \beta$), so we fix this one. A direct computation gives $f''_\beta(u) = \beta(1 - \beta)/(um) > 0$. The Rényi divergence of order $\alpha > 1$ is $D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \sum_y P(y)^\alpha Q(y)^{1 - \alpha}$ ([van Erven and Harremoës, 2014](#)).

Definition 1.2 (Distillation risk; (ε, D, μ) -lossless). For a divergence D on $\Delta(\mathcal{Y})$ and an input law μ , the *sequence-level distillation risk* is $\mathcal{L}_D(\theta; \mu) := \mathbb{E}_{x \sim \mu} D(P^x \| Q_\theta^x)$. The student is (ε, D, μ) -*lossless* if $\mathcal{L}_D(\theta; \mu) \leq \varepsilon$. For a family of prefix laws $\nu = (\nu_t)_{t \leq T}$, the *token-level risk* is $\ell_D(\theta; \nu) := \sum_{t=1}^T \mathbb{E}_{s \sim \nu_t} D(p_t(\cdot | s) \| q_t(\cdot | s))$.

Definition 1.3 (Task-level excess risk). A task is a measurable reward $r : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$; a loss ℓ is handled as $r = 1 - \ell$. Its excess risk under sampling is $\Delta_r(\theta; \mu) := \mathbb{E}_{x \sim \mu} [\mathbb{E}_{y \sim P^x} r(x, y) - \mathbb{E}_{y \sim Q_\theta^x} r(x, y)]$. For a class $\mathcal{R}$ of tasks, $\Delta_{\mathcal{R}}(\theta; \mu) := \sup_{r \in \mathcal{R}} |\Delta_r(\theta; \mu)|$.

1.2 Relations between the definitions

Proposition 1.4 (TV is the worst-case task regret). [proved]

(a) $\sup_{r: \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]} \Delta_r(\theta; \mu) = \sup_r |\Delta_r(\theta; \mu)| = \mathbb{E}_{x \sim \mu} \text{TV}(P^x, Q_\theta^x)$.

- (b) If the task sees y only through a statistic $\phi(x, y)$, for example the final answer, so that $r = g(x, \phi(x, y))$, then $\sup_g |\Delta_r| = \mathbb{E}_{x \sim \mu} \text{TV}(\phi_{\#} P^x, \phi_{\#} Q_{\theta}^x) \leq \mathbb{E}_{x \sim \mu} \text{TV}(P^x, Q_{\theta}^x)$.
- (c) For each x , the minimum error probability of any test that tells, from one draw, whether the draw came from P^x or from Q_{θ}^x (equal priors) is $\frac{1}{2}(1 - \text{TV}(P^x, Q_{\theta}^x))$.

Proof. (a) For fixed x , $\sum_y r(x, y)(P^x(y) - Q_{\theta}^x(y)) \leq \sum_y (P^x(y) - Q_{\theta}^x(y))_+ = \text{TV}$, with equality at $r = \mathbf{1}\{P^x > Q_{\theta}^x\}$. This pointwise maximiser is measurable because $\mathcal{X} \times \mathcal{Y}$ is countable. Taking $1 - r$ covers the absolute value. Integrate over $x \sim \mu$. (b) Apply (a) to the push-forward laws, then use data processing for TV. (c) This is Le Cam's two-point identity: the Bayes test is $\mathbf{1}\{P^x > Q_{\theta}^x\}$. $\square$

Proposition 1.5 (Pinsker, Bretagnolle–Huber, Hellinger). *[known result; proof sketch] For any P, Q and $K \in \{\text{KL}(P\|Q), \text{KL}(Q\|P)\}$: (i) $\text{TV} \leq \sqrt{K/2}$ (Pinsker, 1964; Tsybakov, 2009); (ii) $\text{TV} \leq \sqrt{1 - e^{-K}}$ (Bretagnolle and Huber, 1979), which is the informative bound when $K > 2$; (iii) $H^2 \leq \text{TV} \leq H\sqrt{2 - H^2}$; (iv) $K \geq -2\log(1 - H^2) \geq 2H^2$.*

Proof of (iv). By Jensen, $\text{KL}(P\|Q) = -2\mathbb{E}_P \log \sqrt{Q/P} \geq -2\log \mathbb{E}_P \sqrt{Q/P} = -2\log(1 - H^2)$. The same argument applies with P and Q exchanged. The other parts are textbook; see Tsybakov (2009, Lemmas 2.3–2.6). $\square$

Lemma 1.6 (f -divergences are at most linear in TV). *[proved] For convex f with $f(1) = 0$, $D_f(P\|Q) \leq (f(0) + f^*(0)) \text{TV}(P, Q)$.*

Proof. Write $u = P/Q$. On $[0, 1]$, convexity gives $f(u) \leq (1 - u)f(0) + uf(1) = (1 - u)f(0)$. For $u > 1$ the slope $(f(u) - f(1))/(u - 1)$ is non-decreasing in u and tends to $f^*(0)$, so $f(u) \leq (u - 1)f^*(0)$. Hence $D_f \leq f(0)\sum_y (Q - P)_+ + f^*(0)\sum_y (P - Q)_+ = (f(0) + f^*(0))\text{TV}$. This is classical; see Sason and Verdú (2016). $\square$

Proposition 1.7 (Properties of JSD_{β}). *[proved] [numerically checked] For all P, Q and $\beta \in (0, 1)$: (i) $2\beta(1 - \beta) \text{TV}^2 \leq \text{JSD}_{\beta}(P\|Q) \leq h(\beta) \text{TV} \leq h(\beta)$; (ii) $\text{JSD}_{\beta}(P\|Q) \leq \beta(1 - \beta)[\text{KL}(P\|Q) + \text{KL}(Q\|P)]$; (iii) if $\text{supp } P = \text{supp } Q$ is finite, then $\text{JSD}_{\beta}/\beta \rightarrow \text{KL}(P\|Q)$ as $\beta \rightarrow 0$ and $\text{JSD}_{\beta}/(1 - \beta) \rightarrow \text{KL}(Q\|P)$ as $\beta \rightarrow 1$.*

Proof. (i) Since $\text{TV}(P, M_{\beta}) = (1 - \beta)\text{TV}(P, Q)$ and $\text{TV}(Q, M_{\beta}) = \beta \text{TV}(P, Q)$, Pinsker applied to each term gives $\text{JSD}_{\beta} \geq 2\beta(1 - \beta)^2 \text{TV}^2 + 2(1 - \beta)\beta^2 \text{TV}^2$. For the upper bound, apply Lemma 1.6 with $f_{\beta}(0) = (1 - \beta) \log \frac{1}{1 - \beta}$ and $f_{\beta}^*(0) = \lim_u [\beta \log u - \beta \log(\beta u)] = \beta \log \frac{1}{\beta}$, whose sum is $h(\beta)$. (ii) KL is convex in its second argument, so $\text{KL}(P\|\beta P + (1 - \beta)Q) \leq (1 - \beta)\text{KL}(P\|Q)$ and $\text{KL}(Q\|M_{\beta}) \leq \beta \text{KL}(Q\|P)$. (iii) As $\beta \rightarrow 0$, $M_{\beta} \rightarrow Q$, so $\text{KL}(P\|M_{\beta}) \rightarrow \text{KL}(P\|Q)$. Also $\text{KL}(Q\|Q + \beta(P - Q)) = \frac{\beta^2}{2} \chi^2(P\|Q) + O(\beta^3)$ on a finite common support. Hence $\text{JSD}_{\beta} = \beta \text{KL}(P\|Q) + O(\beta^2)$. The case $\beta \rightarrow 1$ is symmetric. $\square$

Proposition 1.8 (No global comparison; local equivalence). *[proved] [numerically checked]*

- (i) Forward and reverse KL are not comparable in general. For $P = \text{Bern}(\delta)$ and $Q = \text{Bern}(0)$, $\text{KL}(P\|Q) = \infty$ while $\text{KL}(Q\|P) = \log \frac{1}{1 - \delta} \rightarrow 0$. Exchanging the roles gives the opposite.
- (ii) Suppose $a \leq P(y)/Q(y) \leq b$ for all y (necessarily $a \leq 1 \leq b$). For every twice differentiable f with $f(1) = 0$,

$$\frac{1}{2} \inf_{[a, b]} f'' \cdot \chi^2(P\|Q) \leq D_f(P\|Q) \leq \frac{1}{2} \sup_{[a, b]} f'' \cdot \chi^2(P\|Q).$$

In particular $\frac{\chi^2}{2b} \leq \text{KL}(P\|Q) \leq \frac{\chi^2}{2a}$ and $\frac{\chi^2}{2b^2} \leq \text{KL}(Q\|P) \leq \frac{\chi^2}{2a^2}$. Hence $\text{KL}(P\|Q) \leq \frac{b^2}{a} \text{KL}(Q\|P)$ and $\text{KL}(Q\|P) \leq \frac{b}{a^2} \text{KL}(P\|Q)$. Also $\text{JSD}_{\beta} = \frac{\beta(1 - \beta)}{2} \chi^2(1 + O(b - a))$.

Proof. (ii) Replacing f by $\tilde{f}(u) = f(u) - f'(1)(u-1)$ leaves D_f unchanged, because $\sum_y Q(u-1) = 0$. Taylor's theorem gives $\tilde{f}(u) = \frac{1}{2}f''(\xi_u)(u-1)^2$ with $\xi_u \in [a, b]$. Multiply by Q and sum. For forward KL, $f''(u) = 1/u$; for reverse KL, written as $D_{-\log}(P\|Q)$, $f''(u) = 1/u^2$. $\square$

Remark 1.9 (What Proposition 1.8 means for “near-lossless”). In the near-lossless regime with bounded likelihood ratios, every smooth f -divergence equals $\frac{1}{2}f''(1)\chi^2$ to leading order. So forward KL, reverse KL and JSD_β (rescaled by $\beta(1-\beta)$) agree when the student is nearly lossless. The choice of divergence matters only (a) in the lossy, capacity-limited regime, (b) through tails and support mismatch, where likelihood ratios are unbounded, and (c) through *which prefix law* the per-token divergence is averaged over (Section 1.3).

Proposition 1.10 (Coverage versus precision: event bounds). *[proved] [numerically checked] For every event $A \subseteq \mathcal{Y}$:*

- (i) (coverage) $P(A) \leq \frac{\text{KL}(P\|Q) + \log 2}{\log(1/Q(A))}$;
- (ii) (precision) $Q(A) \leq \frac{\text{KL}(Q\|P) + \log 2}{\log(1/P(A))}$;
- (iii) (Rényi, $\alpha > 1$) $Q(A) \leq (e^{D_\alpha(Q\|P)} P(A))^{(\alpha-1)/\alpha}$;
- (iv) (additive) $|P(A) - Q(A)| \leq \text{TV}(P, Q)$.

Proof. (ii) By data processing onto $\{A, A^c\}$, $\text{KL}(Q\|P) \geq \text{kl}(Q(A)\|P(A))$, and $\text{kl}(a\|b) \geq a \log \frac{1}{b} - h(a) \geq a \log \frac{1}{b} - \log 2$. Part (i) is (ii) with the roles exchanged. (iii) By Hölder with exponents α and $\alpha/(\alpha-1)$, $Q(A) = \sum_A P(Q/P) \leq P(A)^{(\alpha-1)/\alpha} (\sum_y P(Q/P)^\alpha)^{1/\alpha}$, and $\sum_y P(Q/P)^\alpha = e^{(\alpha-1)D_\alpha(Q\|P)}$ (Mironov, 2017, Prop. 10). (iv) is the definition of TV. $\square$

Forward KL guarantees *coverage*: the student cannot give negligible probability to events the teacher finds likely. Reverse KL, and Rényi divergences of Q from P , guarantee *precision*: the student cannot put appreciable mass on events the teacher considers essentially impossible (for example hallucinated or unsafe continuations). TV controls neither multiplicatively. A student at $\text{TV} = 10^{-2}$ may emit a teacher-impossible output with probability 10^{-2} , which is invisible to TV but catastrophic at deployment scale.

Proposition 1.11 (Greedy decoding needs margins). *[proved] Let $a^*(s) = \arg \max_a p(a | s)$ have margin $\gamma(s) = p(a^* | s) - \max_{a \neq a^*} p(a | s)$. If $\text{TV}(p_t(\cdot | s), q_t(\cdot | s)) < \gamma(s)/2$ at every state s on the teacher's greedy path from x , then the student's greedy decoding of x equals the teacher's. The constant is tight: moving mass $\gamma/2$ from a^* to the runner-up creates a tie. Section 7 turns this condition into sharp bounds on the probability that the outputs differ.*

Proof. $|p(a | s) - q(a | s)| \leq \text{TV}$ for every a , so $q(a^* | s) - q(a | s) \geq \gamma(s) - 2\text{TV} > 0$. Induct along the path. $\square$

1.3 Sequence level versus token level

Theorem 1.12 (Chain rules: each KL direction has its own prefix law). *[proved] [numerically checked] For every x ,*

$$\text{KL}(P^x\|Q_\theta^x) = \sum_{t=1}^T \mathbb{E}_{s_t \sim P^x} \text{KL}(p_t(\cdot | s_t) \| q_t(\cdot | s_t)), \quad (2)$$

$$\text{KL}(Q_\theta^x\|P^x) = \sum_{t=1}^T \mathbb{E}_{s_t \sim Q_\theta^x} \text{KL}(q_t(\cdot | s_t) \| p_t(\cdot | s_t)). \quad (3)$$

After averaging over $x \sim \mathcal{D}$: $\mathbb{E}_{\mathcal{D}} \text{KL}(P\|Q_\theta) = T \varepsilon_{\text{off}}^{\text{KL}}$ (teacher prefixes) and $\mathbb{E}_{\mathcal{D}} \text{KL}(Q_\theta\|P) = T \varepsilon_{\text{on}}^{\text{KL}}$ (student prefixes).

Proof. $\log \frac{P^x(y)}{Q_\theta^x(y)} = \sum_t \log \frac{p_t(y_t|s_t)}{q_t(y_t|s_t)}$. Take $\mathbb{E}_{y \sim P^x}$ and condition each summand on s_t (tower property). The reverse identity is the same computation with $y \sim Q_\theta^x$. $\square$

Remark 1.13. Supervised distillation on teacher-generated sequences with soft labels at every position (the token-level analogue of [Kim and Rush \(2016\)](#)) minimises (2) *exactly*. On-policy reverse-KL distillation, as in the objectives of [Gu et al. \(2024\)](#) and [Agarwal et al. \(2024\)](#) with $\lambda = 1$ and reverse KL, minimises (3) exactly, provided the gradient is taken through the sampling distribution (Proposition 4.9). The two remaining pairings are forward KL on student prefixes (for example GKD with forward KL on on-policy data) and reverse KL on teacher prefixes. These are genuine discrepancies (Proposition 1.15) but do not equal any sequence-level f -divergence.

Proposition 1.14 (Hybrid argument for TV, and its converse). *[proved] [numerically checked]*
For every x ,

$$\text{TV}(P^x, Q_\theta^x) \leq \sum_{t=1}^T \mathbb{E}_{s_t \sim P^x} \text{TV}_t(s_t) \quad \text{and} \quad \text{TV}(P^x, Q_\theta^x) \leq \sum_{t=1}^T \mathbb{E}_{s_t \sim Q_\theta^x} \text{TV}_t(s_t),$$

where $\text{TV}_t(s) = \text{TV}(p_t(\cdot | s), q_t(\cdot | s))$. Conversely, for each t , $\mathbb{E}_{s_t \sim P^x} \text{TV}_t(s_t) \leq 2 \text{TV}(P^x, Q_\theta^x)$, and the same holds with Q_θ^x .

Proof. Let H_k draw $y_{1:k}$ from p and $y_{k+1:T}$ from q_θ , so that $H_T = P^x$ and $H_0 = Q_\theta^x$. H_k and H_{k-1} share the law of s_k and the continuation kernel after step k . By data processing, $\text{TV}(H_k, H_{k-1}) \leq \mathbb{E}_{s_k \sim P^x} \text{TV}_k(s_k)$. Telescope. For the second bound, exchange the roles of p and q_θ . For the converse, let Π_t^P and Π_t^Q be the laws of (s_t, y_t) and $\Pi^H := d_t^p \otimes q_t$. Then $\mathbb{E}_{d_t^p} \text{TV}_t = \text{TV}(\Pi_t^P, \Pi^H) \leq \text{TV}(\Pi_t^P, \Pi_t^Q) + \text{TV}(d_t^p, d_t^q) \leq 2 \text{TV}(P^x, Q_\theta^x)$, because marginalisation cannot increase TV. $\square$

Proposition 1.15 (The mismatched pairings are divergences with a $\sqrt{T}$ -lossy Pinsker bound). *[proved]* Let $H_{\text{on}}^{\text{KL}} := \sum_t \mathbb{E}_{s \sim Q_\theta^x} \text{KL}(p_t \| q_t)$ and $H_{\text{off}}^{\text{rKL}} := \sum_t \mathbb{E}_{s \sim P^x} \text{KL}(q_t \| p_t)$. Each is ≥ 0 , and each vanishes iff $P^x = Q_\theta^x$. Moreover $\text{TV}(P^x, Q_\theta^x) \leq \sqrt{TH/2}$ for either one. By contrast, the exact pairings give $\text{TV} \leq \sqrt{\text{KL}/2}$ with KL the corresponding sequence-level KL. The factor T is an artefact of the proof, not a proven loss. A sequential Hellinger subadditivity inequality ([Foster et al., 2021](#)) gives $\text{TV} \leq C\sqrt{H \log T}$ instead. Multi-start local search over binary autoregressive pairs finds worst ratios $\text{TV}^2/H \approx 0.50, 0.60, 0.69, 0.74$ for $T = 1, \dots, 4$, far below $T/2$. These are lower estimates of the true worst case.

Proof. If $H_{\text{on}}^{\text{KL}} = 0$, then $p_1 = q_1$ at x . So $d_2^p = d_2^q$, hence $p_2 = q_2$ on the common support, and so on by induction. The bound follows from Proposition 1.14, Pinsker per token, and Cauchy–Schwarz: $\sum_t \mathbb{E} \text{TV}_t \leq \sum_t \sqrt{e_t/2} \leq \sqrt{T \sum_t e_t/2}$. $\square$

Corollary 1.16 (Sequence-level TV from per-token errors). *[proved]*

$$\mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta) \leq \min \left\{ \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}, \sqrt{1 - e^{-T \varepsilon_{\text{off}}^{\text{KL}}}}, \sqrt{T \varepsilon_{\text{on}}^{\text{rKL}}/2}, T \varepsilon_{\text{off}}^{\text{TV}}, T \varepsilon_{\text{on}}^{\text{TV}} \right\}.$$

Proof. Combine Theorem 1.12 and Propositions 1.5 and 1.14 with Jensen. The maps $k \mapsto \sqrt{k}$ and $k \mapsto \sqrt{1 - e^{-k}}$ are concave. $\square$

Example 1.17 (Neither route dominates). *[proved]* (a) *Diffuse errors.* Let $p_t = \text{Bern}(\frac{1}{2})$ and $q_t = \text{Bern}(\frac{1}{2} + \delta)$ be i.i.d. Then $T \varepsilon_{\text{off}}^{\text{TV}} = T\delta$, whereas $\text{KL}(P \| Q) = -\frac{T}{2} \log(1 - 4\delta^2)$, so the Pinsker route gives $\text{TV} \leq \sqrt{T} \delta (1 + o(1))$. This is the correct order, by the central limit theorem for the log-likelihood ratio, and it improves on the hybrid route by $\sqrt{T}$. (b) *Concentrated errors (deterministic teacher).* Let $p_t = \delta_a$ and $q_t(a) = 1 - \delta$. Then $\text{TV} = 1 - (1 - \delta)^T \leq T\delta$ and

$KL = -T \log(1 - \delta)$, so $\sqrt{KL/2} \approx \sqrt{T\delta/2} \gg T\delta$ when $T\delta \ll 1$. Here the hybrid route is tight. In general, if P is a point mass then $TV(P, Q) = 1 - e^{-KL(P\|Q)} \leq KL(P\|Q)$: control is linear rather than square-root.

Proposition 1.18 (Covariate shift). *[proved] For any measurable $g \geq 0$ (for example a per-input divergence), under Assumption A2, $\mathbb{E}_\mu g \leq \rho \mathbb{E}_D g$. If $0 \leq g \leq G$, then without any density-ratio assumption, $|\mathbb{E}_\mu g - \mathbb{E}_D g| \leq G TV(\mu, D)$. Bounded criteria (TV , H^2 , JSD_β) therefore transfer across input shift additively. Unbounded ones (KL) need Assumption A2.*

1.4 Which definition is the right one?

Definition 1.19 (Hierarchy of near-losslessness). At tolerance ε and deployment law μ , the student is:

behaviourally lossless if $\mathbb{E}_\mu TV(P^x, Q_\theta^x) \leq \varepsilon$;

likelihood lossless if $\mathbb{E}_\mu KL(P^x \| Q_\theta^x) \leq \varepsilon$;

two-sided lossless if $\mathbb{E}_\mu [KL(P^x \| Q_\theta^x) + KL(Q_\theta^x \| P^x)] \leq 2\varepsilon$;

decoding lossless if it is behaviourally lossless and the margin condition of Proposition 1.11 holds on the teacher’s greedy paths (Section 7 quantifies this notion).

Position taken in this document.

1. *The target is sequence-level, on the deployment law.* Behavioural losslessness, $\mathbb{E}_\mu TV(P^x, Q_\theta^x) \leq \varepsilon$, is *exactly* the statement that no bounded downstream task evaluated by sampling changes by more than ε (Proposition 1.4). It is also exactly the statistical indistinguishability of the two models from one sample. It is symmetric, bounded, and transfers across covariate shift additively (Proposition 1.18). Token-level criteria are *surrogates*; they relate to the target only through Theorem 1.12 and Proposition 1.14.
2. *TV alone is not enough when probabilities are consumed or events are rare.* If the student’s likelihoods are used (scoring, calibration, uncertainty, perplexity), the relevant quantity is the log-loss regret, which is *exactly* $\mathbb{E}_\mu KL(P^x \| Q_\theta^x)$. If the student is sampled at scale and rare events matter (hallucinations, unsafe outputs), the precision bound of Proposition 1.10(ii–iii) requires the *reverse* direction. Hence two-sided losslessness. If decoding is greedy or beam search, margins matter too (Proposition 1.11). Sampling-level losslessness then controls the decoded output only through the teacher’s probability of its own greedy prefix (Theorem 7.9), which can be exponentially small.
3. *Among trainable surrogates, sequence-level forward KL is canonical.* It dominates behavioural loss via Pinsker and gives log-loss exactly. By Theorem 1.12 its per-token decomposition uses *teacher prefixes only*, so it is estimable without student roll-outs and has no exposure-bias term (Section 2.5). It is convex in the logits (Proposition 2.11). More generally, KL is the only divergence on the simplex that is both an f -divergence and a Bregman divergence (Amari, 2009). That characterisation singles out the KL *family*, not a direction: both argument orders qualify.
4. *But when exactness is impossible, the surrogate must be chosen for the target.* Under misspecification, minimisers of either KL direction can be arbitrarily bad in TV, whereas minimisers of a bounded divergence such as JSD_β are TV-competitive (Proposition 4.13). This is why the objective proposed in Section 4 is a *sequence-level* JSD_β with an optional forward-KL anchor.
5. *In the truly near-lossless regime the question dissolves.* By Proposition 1.8, all smooth divergences agree to second order when likelihood ratios are bounded. The practically important choice is then the *prefix law* and the *estimator*, not the divergence.

In one sentence: a student should be called ε -lossless if it is two-sided lossless at level ε under the deployment input law (behavioural losslessness at level $\sqrt{\varepsilon/2}$ then follows). It should be trained with a sequence-level divergence whose minimiser remains TV-competitive when ε -losslessness is out of reach.

2 Decomposing the student’s gap

2.1 Pipeline and master decomposition

Pipeline. We draw n prompts $x_i \sim \mathcal{D}$ i.i.d. For each prompt the teacher generates $y^{(i)} \sim P^{x_i}$ and supplies either soft labels $p_t(\cdot | s_t^{(i)})$ at every position, or only the sampled tokens (hard labels). On-policy variants additionally sample $y \sim Q_\theta^{x_i}$. Training uses a prefix law $\nu(\theta) = (\nu_t(\theta))_t$: $\nu_t = d_t^p$ (off-policy), $\nu_t = d_t^\theta$ (on-policy), or a mixture $\nu_t = \lambda d_t^\theta + (1 - \lambda)d_t^p$ (Agarwal et al., 2024). The per-state loss is $\ell_\theta(s) = D(p_t(\cdot | s) \| q_t(\cdot | s))$. The population and empirical training objectives are

$$F(\theta) := \mathbb{E}_{x \sim \mathcal{D}} \sum_{t=1}^T \mathbb{E}_{s \sim \nu_t(\theta) | x} \ell_\theta(s), \quad \hat{F}_n(\theta) := \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^T \ell_\theta(s_t^{(i)}).$$

For on-policy terms, $\hat{F}_n$ is defined with fixed sampling noise (for example Gumbel-max reparametrisation), so that it is a fixed random function of θ . Let $\theta^* \in \arg \min_{\Theta_S} F$, $\hat{\theta} \in \arg \min \hat{F}_n$, and let $\tilde{\theta}$ be the output of the optimiser. The *evaluation functional* $\mathcal{E}_\mu(\theta)$ is a sequence-level criterion under the deployment law, for example $\mathbb{E}_\mu \text{KL}(P^x \| Q_\theta^x)$ or a task regret. $\mathcal{E}_\mathcal{D}$ is the same functional under $\mathcal{D}$.

Theorem 2.1 (Master decomposition). *[proved]*

$$\mathcal{E}_\mu(\tilde{\theta}) = \underbrace{F(\theta^*)}_A \text{ approximation} + \underbrace{F(\tilde{\theta}) - F(\theta^*)}_{\text{estimation+optimisation}} + \underbrace{\mathcal{E}_\mathcal{D}(\tilde{\theta}) - F(\tilde{\theta})}_{\mathcal{E}_{\text{shift}} \text{ prefix shift}} + \underbrace{\mathcal{E}_\mu(\tilde{\theta}) - \mathcal{E}_\mathcal{D}(\tilde{\theta})}_{\mathcal{E}_{\text{cov}} \text{ covariate shift}}, \quad (4)$$

$$F(\tilde{\theta}) - F(\theta^*) \leq \underbrace{2 \sup_{\theta} |\hat{F}_n(\theta) - F(\theta)|}_{\mathcal{E}_{\text{est}}} + \underbrace{\hat{F}_n(\tilde{\theta}) - \hat{F}_n(\hat{\theta})}_{\mathcal{E}_{\text{opt}}}. \quad (5)$$

Proof. (4) telescopes. For (5), write $F(\tilde{\theta}) - F(\theta^*) = [F - \hat{F}_n](\tilde{\theta}) + [\hat{F}_n(\tilde{\theta}) - \hat{F}_n(\hat{\theta})] + [\hat{F}_n(\hat{\theta}) - \hat{F}_n(\theta^*)] + [\hat{F}_n - F](\theta^*)$, and note that the third bracket is ≤ 0 . This is the approximation–estimation–optimisation split of Bottou and Bousquet (2008) with two shift terms added. $\square$

When $\mathcal{E}$ is a KL, the behavioural criterion follows from Pinsker: $\mathbb{E}_\mu \text{TV}(P^x, Q_\theta^x) \leq \sqrt{\mathcal{E}_\mu(\tilde{\theta})/2}$. The key structural fact is that $\mathcal{E}_{\text{shift}} \equiv 0$ for exactly two pairings (Theorem 1.12). The table lists what each pairing’s training objective F equals and the resulting bound on sequence-level behaviour:

per-token loss	prefix law	F equals	$\mathbb{E}_\mathcal{D} \text{TV}(P, Q_\theta) \leq$
forward KL	teacher	$\mathbb{E}_\mathcal{D} \text{KL}(P \ Q_\theta)$; $\mathcal{E}_{\text{shift}} = 0$	$\sqrt{F/2}$
reverse KL	student	$\mathbb{E}_\mathcal{D} \text{KL}(Q_\theta \ P)$; $\mathcal{E}_{\text{shift}} = 0$	$\sqrt{F/2}$
forward KL	student	hybrid (Proposition 1.15)	$\sqrt{TF/2}$
reverse KL	teacher	hybrid (Proposition 1.15)	$\sqrt{TF/2}$
TV	either	hybrid (Proposition 1.14)	F
JSD $_\beta$	either	hybrid	$\sqrt{TF/(2\beta(1-\beta))}$

For the hybrid rows, the factor T inside the square root comes from our Cauchy–Schwarz argument. A sequential subadditivity inequality for squared Hellinger distance (Foster et al., 2021) improves it to $O(\log T)$; we have not re-derived that lemma’s constant. For hybrid rows, the on-policy per-token loss can also exceed the training loss. For bounded losses the excess is at most $G \sum_t \sum_{k < t} e_k^{\text{off,TV}}$, which is $O(T^2)$ in the per-token TV error (Proposition 2.19). For reverse KL on teacher prefixes it is unbounded relative to the training loss.

2.2 Approximation error

Lemma 2.2 (Log-ratio variance). *[proved] [numerically checked] Let $\delta = \log(p/q)$ on $\text{supp } p$ and $M := \max\{0, \max_y \delta(y)\}$. Then $\text{KL}(p\|q) \geq \frac{e^{-M}}{2} \mathbb{E}_p[\delta^2] \geq \frac{e^{-M}}{2} \text{Var}_p(\delta)$.*

Proof. $\mathbb{E}_p[e^{-\delta}] = q(\text{supp } p) \leq 1$. Taylor expansion gives $e^{-\delta} = 1 - \delta + \frac{\delta^2}{2}e^{-\xi}$ with ξ between 0 and δ , so $e^{-\xi} \geq e^{-M}$. Taking $\mathbb{E}_p$: $1 \geq 1 - \text{KL}(p\|q) + \frac{e^{-M}}{2} \mathbb{E}_p \delta^2$. $\square$

Theorem 2.3 (Softmax-bottleneck approximation floor). *[proved under Assumption A5] [numerically checked] Let $s_1, \dots, s_N$ be contexts at which the teacher has full support. Let $A \in \mathbb{R}^{N \times V}$ have rows $\log p(\cdot | s_i)$, let $\pi_i := \min_y p(y | s_i)$, and let $\Pi = \text{diag}(\pi_i)$. For every student of width d_S with $\max_{i,y} \log \frac{p(y|s_i)}{q_\theta(y|s_i)} \leq M$,*

$$\frac{1}{N} \sum_{i=1}^N \text{KL}(p(\cdot | s_i) \| q_\theta(\cdot | s_i)) \geq \frac{e^{-M}}{2N} \sum_{j > d_S + 1} \sigma_j^2(\Pi^{1/2} AC).$$

Proof. By Lemma 2.2, $\text{KL}_i \geq \frac{e^{-M}}{2} \text{Var}_{p_i}(\delta_i)$. Also $\text{Var}_{p_i}(\delta_i) = \min_c \mathbb{E}_{p_i}(\delta_i - c)^2 \geq \pi_i \min_c \|\delta_i - c\mathbf{1}\|^2 = \pi_i \|C\delta_i\|^2$. The student’s log-probability rows are $Wh_i + b - \text{lse}_i \mathbf{1}$, so the centred matrix ZC has rows $C(Wh_i + b)$ and rank at most $d_S + 1$. Left-multiplication by the diagonal $\Pi^{1/2}$ does not increase rank. Hence $\sum_i \pi_i \|C\delta_i\|^2 = \|\Pi^{1/2}(A - Z)C\|_F^2 \geq \min_{\text{rank}(Y) \leq d_S + 1} \|\Pi^{1/2}AC - Y\|_F^2$, which equals the tail sum by Eckart–Young–Mirsky (Eckart and Young, 1936). $\square$

Remark 2.4. (i) If the teacher also has a linear read-out of width d_T , then $\text{rank}(AC) \leq d_T + 1$. The floor is then a sum over the teacher’s singular values with indices in $(d_S + 1, d_T + 1]$, and it vanishes if $d_S \geq d_T$. This quantifies the “softmax bottleneck” (Yang et al., 2018). Its relevance to small language models was studied empirically by Godey et al. (2024); we are fairly, but not fully, confident of that reference’s exact claims. (ii) The factor π_i is crude. Replacing $\Pi^{1/2}C$ by any common lower bound $\Gamma^{1/2}$ on the Fisher matrices $\text{diag}(p_i) - p_i p_i^\top$ gives the tail spectrum of $A\Gamma^{1/2}$. In the check of `verification/check_capacity.py`, the row-weighted floor is within a factor 15 of a fitted student’s loss. (iii) The floor is a statement about the *read-out*. When $d_S \geq d_T$ it is silent; information-theoretic floors (Section 3) take over.

Corollary 2.5 (Approximation grows linearly with the horizon). *[proved] If the per-token approximation floor on teacher prefixes is $\geq \varepsilon_0$ at every t , then $\min_\theta \mathbb{E}_\mathcal{D} \text{KL}(P\|Q_\theta) \geq T\varepsilon_0$. Likelihood losslessness at level ε therefore requires $\varepsilon_0 \leq \varepsilon/T$. For TV, Proposition 1.14 gives only $\mathbb{E} \text{TV}(P, Q_\theta) \geq \frac{1}{2} \max_t \mathbb{E}_{d_t^p} \text{TV}_t$. Whether TV grows with T depends on how the errors are structured (Example 1.17).*

2.3 Estimation error

Theorem 2.6 (Uniform deviation, soft-label off-policy KD). *[proved under Assumption A1] With soft labels, $\ell(x, y; \theta) = \sum_t \text{KL}(p_t(s_t) \| q_t(s_t)) \in [0, TB]$, since $\text{KL}(p\|q) \leq \mathbb{E}_p[-\log q] \leq B$. With probability $\geq 1 - \delta$,*

$$\sup_\theta |\hat{F}_n(\theta) - F(\theta)| \leq 2\mathfrak{R}_n(\ell \circ \Theta_S) + TB \sqrt{\frac{\log(2/\delta)}{2n}}.$$

For a finite class (Assumption A4), $\mathfrak{R}_n \leq TB \sqrt{2R \ln 2/n}$ by Massart’s lemma.

Proof. Symmetrisation and McDiarmid (Bartlett and Mendelson, 2002). $\square$

This bound is pessimistic: a slow rate, and linear in T . For the realisable case, a horizon-free fast rate holds even with *hard* labels. The logic is due to Zhang (2006) and Wong and Shen (1995); its horizon-independence in imitation learning was emphasised by Foster et al. (2024).

Theorem 2.7 (Horizon-free rate for sequence-level hard-label KD). *[proved under Assumptions A3 and A4] Let $\hat{\theta}$ maximise $\sum_{i=1}^n \log Q_{\theta}(y^{(i)} | x_i)$ over Θ_S , with $y^{(i)} \sim P^{x_i}$. With probability $\geq 1 - \delta$,*

$$\mathbb{E}_{x \sim \mathcal{D}} H^2(P^x, Q_{\hat{\theta}}^x) \leq \frac{\log(|\Theta_S|/\delta)}{n} \leq \frac{R \ln 2 + \log(1/\delta)}{n},$$

$$\mathbb{E}_{x \sim \mathcal{D}} \text{TV}(P^x, Q_{\hat{\theta}}^x) \leq \sqrt{\frac{2(R \ln 2 + \log(1/\delta))}{n}}.$$

Neither bound depends on T .

Proof. See Appendix A.1. The key lemma is $\mathbb{E}_{y \sim P} \sqrt{Q(y)/P(y)} = 1 - H^2(P, Q) \leq e^{-H^2}$, combined with a Chernoff bound and a union bound over Θ_S .

Proposition 2.8 (Soft labels Rao–Blackwellise hard labels). *[proved] Fix a state law ν with states drawn independently of labels. Let $\hat{h} = -\log q_{\theta}(y | s)$ and $\hat{s} = \text{CE}(p(\cdot | s), q_{\theta}(\cdot | s))$, where $s \sim \nu$ and $y \sim p(\cdot | s)$. Then $\mathbb{E}\hat{h} = \mathbb{E}\hat{s}$ and*

$$\text{Var } \hat{h} - \text{Var } \hat{s} = \mathbb{E}_{s \sim \nu} \text{Var}_{y \sim p(\cdot | s)} \log q_{\theta}(y | s) \geq 0,$$

$$\mathbb{E} \|\nabla \hat{h} - \nabla \mathbb{E}\hat{h}\|^2 - \mathbb{E} \|\nabla \hat{s} - \nabla \mathbb{E}\hat{s}\|^2 = \mathbb{E}_{\nu} \text{tr Cov}_{y \sim p(\cdot | s)} \nabla_{\theta} \log q_{\theta}(y | s).$$

At $\theta = \theta^{\circ}$ (Assumption A3), the gradient gap equals $\mathbb{E}_{\nu} \text{tr } \mathcal{I}(s)$. Here $\mathcal{I}(s) = J(s)^{\top} (\text{diag } p - pp^{\top}) J(s)$ is the per-state Fisher information and $J = \partial z / \partial \theta$. The gap vanishes wherever $p(\cdot | s)$ is a point mass.

Proof. $\hat{s} = \mathbb{E}[\hat{h} | s]$, so the law of total variance applies; likewise for the gradient, since $\nabla \hat{s} = \mathbb{E}[\nabla \hat{h} | s]$. At θ° , $\text{Cov}_{y \sim p} \nabla \log q = J^{\top} \text{Cov}_{y \sim p}(e_y) J = J^{\top} (\text{diag } p - pp^{\top}) J$. $\square$

Remark 2.9. For whole teacher-sampled sequences, later states depend on earlier labels. Position-wise Rao–Blackwellisation still holds, but cross-time covariances can in principle make the variance of the *sum* larger. We therefore claim no sequence-level variance ordering. The variance-reduction view of distillation is developed by Menon et al. (2021). Related analyses include Lopez-Paz et al. (2016), Phuong and Lampert (2019), Dao et al. (2021) and Harutyunyan et al. (2023).

Proposition 2.10 (Soft labels: exact recovery versus $k/2n$). *Let $q_{\theta}(\cdot | s) = \text{softmax}(\Theta \phi(s))$ with $\Theta \in \mathbb{R}^{V \times d}$, and let $p = q_{\theta^{\circ}}$.*

- (a) *[proved] If $\phi(s_1), \dots, \phi(s_n)$ span $\mathbb{R}^d$ (so $n \geq d$), every minimiser of $\sum_i \text{KL}(p(\cdot | s_i) \| q_{\theta}(\cdot | s_i))$ satisfies $q_{\theta} = p$ at every s . Estimation error is zero.*
- (b) *[known result; proof sketch] If $\mathbb{E}[\phi \phi^{\top}] \succ 0$, the hard-label MLE from n i.i.d. pairs (s_i, y_i) satisfies $n \text{KL}(p \| q_{\hat{\theta}}) \Rightarrow \frac{1}{2} \chi_{(V-1)d}^2$ in distribution. The limiting mean is $\frac{(V-1)d}{2}$. This is the classical $k/2n$ law for a regular, well-specified model with k free parameters, the computation behind AIC (Akaike, 1974); see White (1982) for the misspecified case. The expectation $\mathbb{E} \text{KL}$ is infinite for every n , because with positive probability the MLE does not exist or puts zero mass on an observed class. The $k/2n$ law is therefore a statement about the limit law, not about $\mathbb{E} \text{KL}$.*

Proof of (a). The objective is 0 at Θ° . It vanishes at Θ iff $(\Theta - \Theta^{\circ})\phi(s_i) \in \text{span}(\mathbf{1})$ for all i , which means $C(\Theta - \Theta^{\circ})\phi(s_i) = 0$. By spanning, $C(\Theta - \Theta^{\circ}) = 0$. So the logits differ by a constant vector at every s , and the softmax is unchanged. $\square$

In the realisable case, a soft label is therefore worth “infinitely many” hard labels at its state. Once the teacher’s labels are exact, what remains is extrapolation, which is controlled by coverage of the feature space and by approximation, not by label noise.

2.4 Optimisation error

Proposition 2.11 (Curvature in logit space). *[proved] [numerically checked]*

- (i) $z \mapsto \text{KL}(p \parallel \text{softmax } z)$ is convex. Its Hessian satisfies $0 \preceq \text{diag}(q) - qq^\top \preceq \frac{1}{2}C$ (Böhning, 1992), so it is $\frac{1}{2}$ -smooth.
- (ii) $z \mapsto \text{KL}(\text{softmax } z \parallel p)$ is not convex, for any $V \geq 2$ and any p of full support. With $V = 2$, $f(z) = \text{KL}(\sigma(z) \parallel \text{Bern}(p))$ has $f''(z) = \sigma'(z) [1 + (1 - 2\sigma(z))(z - z_p)]$ with $z_p = \text{logit } p$, which is negative whenever $(2\sigma(z) - 1)(z - z_p) > 1$; for example $p = 0.3$, $z = z_p + 3$. For general V , along the ray $z = (t, 0, \dots, 0)$ the second derivative is $-(V - 1)(t - 2)e^{-t} + O(e^{-t}) < 0$ for large t .
- (iii) If p has full support, then for β close to 1, $z \mapsto \text{JSD}_\beta(p \parallel \text{softmax } z)$ is not convex: $\text{JSD}_\beta/(1 - \beta) \rightarrow \text{KL}(q \parallel p)$ together with its second derivatives, uniformly on compacts.
- (iv) Even with logits linear in a scalar θ , reverse KL can have spurious local minima. Take two contexts, $V = 2$, logits $z_1 = 5.2 - \theta/2$ and $z_2 = -1.5 - \theta/3$, and teacher probabilities 0.79 and 0.26 for the first token. The summed reverse KL has strict local minima at $\theta \approx -1.00$ (value 0.2191) and $\theta \approx 7.23$ (value 0.2300), separated by a barrier of height 0.2678 at $\theta \approx 3.61$. The forward KL has a single minimum.

Consequently, for any student whose logits are linear in θ (for example, training only the read-out of Assumption A5), off-policy forward-KL distillation is a convex problem. Reverse-KL objectives can have spurious local minima even then (iv).

Proof. (i) The Hessian of lse is $\text{diag}(q) - qq^\top$; Böhning’s bound gives the smoothness constant. (ii) $f'(z) = \sigma(z)(1 - \sigma(z))(z - z_p)$, because $\log \frac{q}{1-q} = z$; differentiate once more. On the ray, $q_j = e^{-t}(1 + O(e^{-t}))$ for $j > 1$, and the reverse KL equals $\log \frac{1}{p_1} - (V - 1)te^{-t} + O(e^{-t})$; differentiate twice. (iii) Write $\text{JSD}_\beta/(1 - \beta) = \text{KL}(q \parallel M_\beta) + \frac{\beta}{1-\beta} \text{KL}(p \parallel M_\beta)$. Since $p > 0$, $M_\beta \geq \beta p$ is bounded away from 0, so $\text{KL}(q \parallel M_\beta)$ is jointly smooth in (β, z) up to $\beta = 1$, where it equals $\text{KL}(q \parallel p)$. Let $\epsilon = 1 - \beta$ and $g(s) := \text{KL}(p \parallel p + s(q - p))$. Then $g(0) = g'(0) = 0$, so $g(\epsilon)/\epsilon = \epsilon \int_0^1 (1-t) g''(\epsilon t) dt$. This is jointly smooth in (ϵ, z) and vanishes at $\epsilon = 0$ together with its z -derivatives. Hence second z -derivatives converge uniformly on compacts, and the strict negative curvature of (ii) persists for β near 1. (iv) is a direct computation (`check_divergences.py`). $\square$

Theorem 2.12 (Stochastic optimisation). *[known result; proof sketch]* Under Assumption A6, SGD with a tuned constant step size satisfies $\min_{k < K} \mathbb{E} \|\nabla F(\theta_k)\|^2 \lesssim \frac{L\Delta}{K} + \sigma \sqrt{\frac{L\Delta}{K}}$ with $\Delta = F(\theta_0) - \inf F$ (Ghadimi and Lan, 2013). If F is convex (Proposition 2.11), then $\mathbb{E} F(\bar{\theta}_K) - \min F \lesssim \frac{L\|\theta_0 - \theta^*\|^2}{K} + \frac{\sigma\|\theta_0 - \theta^*\|}{\sqrt{K}}$. In both cases σ^2 enters, so the Rao–Blackwell gap of Proposition 2.8 translates directly into lower optimisation error at fixed compute.

Remark 2.13. No non-convex guarantee beyond stationarity is available for deep students. Empirically, optimisation is often the binding constraint on fidelity. Stanton et al. (2021) find that students frequently fail to match their teachers even when they have the capacity to do so. Beyer et al. (2022) find that long, “patient and consistent” schedules are needed. We treat $\mathcal{E}_{\text{opt}}$ as measurable (Section 6.3), not as provably small.

2.5 Distribution shift and exposure bias

Let $c_t : \mathcal{S}_t \times \mathcal{Y} \rightarrow [0, 1]$ be per-step costs and $J_c(\pi) := \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi} \sum_{t=1}^T c_t(s_t, y_t)$. Let $r : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$ be a terminal reward and $J_r(\pi) := \mathbb{E} r(x, y)$. The teacher’s cost-to-go is

$Q_t^p(s, a) := c_t(s, a) + \mathbb{E}_p[\sum_{k>t} c_k \mid s_t = s, y_t = a]$, and its *recoverability constant* is

$$u := \max_t \sup_s \left[\max_a Q_t^p(s, a) - \min_a Q_t^p(s, a) \right] \in [0, T].$$

Theorem 2.14 (Terminal rewards: linear in T). *[proved]* $|J_r(p) - J_r(q_\theta)| \leq \mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta) \leq T \min\{\varepsilon_{\text{off}}^{\text{TV}}, \varepsilon_{\text{on}}^{\text{TV}}\}$. Also $|J_r(p) - J_r(q_\theta)| \leq \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}$.

Proof. Propositions 1.4 and 1.14 and Corollary 1.16. $\square$

Theorem 2.15 (Cumulative costs, off-policy: quadratic, and tight). *[proved] [numerically checked]*

$$|J_c(p) - J_c(q_\theta)| \leq \sum_{t=1}^T (T - t + 1) e_t^{\text{off,TV}} \leq T^2 \varepsilon_{\text{off}}^{\text{TV}}.$$

The order is attained. Take a deterministic teacher emitting a^ in every “on-track” state. Let every deviation lead to absorbing off-track states that cost 1 per step. Let the student deviate with probability ε on track. Then $\varepsilon_{\text{off}}^{\text{TV}} = \varepsilon$ and $J_c(q_\theta) - J_c(p) = \sum_{t=1}^T [1 - (1 - \varepsilon)^t] = \varepsilon \frac{T(T+1)}{2} (1 + O(\varepsilon T))$.*

Proof. $|\mathbb{E}_p c_t - \mathbb{E}_{q_\theta} c_t| \leq \text{TV}$ between the laws of $y_{\leq t}$, which is at most $\sum_{k \leq t} e_k^{\text{off,TV}}$ by the hybrid argument (Proposition 1.14) applied to the first t tokens. Sum over t . In the example, the student is on track at step t with probability $(1 - \varepsilon)^{t-1}$. $\square$

Theorem 2.16 (On-policy: linear with recoverability). *[proved] [numerically checked]* The performance-difference identity (Kakade and Langford, 2002) holds:

$$J_c(q_\theta) - J_c(p) = \sum_{t=1}^T \mathbb{E}_{s \sim d_t^\theta} \left[\mathbb{E}_{a \sim q_t(\cdot|s)} Q_t^p(s, a) - V_t^p(s) \right], \quad V_t^p(s) := \mathbb{E}_{a \sim p_t(\cdot|s)} Q_t^p(s, a).$$

Hence $|J_c(q_\theta) - J_c(p)| \leq u T \varepsilon_{\text{on}}^{\text{TV}}$ and, by Pinsker per token and Jensen, $|J_c(q_\theta) - J_c(p)| \leq u T \sqrt{\varepsilon_{\text{on}}^{\text{KL}}/2}$.

Proof. Telescope $V_t^p(s) := \mathbb{E}_{a \sim p} Q_t^p(s, a)$ along the student’s trajectory. For any f with range at most u , $|\mathbb{E}_q f - \mathbb{E}_p f| \leq u \text{TV}(p, q)$. $\square$

Remark 2.17 (What on-policy training buys). Theorems 2.15 and 2.16 are the distillation versions of Ross and Bagnell (2010) and Ross et al. (2011). On-policy training replaces $T^2 \varepsilon_{\text{off}}$ by $u T \varepsilon_{\text{on}}$. It removes compounding only if (a) the teacher is *recoverable* ($u = O(1)$), and (b) training actually makes ε_{on} small, which DAgger-type no-regret arguments deliver relative to the best student *on its own state distribution*. In the non-recoverable cliff of Theorem 2.15, $u = T$, and no training distribution avoids $\Theta(T^2 \varepsilon)$. For language models, u is the worst-case cost of one bad token followed by the teacher taking over. It is small for fluency-type costs and can be $\Theta(T)$ for commitments such as a wrong early step in a proof. Related analyses of exposure bias include Bengio et al. (2015), Ranzato et al. (2016) and Arora et al. (2022). On-policy distillation is studied by Lin et al. (2020), Gu et al. (2024), Agarwal et al. (2024) and Ko et al. (2024).

Theorem 2.18 (KL chain rules and compounding). *[proved]*

- (i) Off-policy forward KL has $\mathcal{E}_{\text{shift}} = 0$ for $\mathcal{E} = \text{KL}(P \| Q_\theta)$, and $\mathbb{E} \text{TV}(P, Q_\theta) \leq \sqrt{T \varepsilon_{\text{off}}^{\text{KL}}/2}$.
- (ii) For cumulative costs, $|J_c(p) - J_c(q_\theta)| \leq \sum_{t=1}^T \sqrt{\frac{1}{2} \sum_{k \leq t} e_k^{\text{off,KL}}} \leq T^{3/2} \sqrt{\varepsilon_{\text{off}}^{\text{KL}}/2}$.
- (iii) On-policy reverse KL has $\mathcal{E}_{\text{shift}} = 0$ for $\mathcal{E} = \text{KL}(Q_\theta \| P)$. With recoverability, $|J_c(q_\theta) - J_c(p)| \leq u T \sqrt{\varepsilon_{\text{on}}^{\text{KL}}/2}$.

Proof. (i) and (iii) are Theorem 1.12. (ii) The law of $y_{\leq t}$ has KL $\sum_{k \leq t} e_k^{\text{off,KL}}$ by the truncated chain rule; apply Pinsker per t . The last claim is Theorem 2.16. $\square$

Bound (ii) improves on Theorem 2.15 only when errors are *diffuse* (Example 1.17(a)): there $e^{\text{off,TV}} \approx \sqrt{e^{\text{off,KL}}}$, so $T^{3/2}\sqrt{\varepsilon}$ beats $T^2\sqrt{\varepsilon}$. For concentrated errors (deterministic teacher, Example 1.17(b)) the TV route is tighter. In the cliff with $T = 100$ and $\varepsilon = 10^{-3}$, the true gap is 4.9, Theorem 2.15 gives 5.05, and (ii) gives 15.0.

Proposition 2.19 (Exposure-bias gap for mismatched pairings). *[proved] [numerically checked] Let $g_t : \mathcal{S}_t \rightarrow [0, G]$ be any bounded per-state loss, for example per-token TV ($G = 1$), JSD $_\beta$ ($G = h(\beta)$), or clipped KL. Then*

$$\left| \sum_{t=1}^T (\mathbb{E}_{d_t^\theta} - \mathbb{E}_{d_t^p}) g_t \right| \leq G \sum_{t=1}^T \text{TV}(d_t^\theta, d_t^p) \leq G \sum_{t=1}^T \sum_{k < t} e_k^{\text{off,TV}} \leq G \frac{T(T-1)}{2} \max_k e_k^{\text{off,TV}}.$$

Also $\leq G \sum_t \sqrt{\frac{1}{2} \sum_{k < t} e_k^{\text{off,KL}}}$. For a mixture prefix law $\nu_t = \lambda d_t^\theta + (1-\lambda)d_t^p$, the gap is multiplied by $1-\lambda$. For unbounded g_t , for example reverse KL on teacher prefixes, no bound in terms of the training loss exists. The training loss can be arbitrarily small while the on-policy loss is arbitrarily large, on states the teacher visits rarely but the student visits often. With $T = 2$, $p_1 = (1-\eta, \eta)$, $q_1 = (1-r\eta, r\eta)$, reverse KL $r^{3/2}$ at the rare state and $\eta = r^{-2}$, the training loss is $O(r^{-1/2})$ while the on-policy loss is $\Theta(r^{1/2})$. Exactly zero training loss does force $Q_\theta^x = P^x$ (Proposition 1.15).

Proof. Bounded-function duality for TV, then the hybrid argument for $\text{TV}(d_t^\theta, d_t^p)$ as in Proposition 1.14. $\square$

Proposition 2.20 (Covariate-shift term). *[proved] Let $\mathcal{E}(\theta) = \mathbb{E}_x e_\theta(x)$ with $e_\theta \geq 0$ pointwise, for example a per-input divergence. Under Assumption A2, $\mathcal{E}_{\text{cov}} \leq (\rho - 1)\mathcal{E}_\mathcal{D}(\tilde{\theta})$. If $0 \leq e_\theta \leq G$, then $\mathcal{E}_{\text{cov}} \leq G \text{TV}(\mu, \mathcal{D})$ (Proposition 1.18). Pointwise non-negativity is needed: a task regret can be negative at some inputs, and then the first bound fails.*

2.6 Assembling the bound, and which term dominates

Corollary 2.21 (End-to-end bound for off-policy soft-label forward-KL distillation). *[proved under Assumptions A1 and A2] Let $\mathcal{A} = \min_\theta \mathbb{E}_\mathcal{D} \text{KL}(P \| Q_\theta)$. With probability $\geq 1 - \delta$,*

$$\mathbb{E}_\mu \text{TV}(P^x, Q_\theta^x) \leq \sqrt{\frac{\rho}{2} \left[\mathcal{A} + 2 \left(2\mathfrak{R}_n + TB \sqrt{\frac{\log(2/\delta)}{2n}} \right) + \mathcal{E}_{\text{opt}} \right]}.$$

Under Assumptions A3 and A4, with hard labels and exact ERM, instead $\mathbb{E}_\mu \text{TV} \leq \min \{ \rho \eta_n, \eta_n + \text{TV}(\mu, \mathcal{D}) \}$ with $\eta_n = \sqrt{2(R \ln 2 + \log(1/\delta))/n}$.

Proof. Theorem 2.1 with $\mathcal{E}_{\text{shift}} = 0$ (Theorem 1.12), Theorem 2.6, Proposition 2.20, and Pinsker. For the second claim use Theorem 2.7 and Proposition 1.18. $\square$

Heuristic 2.22 (The dominant term in large-scale LLM distillation). *(1) $\mathcal{E}_{\text{est}}$ is small. Teacher data is effectively unlimited, soft labels remove label noise (Propositions 2.8 and 2.10), and realisable rates are horizon-free (Theorem 2.7). (2) $\mathcal{E}_{\text{opt}}$ is real but shrinks with compute (Theorem 2.12); it dominates only for undertrained students. (3) $\mathcal{A}$ is the floor. It dominates for small students (Theorem 2.3; Section 3) and, at sequence level, grows linearly in T (Corollary 2.5). (4) $\mathcal{E}_{\text{shift}}$ is zero for the exact pairings, but for mismatched pairings and cumulative criteria it amplifies $\mathcal{A}$ by up to $O(T)$ (Theorem 2.15 and Proposition 2.19). Hence in the capacity-limited regime the dominant term is approximation error as seen by the deployment criterion: the irreducible floor, amplified by prefix shift and misallocated by the choice of divergence (Proposition 4.13). Section 4 designs an objective against exactly this term.*

3 Lower bounds: is lossless even possible?

Framing. For a *fixed* teacher, whether lossless distillation is possible is purely a question of approximation theory: is p in the closure of $\{q_\theta\}$? Information theory enters once we ask what a class of 2^R students can do *uniformly over teachers*, either on average under a prior π over teachers or in the worst case over a class. We then view the student as a code of R bits for the teacher’s conditional law, with distortion $\mathbb{E}_x \text{KL}(p(\cdot | x) \| q_\theta(\cdot | x))$. The random teacher is denoted $\mathbf{T} \sim \pi$, and its conditionals $p_{\mathbf{T}}(\cdot | x)$.

3.1 An information-theoretic capacity floor

Theorem 3.1 (Capacity floor). *[proved under Assumption A4] [numerically checked] Let $\theta^*(\mathbf{T})$ be any measurable choice of student from a class of at most 2^R elements. Fix a design $x_{1:N} \in \mathcal{X}^N$ and a replication number $m \geq 1$. Let $Y = (Y_{ij})_{i \leq N, j \leq m}$ with $Y_{ij} \sim p_{\mathbf{T}}(\cdot | x_i)$ independent given $\mathbf{T}$. Then*

$$\mathbb{E}_\pi \frac{1}{N} \sum_{i=1}^N \text{KL}(p_{\mathbf{T}}(\cdot | x_i) \| q_{\theta^*(\mathbf{T})}(\cdot | x_i)) \geq \frac{I(\mathbf{T}; Y) - R \ln 2}{mN}.$$

In particular: (a) if $\mathcal{D}$ is uniform on $\{x_1, \dots, x_N\}$, the left side is $\mathbb{E}_\pi \mathbb{E}_{x \sim \mathcal{D}} \text{KL}$; (b) for general $\mathcal{D}$, $\mathbb{E}_\pi \mathbb{E}_{x \sim \mathcal{D}} \text{KL} \geq \sup_{N,m} \frac{I(\mathbf{T}; Y | X_{1:N}) - R \ln 2}{mN}$ with $X_i \stackrel{iid}{\sim} \mathcal{D}$.

Proof. Write $\Theta = \theta^*(\mathbf{T})$. KL tensorises, so $m \sum_i \text{KL}(p_{\mathbf{T}}(\cdot | x_i) \| q_\Theta(\cdot | x_i)) = \text{KL}(\otimes_{ij} p_{\mathbf{T}}(\cdot | x_i) \| \otimes_{ij} q_\Theta(\cdot | x_i)) = \mathbb{E}[-\log q_\Theta(Y) | \mathbf{T}] - H(Y | \mathbf{T})$. Average over $\mathbf{T}$. Since q_Θ is a valid conditional law of Y given Θ , the cross-entropy satisfies $\mathbb{E}[-\log q_\Theta(Y)] \geq H(Y | \Theta)$. Hence

$$m \mathbb{E} \sum_i \text{KL}_i \geq H(Y | \Theta) - H(Y | \mathbf{T}) = I(Y; \mathbf{T}) - I(Y; \Theta) \geq I(Y; \mathbf{T}) - H(\Theta) \geq I(Y; \mathbf{T}) - R \ln 2,$$

where the equality uses $H(Y)$ on both sides and the fact that Θ is a function of $\mathbf{T}$. For (b), apply the bound conditionally on $X_{1:N}$ and average, since $\mathbb{E}_{X_{1:N}} \frac{1}{N} \sum_i \text{KL}(\cdot; X_i) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}$. $\square$

Let $I_\infty := \sup_{N,m} I(\mathbf{T}; Y | X_{1:N})$ be the *functional information* of the teacher on $\mathcal{D}$: how much its input–output behaviour reveals about which teacher it is. Theorem 3.1 says that on average over π , lossless distillation is *impossible* whenever $R \ln 2 < I_\infty$. The student’s description length must be at least the teacher’s functional information. This is not the teacher’s parameter count: trained networks are highly redundant as functions, which is what makes pruning, quantisation and distillation work at all.

Corollary 3.2 (K independent facts; the floor is tight). *[proved] [numerically checked] Let the teacher be a uniformly random function $F : [K] \rightarrow \mathcal{V}$ with $p(\cdot | i) = \delta_{F(i)}$, and let $\mathcal{D}$ be uniform on $[K]$. Every R -bit student class satisfies $\mathbb{E} \mathbb{E}_{x \sim \mathcal{D}} \text{KL} \geq (\ln V - \frac{R \ln 2}{K})_+$. The student that memorises $\lfloor R / \log_2 V \rfloor$ facts and is uniform elsewhere attains this up to rounding.*

Proof. Take $N = K$, $x_i = i$, $m = 1$: $I(F; Y) = K \ln V$. The memorising student has $\text{KL} \ln V$ on each fact it does not store and 0 on the others. $\square$

Corollary 3.3 (Behavioural floor for K facts). *[proved] In the setting of Corollary 3.2 with $V \geq 2$, let $\bar{e} := \mathbb{E} \mathbb{E}_{x \sim \mathcal{D}} \text{TV}(p(\cdot | x), q_{\theta^*}(\cdot | x))$. Every R -bit student class satisfies*

$$h(\bar{e}) + \bar{e} \ln(V - 1) \geq \ln V - \frac{R \ln 2}{K}.$$

The left side increases from 0 to $\ln V$ as $\bar{e}$ goes from 0 to $1 - \frac{1}{V}$. So $\bar{e} > 0$ whenever $R < K \log_2 V$, and $\bar{e} \geq 1 - \frac{1}{V}$ at $R = 0$, which is tight. In particular $\bar{e} \geq (\ln V - \ln 2 - R \ln 2 / K) / \ln(V - 1)$ for $V \geq 3$. The KL floor of Corollary 3.2 is therefore not an artefact of KL: behavioural losslessness also fails once $R < K \log_2 V$.

Proof. Let $\hat{G}_i \sim q_{\theta^*}(\cdot | i)$ independently given θ^* . Then $e_i := \mathbb{P}(\hat{G}_i \neq F(i)) = \mathbb{E}TV_i$, because the teacher is a point mass. Since the $F(i)$ are independent and $\hat{G}$ depends on F only through θ^* , $\sum_i I(F(i); \hat{G}_i) \leq I(F; \hat{G}) \leq H(\theta^*) \leq R \ln 2$. Fano gives $I(F(i); \hat{G}_i) \geq \ln V - h(e_i) - e_i \ln(V-1)$. Sum over i and use concavity of h : $K[\ln V - h(\bar{e}) - \bar{e} \ln(V-1)] \leq R \ln 2$. Monotonicity follows from $\frac{d}{de}[h(e) + e \ln(V-1)] = \ln \frac{(1-e)(V-1)}{e} \geq 0$ for $e \leq 1 - \frac{1}{V}$. The explicit bound uses $h \leq \ln 2$. $\square$

Corollary 3.4 (Metric-entropy version). *[proved] Suppose Θ_S has an ε -cover Θ_ε in $\|\log q_\theta - \log q_{\theta'}\|_\infty$ (over the design and $\mathcal{V}^T$). Then $\text{KL}(p\|q_{\theta'}) \leq \text{KL}(p\|q_\theta) + \varepsilon$ whenever θ' is ε -close to θ . Hence*

$$\mathbb{E}_\pi \min_{\theta \in \Theta_S} \frac{1}{N} \sum_i \text{KL}_i \geq \frac{I(\mathbf{T}; Y) - \log \mathcal{N}(\varepsilon)}{mN} - \varepsilon.$$

For P parameters in $[-B_\theta, B_\theta]^P$ with $\theta \mapsto \log q_\theta$ being L -Lipschitz in ℓ_∞ , $\log \mathcal{N}(\varepsilon) \leq P \log[LB_\theta/\varepsilon]$, so the effective rate is $R_{\text{eff}}(\varepsilon) \approx P \log_2(LB_\theta/\varepsilon)$ bits. Pseudo-dimension and Rademacher bounds enter through the usual covering-number estimates.

Proof. $\text{KL}(p\|q_{\theta'}) - \text{KL}(p\|q_\theta) = \mathbb{E}_p[\log q_\theta - \log q_{\theta'}] \leq \varepsilon$. Apply Theorem 3.1 to the finite class Θ_ε . $\square$

Remark 3.5 (Bits per parameter). The rate that matters is *usable* bits. Allen-Zhu and Li (2024) report, in controlled synthetic settings, a knowledge capacity of about 2 bits per parameter for language models. Kumar et al. (2024) model how precision changes the effective parameter count. Combining such estimates with Theorem 3.1 gives a heuristic necessary condition. A student with N_S parameters cannot be lossless on content whose functional information exceeds roughly $2N_S$ bits. Treat this as an order-of-magnitude guide; the constant is an empirical estimate for a particular kind of knowledge.

3.2 Distillation as rate–distortion

Definition 3.6 (Distillation distortion–rate function). For a prior π on teachers, a reproduction alphabet of student functions, and distortion $d(\mathbf{T}, \hat{\mathbf{T}}) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(p_{\mathbf{T}}(\cdot | x) \| p_{\hat{\mathbf{T}}}(\cdot | x))$, define $D_\pi(R) := \inf\{\mathbb{E} d(\mathbf{T}, \hat{\mathbf{T}}) : I(\mathbf{T}; \hat{\mathbf{T}}) \leq R \ln 2\}$.

Theorem 3.7 (One-shot converse). *[proved] Every class of 2^R students satisfies $\mathbb{E}_\pi \min_\theta d(\mathbf{T}, q_\theta) \geq D_\pi(R)$.*

Proof. $\theta^*(\mathbf{T})$ is a deterministic function of $\mathbf{T}$ taking at most 2^R values, so $I(\mathbf{T}; \theta^*) \leq H(\theta^*) \leq R \ln 2$. It is therefore feasible in the infimum. $\square$

Unlike Shannon’s asymptotic theorem (Shannon, 1959), no block coding is needed. The converse holds for a single teacher draw because it uses only $I \leq H \leq \log(\#\text{codewords})$ (Cover and Thomas, 2006; Polyanskiy and Wu, 2025). For log-loss distortion the rate–distortion function has the simple form $R(D) = H - D$ (Courtade and Weissman, 2014). Corollary 3.2 is an instance of this.

Proposition 3.8 (Power-law spectrum gives a power-law floor). *[proved within the stated model] [numerically checked] Model. Near-lossless regime (Proposition 1.8), in Fisher-whitened centred log-probability coordinates, where $\text{KL} \approx \frac{1}{2} \|z - \hat{z}\|^2$. The teacher’s whitened log-probability function is a centred Gaussian vector with covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$. Reverse water-filling (Cover and Thomas, 2006, Thm. 10.3.3) gives*

$$D(R) = \frac{1}{2} \sum_i \min(w, \lambda_i), \quad \sum_i \frac{1}{2} \log_+ \frac{\lambda_i}{w} = R \ln 2.$$

If $\lambda_i = c i^{-\alpha}$ with $\alpha > 1$, then $D(R) = \frac{c}{2} \frac{\alpha}{\alpha-1} \left(\frac{2R \ln 2}{\alpha}\right)^{-(\alpha-1)} (1 + o(1))$, so $D(R) \asymp R^{-(\alpha-1)}$.

Proof sketch. With k active components, $w = \lambda_k$. The rate is $\frac{\alpha}{2} \sum_{i \leq k} \log \frac{k}{i} = \frac{\alpha}{2} (k - O(\log k))$ by Stirling, so $k \approx 2R \ln 2 / \alpha$. The distortion is $\frac{1}{2} [kw + \sum_{i > k} \lambda_i] = \frac{\epsilon}{2} k^{1-\alpha} (1 + \frac{1}{\alpha-1})(1 + o(1))$. The fitted exponents in `check_capacity.py` match $-(\alpha - 1)$ to within 0.04 for $\alpha \in \{1.5, 2, 3\}$. $\square$

Proposition 3.9 (Finite teacher dimension: from power law to exponential). *[proved within the stated model] [numerically checked] If $\lambda_i = 0$ for $i > k_T$, then as soon as $w < \lambda_{k_T}$, $D(R) = \frac{k_T}{2} (\prod_{i \leq k_T} \lambda_i)^{1/k_T} e^{-2R \ln 2 / k_T}$. The distortion decays exponentially in the student’s rate once the student can resolve every component the teacher has. For $k_T = 1000$ and $\alpha = 2$, D drops by a factor $e^4 \approx 55$ when $R \ln 2$ doubles from 2000 to 4000 nats. Below $R \ln 2 \approx k_T$ nats it follows the power law.*

Proposition 3.10 (No capacity gap under optimal rate-limited allocation). *[proved within the stated model] Let the ground truth have spectrum $(\lambda_i)_{i \geq 1}$ and let a teacher of size k_T know the first k_T components exactly. The best R -bit student’s distortion with respect to the ground truth is non-increasing in k_T .*

Proof. Let $E(k_T) = D_{k_T}(R) + \frac{1}{2} \sum_{i > k_T} \lambda_i$. For $k' > k_T$, one feasible code for the larger teacher uses the optimal code for the first k_T components and reproduces 0 elsewhere. Hence $D_{k'}(R) \leq D_{k_T}(R) + \frac{1}{2} \sum_{k_T < i \leq k'} \lambda_i$, so $E(k') \leq E(k_T)$. $\square$

3.3 What this implies for distillation scaling laws

Supervised pre-training follows power laws in parameters N and tokens D (Kaplan et al., 2020). A widely used joint form is the additive law $L(N, D) = E + A/N^a + B/D^b$ of Hoffmann et al. (2022). Busbridge et al. (2025) fit a distillation scaling law in which the student’s cross-entropy depends on the teacher’s cross-entropy and on the student’s supervised-scaling loss. That law captures a “capacity gap”: beyond some point, stronger teachers yield worse students (see also Cho and Hariharan, 2019; Mirzadeh et al., 2020; Zhang et al., 2023). We are confident of this qualitative content but not of the exact parametrisation in Busbridge et al. (2025). The claim of Zhang et al. (2023) that the optimal teacher is about 2.5× the student is quoted from memory and should be checked. The results above imply the following.

Heuristic 3.11 (Student-size exponent). *With a constant number of usable bits per parameter, $R_S \propto N_S$. The approximation part of the gap should then scale as $N_S^{-(\alpha-1)}$, where α is the decay exponent of the teacher’s Fisher-weighted centred log-probability spectrum. This spectrum is measurable (Section 6.3). It should cross over to faster (exponential) decay once $R_S \ln 2$ approaches the teacher’s functional information (Proposition 3.9).*

Heuristic 3.12 (Data exponent: soft labels are steeper). *Consider a projection-estimator model. The target has coefficients of energy $\lambda_i = ci^{-\alpha}$ in an orthonormal basis under $\mathcal{D}$, and the student estimates the first k from n samples. The error is approximately $\sigma^2 \frac{k}{n} + \sum_{i > k} \lambda_i$, where σ^2 is the effective noise. With hard labels, σ^2 includes label noise $\sigma_0^2 > 0$. The optimum is $k \asymp n^{1/\alpha}$, giving error $\asymp n^{-(\alpha-1)/\alpha}$. With soft labels, the only noise is the aliased tail, $\sigma^2 \asymp k^{1-\alpha}$. Then $k \asymp n$ and the error is $\asymp n^{-(\alpha-1)}$. The rigorous version of this noiseless/noisy crossover in kernel regression is due to Cui et al. (2021). The prediction is that the distillation-token exponent should be steeper for full-logit distillation than for sequence-level (hard-label) distillation, with top- k logits in between.*

Heuristic 3.13 (Capacity gap as a finite-sample aliasing effect). *In the same model, let the teacher have k_T components and the student $k < k_T$. With soft labels, the student’s error with respect to the ground truth is approximately*

$$\frac{k}{n} \sum_{k < i \leq k_T} \lambda_i + \sum_{i > k} \lambda_i.$$

The first term increases in k_T (teacher content the student cannot represent aliases into the components it estimates). For $k_T \leq k$ the error is $\sum_{i>k_T} \lambda_i$, which decreases in k_T . Hence the optimal teacher has $k_T \approx k$, and the penalty for a stronger teacher is at most a factor $(1 + k/n)$ on the tail. It vanishes as $n \rightarrow \infty$. Together with Proposition 3.10, this predicts that capacity gaps are finite-data and optimisation effects, and should shrink as distillation tokens grow at fixed student size.

Conjecture 3.14 (Distillation gap law). *For a teacher whose Fisher-weighted log-probability spectrum decays as $i^{-\alpha}$, the per-token forward-KL gap of a well-optimised student with R_S usable bits and n distillation tokens satisfies*

$$\varepsilon_{\text{off}}^{\text{KL}} \asymp A R_S^{-(\alpha-1)} + B n^{-b} + \frac{k(R_S)}{n} \text{Tail}_T(k(R_S)),$$

with $b = \alpha - 1$ for full soft labels and $b = (\alpha - 1)/\alpha$ for hard labels. The first term switches to exponential decay once $R_S \ln 2 \gtrsim I_\infty$.

4 Objective design

4.1 Gradients with respect to student logits

Proposition 4.1 (Logit gradients of f -divergences). *[proved] [numerically checked] Let $q = \text{softmax}(z)$, $u = p/q$, and $\psi_f(u) := f(u) - u f'(u)$. Then $\nabla_z D_f(p||q) = q \odot (\psi_f(u) - \mathbb{E}_q \psi_f(u))$. In particular:*

$$\begin{aligned} \nabla_z \text{KL}(p||q) &= q - p, \\ \nabla_z \text{KL}(q||p) &= q \odot (\log \frac{q}{p} - \text{KL}(q||p) \mathbf{1}), \\ \nabla_z \text{JSD}_\beta(p||q) &= (1 - \beta) q \odot (\log \frac{q}{m} - \mathbb{E}_q \log \frac{q}{m} \mathbf{1}), \quad m = \beta p + (1 - \beta) q. \end{aligned}$$

Proof. $\partial_{q(y)} \sum_{y'} q(y') f(p(y')/q(y')) = f(u_y) - u_y f'(u_y)$. The softmax Jacobian is $\partial q_j / \partial z_i = q_j (\delta_{ij} - q_i)$. For KL, $\psi = -u$. For reverse KL, written as $D_{-\log}$, $\psi = 1 - \log u$. For JSD_β we have $f_\beta(u) - u f'_\beta(u) = -(1 - \beta) \log(\beta u + 1 - \beta) = (1 - \beta) \log(q/m)$. $\square$

4.2 Mode covering versus mode seeking

Proposition 4.2 (Support behaviour). *[proved] If $\text{KL}(p||q) < \infty$ then $\text{supp } p \subseteq \text{supp } q$ (zero-avoiding, mode covering). If $\text{KL}(q||p) < \infty$ then $\text{supp } q \subseteq \text{supp } p$ (zero-forcing, mode seeking). $\text{JSD}_\beta \leq h(\beta)$ is always finite and imposes neither constraint.*

Proposition 4.3 (Missing-mode gradients). *[proved] [numerically checked] Fix a token i with $p_i > 0$ and let $q_i \rightarrow 0$, the other coordinates staying bounded away from 0.*

- (i) *Forward KL: $\partial_{z_i} \text{KL}(p||q) = q_i - p_i \rightarrow -p_i$. The restoring force equals the missing mass.*
- (ii) *Reverse KL (assuming $\text{supp } p = \mathcal{V}$, otherwise $\text{KL}(q||p) = \infty$): $|\partial_{z_i} \text{KL}(q||p)| \leq q_i (|\log \frac{q_i}{p_i}| + \text{KL}(q||p)) = O(q_i \log \frac{1}{q_i}) \rightarrow 0$. A mode the student has dropped exerts no pull.*
- (iii) *JSD_β : exactly, the pull is $-\partial_{z_i} \text{JSD}_\beta = (1 - \beta) q_i [\log(1 - \beta + \beta p_i/q_i) + c]$ with $c = \text{KL}(q||m) \in [0, \log \frac{1}{1-\beta}]$. For a missing mode ($p_i > q_i$):*
 - *it is at most $\beta(1 - \beta)(p_i - q_i) + (1 - \beta) q_i \log \frac{1}{1-\beta}$, at most a quarter of the forward-KL pull when $q_i \ll p_i$;*
 - *while $q_i \gg \beta p_i$ it equals $\beta p_i - (1 - \beta) [\log \frac{1}{1-\beta} - c] q_i + O(\beta^2 p_i^2/q_i)$, which is forward-like with weight β ;*
 - *as $q_i \rightarrow 0$ it equals $(1 - \beta) q_i \log \frac{\beta p_i}{q_i} (1 + o(1)) \rightarrow 0$, which is reverse-like.*

A teacher mode is effectively recovered only while the student's mass on it is not $\ll \beta p_i$.

Proof. Read off Proposition 4.1. For (iii), $m_i/q_i = 1 - \beta + \beta p_i/q_i$, and $0 \leq c = \mathbb{E}_q \log \frac{q}{m} \leq \log \frac{1}{1-\beta}$ because $m \geq (1 - \beta)q$. The upper bound uses $\log(1 - \beta + \beta r) \leq \beta(r - 1)$. The middle regime expands $\log(1 - \beta + \beta r) = \log(1 - \beta) + \frac{\beta r}{1-\beta} + O(\beta^2 r^2)$. The last uses $\log(1 - \beta + \beta p_i/q_i) = \log \frac{\beta p_i}{q_i} + o(1)$. $\square$

Proposition 4.4 (M-projection versus I-projection). *[proved] (a), [known result; proof sketch] (b). (a) For an exponential-family student $q_\eta \propto e^{\eta^\top \phi}$, $\eta \mapsto \text{KL}(p \| q_\eta)$ is convex, and its stationarity condition is moment matching, $\mathbb{E}_{q_\eta} \phi = \mathbb{E}_p \phi$. Stationarity of $\text{KL}(q_\eta \| p)$ is $\text{Cov}_{q_\eta}(\phi, \log \frac{q_\eta}{p}) = 0$, which is generally non-convex. (b) Let $p = w\mathcal{N}(-a, 1) + (1 - w)\mathcal{N}(a, 1)$ with $w < \frac{1}{2}$, and let $q = \mathcal{N}(\mu, s^2)$. The forward minimiser is $\mu = (1 - 2w)a$, $s^2 = 1 + 4w(1 - w)a^2$. At it, $\text{KL}(q \| p) = \Theta(a^2)$ and $\text{TV}(p, q) \rightarrow 1$ as $a \rightarrow \infty$. The reverse KL has local minimisers near $\mu = \pm a$ with $s \rightarrow 1$, of value $\rightarrow \log \frac{1}{w}$ and $\log \frac{1}{1-w}$ respectively; the global one is the heavier mode. At it, $\text{KL}(p \| q) = \Theta(a^2)$ but $\text{TV}(p, q) \rightarrow w$.*

Proof. (a) $\nabla_\eta \text{KL}(p \| q_\eta) = \mathbb{E}_{q_\eta} \phi - \mathbb{E}_p \phi$, and the Hessian is $\text{Cov}_{q_\eta} \phi \succeq 0$. For the reverse direction, differentiate $\mathbb{E}_{q_\eta} [\log q_\eta - \log p]$. (b) The forward case is moment matching. The mixture has mean $(1 - 2w)a$ and variance $1 + a^2 - (1 - 2w)^2 a^2$. The reverse values follow because $p \approx (1 - w)\mathcal{N}(a, 1)$ on the bulk of $\mathcal{N}(a, 1)$. Classical discussions are Minka (2005) and Bishop (2006). $\square$

Each direction’s minimiser is catastrophic in the other direction. In this example, the mode-seeking solution is *behaviourally* better ($\text{TV} \approx w$ versus ≈ 1): it produces valid samples and drops a mode, whereas the covering solution produces mostly invalid samples. This is the formal content of “mode covering versus mode seeking”. Which is preferable depends on the deployment criterion (Section 1.4). Under misspecification, neither is TV-optimal (Proposition 4.13).

4.3 Temperature

Let $p^\tau = \text{softmax}(v/\tau)$, $q^\tau = \text{softmax}(z/\tau)$, and $\mathcal{L}_\tau(z) := \tau^2 \text{KL}(p^\tau \| q^\tau)$, the loss of Hinton et al. (2015).

Proposition 4.5 (Temperature and the τ^2 factor). *[proved] [numerically checked]*

- (i) $\nabla_z \mathcal{L}_\tau = \tau(q^\tau - p^\tau)$. Without the factor τ^2 the gradient is $\frac{1}{\tau}(q^\tau - p^\tau) = O(\tau^{-2})$ as $\tau \rightarrow \infty$. The factor τ^2 keeps gradient magnitudes $O(1)$, so the relative weighting against a hard-label term is stable across τ .
- (ii) (High temperature: logit matching.) $\tau(q_i^\tau - p_i^\tau) = \frac{1}{V}(\tilde{z}_i - \tilde{v}_i) + \frac{1}{2V\tau}[(\tilde{z}_i^2 - \tilde{v}_i^2) - \frac{1}{V}(\|\tilde{z}\|^2 - \|\tilde{v}\|^2)] + O(\tau^{-2})$, so $\mathcal{L}_\tau(z) = \frac{1}{2V}\|\tilde{z} - \tilde{v}\|^2 + O(1/\tau)$. This is centred-logit regression, as in Ba and Caruana (2014).
- (iii) (Low temperature: argmax matching.) If v has a unique maximiser i^* , then $\tau \text{KL}(p^\tau \| q^\tau) \rightarrow \max_j z_j - z_{i^*}$, a perceptron/hinge loss on the teacher’s argmax, and $\mathcal{L}_\tau \rightarrow 0$.
- (iv) (Geometry, not solution.) The global minimisers $\{v + c\mathbf{1}\}$ do not depend on τ , and $\nabla^2 \mathcal{L}_\tau|_{z=v} = \text{diag}(p^\tau) - p^\tau p^{\tau\top} \rightarrow \frac{1}{V}C$ as $\tau \rightarrow \infty$. Temperature changes which errors are cheap, and hence the misspecified solution and the conditioning, but not the realisable solution.

Proof. (i) Chain rule: $\partial_z \text{KL}(p^\tau \| \text{softmax}(z/\tau)) = \frac{1}{\tau}(q^\tau - p^\tau)$. (ii) Expand $e^{\tilde{z}_i/\tau} = 1 + \tilde{z}_i/\tau + \tilde{z}_i^2/2\tau^2 + O(\tau^{-3})$, with $\sum_j e^{\tilde{z}_j/\tau} = V + \|\tilde{z}\|^2/2\tau^2 + O(\tau^{-3})$, and similarly for v . (iii) $p^\tau \rightarrow \delta_{i^*}$, and $-\tau \log q_{i^*}^\tau = \tau \text{lse}(z/\tau) - z_{i^*} \rightarrow \max_j z_j - z_{i^*}$; the entropy terms of p^τ vanish. (iv) The Hessian of $\tau^2 \text{lse}(z/\tau)$ is $\text{diag}(q^\tau) - q^\tau q^{\tau\top}$. $\square$

4.4 Dark knowledge as Fisher information

Proposition 4.6 (Low-probability tokens are invisible at $\tau = 1$). *[proved] [numerically checked]* For $\delta \in \mathbb{R}^V$ supported on $A \subseteq \mathcal{V}$, the curvature satisfies $\delta^\top (\text{diag } p - pp^\top) \delta \leq \max_{i \in A} p_i \|\delta\|^2 \leq$

$p(A)\|\delta\|^2$. The exact cost of perturbing the matched logits v is

$$\text{KL}(p \parallel \text{softmax}(v + \delta)) = \log \left(1 + \sum_{i \in A} p_i (e^{\delta_i} - 1) \right) - \sum_{i \in A} p_i \delta_i \leq \frac{1}{2} \sum_{i \in A} p_i \delta_i^2 e^{(\delta_i)_+}.$$

So perturbing the logits of tokens of total teacher mass 10^{-6} by δ costs at most $\frac{1}{2} \cdot 10^{-6} e^{\max_i(\delta_i)_+} \|\delta\|^2$. Their relative logits, which is the “dark knowledge”, are essentially unconstrained at $\tau = 1$. At temperature τ , p is replaced by the flatter p^τ . (Proof: $\log(1+x) \leq x$ and $e^u - 1 - u \leq \frac{1}{2}u^2 e^{u_+}$.)

Proposition 4.7 (What soft labels add to hard labels). *[proved]*

- (a) At θ° , one hard label at s carries Fisher information $\mathcal{I}(s) = J^\top (\text{diag } p - pp^\top) J$, and m i.i.d. hard labels carry $m\mathcal{I}(s)$. The soft label is the $m \rightarrow \infty$ limit. It determines the centred teacher logits $\tilde{v}(s)$ exactly, because $p \mapsto C \log p$ is a bijection from the open simplex onto range C .
- (b) The per-state gradient-variance reduction from using the soft label is $\text{tr } \mathcal{I}(s)$ (Proposition 2.8). It is zero iff the teacher is deterministic at s (for J of full rank on range C). In the realisable linear case it upgrades a $k/2n$ error to exact recovery (Proposition 2.10).
- (c) For any $\tau > 0$, p^τ is a bijective function of p . Temperature adds no information; it reweights existing information in the loss (Proposition 4.5(iv), Proposition 4.6). It matters only when the channel is lossy, for example with top- k logits, finite precision, or finite optimisation budgets.
- (d) Under Assumption A5 for the teacher, $\tilde{v}(s) = CW_\top h_\top(s) + Cb_\top$. The soft label therefore reveals the teacher’s final hidden state up to $\ker(CW_\top)$ and a known offset: dark knowledge is an affine image of the teacher’s representation. This links to Section 5.

So soft labels carry information beyond hard labels *exactly when the teacher is non-degenerate*. The amount is unbounded in the noiseless realisable case and equals the Rao–Blackwell gap otherwise. Temperature is a preconditioner, not an information source. This sharpens the “dark knowledge” account of Hinton et al. (2015) and the gradient decompositions of Furlanello et al. (2018) and Tang et al. (2020). Relations to label smoothing are discussed by Müller et al. (2019).

4.5 Sequence-level objectives and unbiased gradient estimators

Proposition 4.8 (Sequence-level f -divergence gradients). *[proved] [numerically checked]* For sequence laws P and Q_θ , $\nabla_\theta D_f(P \| Q_\theta) = \mathbb{E}_{y \sim Q_\theta} [\nabla_\theta \log Q_\theta(y) \psi_f(P(y)/Q_\theta(y))]$. This gives $\nabla \text{KL}(Q_\theta \| P) = \mathbb{E}_{Q_\theta} [\nabla \log Q_\theta \cdot \log \frac{Q_\theta}{P}]$, $\nabla \text{JSD}_\beta(P \| Q_\theta) = (1 - \beta) \mathbb{E}_{Q_\theta} [\nabla \log Q_\theta \cdot \log \frac{Q_\theta}{M_\beta}]$ with M_β held fixed inside the logarithm, and $\nabla \text{KL}(P \| Q_\theta) = -\mathbb{E}_P [\nabla \log Q_\theta]$ (off-policy).

Proof. $\nabla D_f = \sum_y \nabla Q_\theta(y) \partial_{Q(y)} D_f$ with $\partial_{Q(y)} D_f = \psi_f(u_y)$ as in Proposition 4.1. Constants integrate to zero against ∇Q_θ . The divergence-minimisation view of imitation learning is developed by Ghasemipour et al. (2019) and Ke et al. (2020), and for sequence-level distillation by Wen et al. (2023). $\square$

Proposition 4.9 (Rao–Blackwellised unbiased estimator for sequence-level reverse KL). *[proved] [numerically checked]* Let $y \sim Q_\theta^x$, $g_t(s) := \text{KL}(q_\theta(\cdot | s) \| p(\cdot | s))$, and let b_t be any function of s_t (a baseline). Then

$$\hat{g} = \sum_{t=1}^T \left[\nabla_\theta g_t(s_t) \Big|_{s_t \text{ fixed}} + \nabla_\theta \log q_\theta(y_t | s_t) \left(\sum_{k>t} g_k(s_k) - b_t(s_t) \right) \right]$$

satisfies $\mathbb{E} \hat{g} = \nabla_\theta \text{KL}(Q_\theta^x \| P^x)$. The “stop-gradient” estimator that keeps only the first term is biased. Its bias is $-\sum_t \mathbb{E} [\nabla \log q_\theta(y_t | s_t) \sum_{k>t} g_k(s_k)]$, the effect of the student’s actions on

its own future prefixes. Under Assumption A3 both estimators vanish in expectation at θ° . In general the stop-gradient fixed points are the θ that are stationary for $\theta' \mapsto \sum_t \mathbb{E}_{d_t^\theta} \text{KL}(q_{\theta'} \| p)$. This set includes the fixed points of the DAgger-type map $\theta \mapsto \arg \min_{\theta'} \sum_t \mathbb{E}_{d_t^\theta} \text{KL}(q_{\theta'} \| p)$, and is larger when that inner objective is non-convex (Proposition 2.11). These are not, in general, stationary points of the sequence-level KL.

Proof. $F(\theta) = \sum_t \mathbb{E}_{s \sim d_t^\theta} g_t(\theta; s)$ by Theorem 1.12. Differentiating gives a direct term and a score term through d_t^θ : $\sum_t \mathbb{E}[(\sum_{k < t} \nabla \log q(y_k | s_k)) g_t(s_t)] = \sum_k \mathbb{E}[\nabla \log q(y_k | s_k) \sum_{t > k} g_t(s_t)]$. Replacing the sampled per-token log-ratios by g_k is exact by the tower property, since $\mathbb{E}[\log \frac{q}{p}(y_k | s_k) | s_k] = g_k(s_k)$, and s_k contains y_t for $k > t$. Baselines are unbiased because $\mathbb{E}[\nabla \log q(y_t | s_t) | s_t] = 0$. The analytic per-token term is itself the Rao–Blackwellisation of $\nabla \log q(y_t | s_t) \log \frac{q}{p}(y_t | s_t)$. Gu et al. (2024) use a policy gradient for this objective (with further variance-reduction devices). Agarwal et al. (2024) do not differentiate through the student’s sampling; we believe this is their stated choice, but we have not re-verified the exact wording. $\square$

Proposition 4.10 (Sequence-level JSD_β decomposes with prefix-adaptive mixtures). *[proved] The mixture $M_\beta = \beta P + (1 - \beta)Q_\theta$ is autoregressive, with conditionals*

$$m_t(\cdot | s) = w_t(s) p_t(\cdot | s) + (1 - w_t(s)) q_t(\cdot | s), \quad w_t(s) = \sigma\left(\log \frac{\beta}{1-\beta} + \log P(s) - \log Q_\theta(s)\right),$$

where $P(s) = \prod_{k < t} p(y_k | s_k)$ and $Q_\theta(s)$ are prefix probabilities. Consequently

$$\text{JSD}_\beta(P^x \| Q_\theta^x) = \beta \sum_{t=1}^T \mathbb{E}_{s \sim P^x} \text{KL}(p_t(s) \| m_t(s)) + (1 - \beta) \sum_{t=1}^T \mathbb{E}_{s \sim Q_\theta^x} \text{KL}(q_t(s) \| m_t(s)).$$

The token-level JSD_β of Agarwal et al. (2024) differs in two ways. It fixes $w_t \equiv \beta$, and it evaluates both terms on one prefix law (student or mixed) rather than the β -term on teacher prefixes. Both differences vanish when $P(s) = Q_\theta(s)$ on all prefixes, since then the prefix laws also coincide. The token-level objective is therefore the near-lossless linearisation of the sequence-level one. Because $\log P(s_t) - \log Q_\theta(s_t)$ is a sum of $t - 1$ per-token log-ratios, the two diverge along long prefixes unless the student is already nearly lossless there.

Proof. $M_\beta(y_{\leq t}) = \beta P(y_{\leq t}) + (1 - \beta)Q_\theta(y_{\leq t})$. Dividing by $M_\beta(y_{< t})$ gives m_t . Now apply the chain rule (Theorem 1.12), which holds for any pair of autoregressive laws, to $\text{KL}(P \| M_\beta)$ and to $\text{KL}(Q_\theta \| M_\beta)$. $\square$

Proposition 4.11 (Unbiased per-token estimator for sequence-level JSD_β). *[proved] Let $y \sim Q_\theta^x$, let $\rho_t(a; s) := \log \frac{q_t(a|s)}{m_t(a|s)}$ be evaluated with stop-gradient on m_t and w_t , and let b_t be any function of s_t . Then*

$$\begin{aligned} \hat{g}_{\text{JSD}} = (1 - \beta) \sum_{t=1}^T \bigg[\sum_{a \in \mathcal{V}} \nabla_\theta q_\theta(a | s_t) \rho_t(a; s_t) \\ + \nabla_\theta \log q_\theta(y_t | s_t) \left(\sum_{k > t} \text{KL}(q_k(s_k) \| m_k(s_k)) - b_t(s_t) \right) \bigg] \end{aligned}$$

is unbiased for $\nabla_\theta \text{JSD}_\beta(P^x \| Q_\theta^x)$. It uses student samples only. The teacher enters through m_t .

Proof. By Proposition 4.8 and Proposition 4.10, $\nabla \text{JSD}_\beta = (1 - \beta) \mathbb{E}_Q[\sum_t \nabla \log q(y_t | s_t) \sum_k \rho_k(y_k; s_k)]$. Terms with $k < t$ vanish, because ρ_k is s_t -measurable. The $k = t$ term is replaced by its conditional expectation over y_t , and $k > t$ terms by $\mathbb{E}[\rho_k | s_k] = \text{KL}(q_k \| m_k)$. Baselines are unbiased as before. $\square$

Proposition 4.12 (A bounded, unbiased, horizon-free value estimator). *[proved] [numerically checked] Draw $y \sim M_\beta^x := \beta P^x + (1 - \beta)Q_\theta^x$, that is, a teacher sample with probability β and a student sample otherwise. Set $r = P^x(y)/M_\beta^x(y)$, $s = Q_\theta^x(y)/M_\beta^x(y)$ and*

$$\varphi(y) := \beta r \log r + (1 - \beta) s \log s.$$

Then $\mathbb{E}_{y \sim M_\beta^x} \varphi = \text{JSD}_\beta(P^x \| Q_\theta^x)$ and $0 \leq \varphi \leq \bar{h}(\beta) := \log \frac{1}{\min(\beta, 1-\beta)}$ for every y and every horizon T . The two coincide at $\beta = \frac{1}{2}$, where $\bar{h}(\frac{1}{2}) = h(\frac{1}{2}) = \log 2$. Only sequence log-probabilities under the two models are needed.

Proof. $\sum_y M(y) [\beta \frac{P}{M} \log \frac{P}{M} + (1 - \beta) \frac{Q}{M} \log \frac{Q}{M}] = \beta \text{KL}(P \| M) + (1 - \beta) \text{KL}(Q \| M)$. For the range, note $\beta r + (1 - \beta) s = 1$ with $r, s \geq 0$. Convexity of $u \log u$ gives $\varphi \geq 1 \cdot \log 1 = 0$. On the segment $\{\beta r + (1 - \beta) s = 1\}$, φ is convex, so it is maximised at an endpoint: $(1/\beta, 0)$ gives $\log \frac{1}{\beta}$ and $(0, 1/(1 - \beta))$ gives $\log \frac{1}{1 - \beta}$. $\square$

By contrast, summing the per-token KLs of Proposition 4.10 along sampled prefixes is unbiased but has a range growing linearly in T . For example, $p_t \equiv (\frac{1}{2}, \frac{1}{2}, 0)$ and $q_t \equiv (\frac{1}{2}, 0, \frac{1}{2})$ give $\frac{\beta T}{2} \log \frac{1}{\beta}$ on the all- a teacher path. That representation is used for gradients (Proposition 4.11), not for value estimates.

4.6 Bounded divergences are agnostically TV-competitive; KL is not

Proposition 4.13 (Agnostic TV-competitiveness). *[proved]*

(a) *For any student class, every $\theta_\beta \in \arg \min_\theta \mathbb{E}_x \text{JSD}_\beta(P^x \| Q_\theta^x)$ satisfies*

$$\mathbb{E}_x \text{TV}(P^x, Q_{\theta_\beta}^x) \leq \sqrt{\frac{h(\beta)}{2\beta(1 - \beta)}} \inf_\theta \mathbb{E}_x \text{TV}(P^x, Q_\theta^x).$$

The constant is minimised at $\beta = \frac{1}{2}$, giving $\sqrt{2 \ln 2 \cdot \text{TV}^} \approx 1.18 \sqrt{\text{TV}^*}$.*

(b) *No bound of the form $\text{TV}(\theta_{\text{KL}}) \leq \phi(\inf_\theta \text{TV})$ with $\phi(0^+) = 0$ holds for either KL direction. Let $P = (1 - \eta)\delta_a + \eta\delta_b$. Forward: for the class $\{\delta_a, \text{Unif}(\mathcal{V})\}$, the KL minimiser is the uniform law, with $\text{TV} \geq 1 - \eta - \frac{1}{V}$, while $\inf \text{TV} = \eta$. Reverse: for the class $\{(1 - \eta')\delta_a + \eta'\delta_c, \delta_b\}$ with $c \notin \text{supp } P$, the reverse-KL minimiser is δ_b (value $\log \frac{1}{\eta}$), with $\text{TV} = 1 - \eta$, while $\inf \text{TV} \leq \eta + \eta'$.*

Proof. (a) Let θ_{TV} attain (or approach) $\inf \mathbb{E} \text{TV}$. Using Proposition 1.7(i) twice and Jensen,

$$\mathbb{E} \text{TV}(\theta_\beta) \leq \sqrt{\frac{\mathbb{E} \text{JSD}_\beta(\theta_\beta)}{2\beta(1 - \beta)}} \leq \sqrt{\frac{\mathbb{E} \text{JSD}_\beta(\theta_{\text{TV}})}{2\beta(1 - \beta)}} \leq \sqrt{\frac{h(\beta) \mathbb{E} \text{TV}(\theta_{\text{TV}})}{2\beta(1 - \beta)}}.$$

The function $h(\beta)/\beta(1 - \beta)$ is symmetric about $\frac{1}{2}$ and is minimised there (direct check). (b) Direct computation. $\text{KL}(P \| \delta_a) = \infty$, and $\text{KL}(\cdot \| P) = \infty$ for any law charging c . $\square$

Proposition 4.13 is the formal reason to prefer a bounded divergence when the student is capacity-limited. KL-based objectives can place their unavoidable error where TV, and hence every bounded downstream task, is maximally hurt.

4.7 Proposed objective

Definition 4.14 (Prefix-adaptive sequence-level JSD distillation, PA-JSD).

$$\mathcal{J}_\beta(\theta) := \mathbb{E}_{x \sim \mathcal{D}} \text{JSD}_\beta(P^x \| Q_\theta^x), \quad \beta = \frac{1}{2} \text{ by default,}$$

Values are estimated with the bounded mixture estimator of Proposition 4.12. Gradients use Proposition 4.11, which is already unbiased for the *full* gradient because its score form

accounts for θ inside M_β . An equivalent unbiased split, based on the exact decomposition of Proposition 4.10, is the pathwise gradient of the teacher-prefix term $\beta\text{KL}(P\|M_\theta)$ plus the score and pathwise gradients of the student-prefix term $(1 - \beta)\text{KL}(Q_\theta\|M_\theta)$. The two recipes are alternatives: adding the teacher-prefix pathwise term to Proposition 4.11 double-counts it and biases the estimator. Optionally add $\gamma\mathcal{R}_F$, the Fisher-aligned representation term of Definition 5.8.

Theorem 4.15 (Properties of PA-JSD). *[proved] (except where marked)*

- (i) Exactness (no exposure bias). $\mathcal{J}_\beta$ is a sequence-level divergence, and its per-token form is an identity (Proposition 4.10). With $\mathcal{E} = \mathbb{E}\text{JSD}_\beta$, Theorem 2.1 holds with $\mathcal{E}_{\text{shift}} \equiv 0$.
- (ii) Behavioural control. $\mathbb{E}_\mathcal{D}\text{TV}(P, Q_\theta) \leq \sqrt{\mathcal{J}_\beta(\theta)/(2\beta(1 - \beta))}$. Under input shift, $\mathbb{E}_\mu\text{JSD}_\beta \leq \mathcal{J}_\beta + h(\beta)\text{TV}(\mu, \mathcal{D})$; no density-ratio assumption is needed.
- (iii) Agnostic competitiveness. The population minimiser is TV-competitive (Proposition 4.13).
- (iv) Fisher consistency. $\mathcal{J}_\beta(\theta) = 0$ iff $Q_\theta^x = P^x$ for $\mathcal{D}$ -a.e. x . Under Assumption A3, the minimisers are exactly the lossless students.
- (v) No loss in the near-lossless regime. If $a \leq P^x/Q_\theta^x \leq b$, then $\mathcal{J}_\beta = \frac{\beta(1-\beta)}{2}\mathbb{E}\chi^2(1 + O(b-a)) = \beta(1 - \beta)\mathbb{E}\text{KL}(P\|Q_\theta)(1 + O(b-a))$.
- (vi) Unbiased, Rao-Blackwellised gradients (Proposition 4.11).
- (vii) Horizon-free estimation [proved under Assumption A4]. Let $\hat{\mathcal{J}}_n(\theta)$ average the estimator φ of Proposition 4.12 over n i.i.d. prompts, one mixture draw each, generated with common random numbers across θ (for example $y = G_\theta(x, \xi)$ with the noise ξ fixed). With probability $\geq 1 - \delta$, $\sup_\theta |\hat{\mathcal{J}}_n(\theta) - \mathcal{J}_\beta(\theta)| \leq \bar{h}(\beta)\sqrt{(R\ln 2 + \log(2/\delta))/(2n)}$, with no factor T . Compare the factor TB for the per-sequence KL loss of Theorem 2.6.

Proof. (i) Proposition 4.10. (ii) Proposition 1.7(i) with Jensen, and Proposition 1.18 with $G = h(\beta)$. (iii) Proposition 4.13. (iv) $\text{JSD}_\beta \geq 2\beta(1 - \beta)\text{TV}^2$. (v) Proposition 1.8(ii) applied to sequence laws, and $f''_\beta(1) = \beta(1 - \beta)$. (vi) Proposition 4.11. (vii) For each fixed θ the summands $\varphi_\theta(x_i, \xi_i)$ are i.i.d., unbiased, and $[0, \bar{h}(\beta)]$ -valued (Proposition 4.12). Apply Hoeffding, then a union bound over the 2^R students. $\square$

Remark 4.16 (Why this targets the dominant term). By Heuristic 2.22, the dominant term under capacity limits is approximation error as seen by the deployment criterion. PA-JSD attacks all three ways that term is inflated. Exactness removes its amplification by prefix shift. Agnostic competitiveness prevents its misallocation. Boundedness makes the estimation and covariate-shift terms horizon-free and additive. In the near-lossless regime it reduces to KL up to the constant $\beta(1 - \beta)$, so nothing is lost when losslessness is reachable. The price is on-policy sampling and the variance of a sequence-level score function. The saturation $w_t \in \{0, 1\}$ on prefixes that are already distinguishable limits the effective horizon of that variance, but does not remove it.

Proposition 4.17 (Likelihood-critical alternative: skew-Jeffreys). *[proved] [numerically checked]* Let $\mathcal{J}_\lambda^{\text{SJ}}(\theta) := (1 - \lambda)\sum_t \mathbb{E}_{d_t^p}\text{KL}(p_t\|q_t) + \lambda\sum_t \mathbb{E}_{d_t^q}\text{KL}(q_t\|p_t)$. It is estimated by off-policy soft-label forward KL plus on-policy reverse KL (Proposition 4.9). Then $\mathcal{J}_\lambda^{\text{SJ}} = \mathbb{E}[(1 - \lambda)\text{KL}(P\|Q_\theta) + \lambda\text{KL}(Q_\theta\|P)]$ exactly, and $\mathbb{E}\text{TV} \leq \sqrt{\mathcal{J}_\lambda^{\text{SJ}}/2}$. The log-loss regret is at most $\mathcal{J}_\lambda^{\text{SJ}}/(1 - \lambda)$. Both event bounds of Proposition 1.10 hold with KL replaced by $\mathcal{J}_\lambda^{\text{SJ}}/(1 - \lambda)$ (coverage) and $\mathcal{J}_\lambda^{\text{SJ}}/\lambda$ (precision), per input. For no $\lambda \in [0, 1]$ is it agnostically TV-competitive. Take $P = (1 - \eta, \eta)$ and the class $\{Q_1 = (1 - \varepsilon, \varepsilon), U = (\frac{1}{2}, \frac{1}{2})\}$. Then $\text{TV}(P, Q_1) \leq \eta$, but $\mathcal{J}_\lambda^{\text{SJ}}(Q_1) \geq (1 - \lambda)\eta(\log \frac{\eta}{\varepsilon} - 1)$, which exceeds the finite value $\mathcal{J}_\lambda^{\text{SJ}}(U)$ once ε is small enough. The minimiser is then U , with $\text{TV} = \frac{1}{2} - \eta$. For example, with $\eta = 0.01$, $\lambda = \frac{1}{2}$ and $\varepsilon = 10^{-120}$: $\mathcal{J}^{\text{SJ}}(Q_1) \approx 1.36 > \mathcal{J}^{\text{SJ}}(U) \approx 1.13$. For $\lambda = 1$ use Proposition 4.13(b).

Recommendation. If two-sided losslessness is within reach, the two objectives agree to second order (Proposition 1.8), and skew-Jeffreys additionally certifies likelihood-level guarantees. If

the student is capacity-limited, use PA-JSD to allocate its error where the deployment criterion is least hurt. A natural schedule is PA-JSD to convergence followed by a skew-Jeffreys fine-tune, monitoring both sequence-level quantities (Section 6.3).

5 Beyond output matching: intermediate representations

5.1 Formalisation

Let $h_T(s) \in \mathbb{R}^{d_T}$ be a teacher representation at some layer and $h_S(s) \in \mathbb{R}^{d_S}$ a student representation, both centred under a prefix law ν , with covariances $\Sigma_{TT}, \Sigma_{SS} \succ 0$ and cross-covariance Σ_{TS} . Representation matching adds a constraint to the output objective $\mathcal{J}$:

$$\min_{\theta} \mathcal{J}(\theta) \quad \text{s.t.} \quad \mathcal{R}(\theta) \leq \eta, \quad \text{or its Lagrangian} \quad \mathcal{J}(\theta) + \gamma \mathcal{R}(\theta).$$

Two standard choices are a learned linear map, as in FitNets (Romero et al., 2015),

$$\mathcal{R}_{LS}(\theta) := \min_{A \in \mathbb{R}^{d_T \times d_S}} \mathbb{E}_{\nu} \|Ah_S(s) - h_T(s)\|^2,$$

and linear CKA (Kornblith et al., 2019),

$$\mathcal{R}_{CKA}(\theta) := 1 - \frac{\|\Sigma_{ST}\|_F^2}{\|\Sigma_{SS}\|_F \|\Sigma_{TT}\|_F}.$$

Relation-based variants match attention maps or value relations (Zagoruyko and Komodakis, 2017; Jiao et al., 2020; Wang et al., 2020; Sun et al., 2020). Contrastive variants maximise a mutual-information bound (Tian et al., 2020).

Proposition 5.1 (The learned-map penalty is CCA; CKA constrains geometry). *[proved] [numerically checked] $\mathcal{R}_{LS} = \text{tr} \Sigma_{TT} - \text{tr}(\Sigma_{TS} \Sigma_{SS}^{-1} \Sigma_{ST})$. If the teacher features are whitened ($\Sigma_{TT} = I$), then $\mathcal{R}_{LS} = d_T - \sum_i \rho_i^2$, where the ρ_i are the canonical correlations. $\mathcal{R}_{LS}$ is invariant under $h_S \mapsto Gh_S$ for every invertible G , so it constrains only the linear information content of h_S . $\mathcal{R}_{CKA}$ is invariant only under orthogonal maps and isotropic scaling, so it also constrains the geometry (spectrum) of h_S . For example, with $d_S = d_T$ and $h_S = \Sigma_{TT}^{-1/2} h_T$: $\mathcal{R}_{LS} = 0$, but $1 - \mathcal{R}_{CKA} = \text{tr} \Sigma_{TT} / (\sqrt{d_T} \|\Sigma_{TT}\|_F) < 1$ unless $\Sigma_{TT} \propto I$.*

Proof. Normal equations give $A^* = \Sigma_{TS} \Sigma_{SS}^{-1}$. With $\Sigma_{TT} = I$, $\text{tr}(\Sigma_{TS} \Sigma_{SS}^{-1} \Sigma_{ST}) = \|\Sigma_{SS}^{-1/2} \Sigma_{ST}\|_F^2 = \sum_i \rho_i^2$. In the example, $\Sigma_{SS} = I$ and $\Sigma_{ST} = \Sigma_{TT}^{1/2}$. The CKA inequality is Cauchy–Schwarz on the eigenvalues. $\square$

5.2 When representation matching is redundant

Proposition 5.2 (Last-layer matching is implied by high-temperature logit matching). *[proved under Assumption A5] If CW_T has full column rank d_T (which requires $V - 1 \geq d_T$, true for language models), then*

$$\mathcal{R}_{LS} \leq \mathbb{E} \|(CW_T)^+(\tilde{z}_S - \tilde{z}_T)\|^2 \leq \frac{\mathbb{E} \|\tilde{z}_S - \tilde{z}_T\|^2}{\sigma_{\min}(CW_T)^2} = \frac{2V}{\sigma_{\min}(CW_T)^2} \lim_{\tau \rightarrow \infty} \mathbb{E} \mathcal{L}_{\tau}.$$

Final-layer feature matching therefore adds nothing to full-logit matching at high temperature. It is informative only when outputs pass through a lossy channel: hard labels, top-k logits, or $\tau = 1$ with a finite optimisation budget (Proposition 4.6).

Proof. Take $A = (CW_T)^+ CW_S$. With read-out biases, $\tilde{z} = C(Wh + b)$, and $(CW_T)^+ CW_T = I$ gives $Ah_S - h_T = Y - c$ with $Y := (CW_T)^+(\tilde{z}_S - \tilde{z}_T)$ and the constant $c := (CW_T)^+ C(b_S - b_T)$. The features are centred, so $\mathbb{E}[Ah_S - h_T] = 0$, hence $c = \mathbb{E}Y$ and $\mathbb{E}\|Ah_S - h_T\|^2 = \mathbb{E}\|Y - \mathbb{E}Y\|^2 \leq \mathbb{E}\|Y\|^2$. The last equality in the statement is Proposition 4.5(ii). $\square$

Proposition 5.3 (No identifiability gain in the realisable noiseless linear model). *[proved within the stated model] Let $u(s) \in \mathbb{R}^D$ be fixed input features, with teacher $h_T = B^\circ u \in \mathbb{R}^k$, $z_T = W^\circ h_T$, and student $h_S = Bu \in \mathbb{R}^k$, $z_S = WBu$ (realisable, $d_S = d_T = k$). With exact soft labels on states whose features span a subspace $\mathcal{U} \subseteq \mathbb{R}^D$, output matching identifies $CW^\circ B^\circ$ on $\mathcal{U}$. Adding exact representation targets identifies B° on the same $\mathcal{U}$. Neither yields any extrapolation beyond $\mathcal{U}$, and neither needs fewer states. Any gain must therefore come from noise, conditioning, or computation.*

5.3 When it provably helps

Proposition 5.4 (Noisy output supervision: fewer parameters learned from noise). *[known result; proof sketch] In the model of Proposition 5.3 (so $d_T = k$) with $\mathcal{U} = \mathbb{R}^D$, suppose outputs are observed only through hard labels $y \sim \text{softmax}(W^\circ B^\circ u)$ and $\text{rank}(W^\circ B^\circ) = k$. The MLE over rank- k factorisations satisfies $n \text{KL} \Rightarrow \frac{1}{2} \chi_k^2(V-1+D-k)$, with limiting mean $\frac{1}{2}k(V-1+D-k)$: the dimension of the manifold of rank- k $(V-1) \times D$ matrices, by the regular-model asymptotics of Proposition 2.10(b). If B° is instead supplied exactly by noiseless representation targets, only W is estimated, and the limiting mean drops to $\frac{1}{2}k(V-1)$. The relative reduction is $(D-k)/(V-1+D-k)$.*

In general, matching at a layer turns estimation of everything *below* that layer into noiseless regression. The statistical gain is the complexity of the sub-network below the matched layer relative to the whole. The gain disappears when the output channel is itself noiseless (Proposition 5.3).

Proposition 5.5 (Representation matching is a preconditioner). *[proved within the stated model] In the model of Proposition 5.3, near the optimum, with $W = W^\circ$ fixed and the near-lossless quadratic output loss, $\mathcal{L}_{\text{out}}(B) = \frac{1}{2} \text{tr}((B - B^\circ)\Sigma_u(B - B^\circ)^\top K)$ with $K = W^{\circ\top} F W^\circ$. Its Hessian is $\Sigma_u \otimes K$. Adding $\gamma \mathbb{E}\|Bu - B^\circ u\|^2$ gives $\Sigma_u \otimes (K + 2\gamma I)$, with the same minimiser. Gradient descent converges linearly at rate $1 - 1/\kappa$, where*

$$\kappa_\gamma = \kappa(\Sigma_u) \frac{\lambda_{\max}(K) + 2\gamma}{\lambda_{\min}(K) + 2\gamma} \searrow \kappa(\Sigma_u) \quad (\gamma \rightarrow \infty).$$

The ratio κ_0/κ_γ can be arbitrarily large, and $\kappa_0 = \infty$ when K is singular, without any change to the realisable solution. Heuristic: in language models K is badly conditioned, because $F = \text{diag } p - pp^\top$ is dominated by a few tokens (Proposition 4.6), so the speed-up can be large in practice.

Proposition 5.6 (Compositional hardness: a computational separation). *[known result; proof sketch] There are targets, for example k -sparse parities over D bits computed through intermediate partial parities, for which every statistical-query learner using only end-to-end labels needs $D^{\Omega(k)}$ queries. This follows from the statistical-query dimension of parities (Blum et al., 1994; Kearns, 1998). Supervising the intermediate partial parities makes each step learnable in polynomial time (Wies et al., 2023). The separation is computational. Information-theoretically, $\log \binom{D}{k} \approx k \log D$ samples suffice end-to-end (Theorem 2.7). Representation matching can therefore reduce the optimisation term $\mathcal{E}_{\text{opt}}$ exponentially, even when it cannot reduce $\mathcal{E}_{\text{est}}$.*

5.4 When it over-constrains, and a Fisher-aligned fix

Theorem 5.7 (Over-constraint in a linear-Gaussian model). *[proved within the stated model] [numerically checked] Let $h_T \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^{d_T}$, with teacher centred logits Wh_T and a constant Fisher matrix F . Let the student be $h_S = Bh_T$ with $B \in \mathbb{R}^{d_S \times d_T}$ and $d_S < \text{rank } \Sigma$, and let $\mathcal{L}_{\text{out}} = \frac{1}{2} \mathbb{E}\|F^{1/2}(W_S B - W)h_T\|^2$. Let $G := F^{1/2}W\Sigma^{1/2}$ have singular values $s_1 \geq s_2 \geq \dots$, and let (λ_i, u_i) be the eigenpairs of Σ .*

- (a) Output only ($\gamma = 0$): $\min \mathcal{L}_{\text{out}} = \frac{1}{2} \sum_{i>d_S} s_i^2$.
(b) Full feature matching ($\gamma \rightarrow \infty$ with $\mathcal{R}_{\text{LS}}$): the row space of B is forced onto $\text{span}(u_1, \dots, u_{d_S})$, provided $\lambda_{d_S} > \lambda_{d_S+1}$. Then $\min \mathcal{L}_{\text{out}} \rightarrow \frac{1}{2} \sum_{i>d_S} \lambda_i \|F^{1/2} W u_i\|^2$.
(c) The cost (b)–(a) is ≥ 0 . It is 0 iff the top- d_S principal subspace of Σ is output-optimal, and it can be arbitrarily large. For example, with $d_T = 2$, $d_S = 1$, $\Sigma = \text{diag}(\lambda_1, \lambda_2)$, $\lambda_1 > \lambda_2$, and $W = w e_2^\top$: (a) is 0 and (b) is $\frac{1}{2} \lambda_2 w^\top F w$. The student is forced to spend its only dimension on a high-variance nuisance direction.

Proof. (a) $\mathcal{L}_{\text{out}} = \frac{1}{2} \|F^{1/2} W_S B \Sigma^{1/2} - G\|_F^2$, and the first matrix has rank $\leq d_S$; apply Eckart–Young. (b) $\min_A \mathbb{E} \|A B h_T - h_T\|^2 \geq \sum_{i>d_S} \lambda_i$, with equality iff $B = M U_{d_S}^\top$ for invertible M (PCA, when $\lambda_{d_S} > \lambda_{d_S+1}$). Given this B , the best read-out reproduces W on the retained subspace, leaving $\frac{1}{2} \mathbb{E} \|F^{1/2} W (I - U_{d_S} U_{d_S}^\top) h_T\|^2$. (c) (a) is the unconstrained minimum. $\square$

Definition 5.8 (Fisher-aligned representation matching). Let $J(s) = \partial \tilde{z}_T(s) / \partial h_T(s)$ be the teacher’s centred-logit Jacobian at the matched layer; at the last layer $J = C W_T$. Let $K := \mathbb{E}_\nu [J^\top F J]$ be the output-relevance matrix and Σ the covariance of h_T . Let U_F collect the top- d_S eigenvectors of $\Sigma^{1/2} K \Sigma^{1/2}$. Match the *output-relevant projection* of the teacher:

$$\mathcal{R}_F(\theta) := \min_A \mathbb{E}_\nu \|A h_S(s) - U_F^\top \Sigma^{-1/2} h_T(s)\|^2.$$

Proposition 5.9 (Fisher alignment has no approximation cost). *[proved within the stated model] [numerically checked] In the model of Theorem 5.7, assume $s_{d_S} > s_{d_S+1}$. For every $\gamma \geq 0$, the minimisers of $\mathcal{L}_{\text{out}} + \gamma \mathcal{R}_F$ are exactly the output-optimal students of case (a). The constraint costs nothing in approximation. Heuristic: it should retain the preconditioning effect of Proposition 5.5 in the retained directions.*

Proof. Here $\Sigma^{1/2} K \Sigma^{1/2} = G^\top G$, so U_F are the top right-singular vectors of G . The Eckart–Young optimum is $F^{1/2} W_S B \Sigma^{1/2} = G U_F U_F^\top$. It is attained by $B = M U_F^\top \Sigma^{-1/2}$ and $W_S = W \Sigma^{1/2} U_F M^{-1}$, for which $h_S = M U_F^\top \Sigma^{-1/2} h_T$ and so $\mathcal{R}_F = 0$. Both terms are at their minima simultaneously, so every minimiser of the sum has $\mathcal{R}_F = 0$ and optimal $\mathcal{L}_{\text{out}}$. The spectral gap gives uniqueness up to M . $\square$

Heuristic 5.10 (Deep networks). *For intermediate layers $J(s)$ varies with s , and K averages over it. Alignment is then approximate, but K can be estimated with a few Jacobian–vector products per batch (Hutchinson-type probes). The theory predicts the following. Full-feature matching helps when the teacher’s high-variance directions are output-relevant. It hurts when they carry nuisance variance, for example the outlier or “massive” activation dimensions reported in large transformers (we are unsure of the canonical reference for this phenomenon). CKA matching over-constrains more than $\mathcal{R}_{\text{LS}}$ when $d_S < d_T$. Fisher-aligned matching should dominate both.*

regime	effect of representation matching
last layer, full logits, high τ	redundant (Proposition 5.2)
noiseless, realisable, linear	no identifiability gain (Proposition 5.3)
hard labels / top- k / sampled sequences	fewer parameters learned from noise (Proposition 5.4)
ill-conditioned output Fisher	preconditioning, faster optimisation (Proposition 5.5)
compositional targets	exponential computational gain (Proposition 5.6)
$d_S < \text{rank } \Sigma$, misaligned variance	over-constraint, unbounded cost (Theorem 5.7)
Fisher-aligned target	no approximation cost in the model (Proposition 5.9)

6 Synthesis

6.1 The full pipeline

Theorem 6.1 (End-to-end sandwich). (i) Necessity [proved under Assumption A4] . For any prior π over teachers, if $R \ln 2 < I_\infty(\pi, \mathcal{D})$ then

$$\mathbb{E}_\pi \min_\theta \mathbb{E}_{\mathcal{D}} \text{KL}(P \| Q_\theta) \geq \sup_{N,m} \frac{I(\mathbf{T}; Y \mid X_{1:N}) - R \ln 2}{mN} > 0.$$

In the K -facts model the behavioural error is bounded below too: $\bar{e} = \mathbb{E} \text{TV}$ satisfies $h(\bar{e}) + \bar{e} \ln(V - 1) \geq \ln V - R \ln 2/K$, so $\bar{e} > 0$ whenever $R < K \log_2 V$ (Theorem 3.1, Corollary 3.3).

- (ii) Sufficiency, realisable [proved under Assumptions A3 and A4] . Sequence-level hard-label KD with $n \geq 2(R \ln 2 + \log \frac{1}{\delta})/\varepsilon^2$ teacher sequences achieves $\mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_{\hat{\theta}}) \leq \varepsilon$ with probability $\geq 1 - \delta$, for every horizon T . Under input shift, $\mathbb{E}_\mu \text{TV} \leq \min\{\rho\varepsilon, \varepsilon + \text{TV}(\mu, \mathcal{D})\}$ (Theorem 2.7, Proposition 1.18). With exact soft labels and a linear-softmax student, $n \geq d$ states in general position give exact recovery (Proposition 2.10).
- (iii) Misspecified, behavioural [proved under Assumption A4] . Let $\hat{\theta}$ minimise, up to optimisation error $\mathcal{E}_{\text{opt}}$, the empirical PA-JSD objective built from the bounded mixture estimator (Proposition 4.12) with common random numbers, as in Theorem 4.15(vii). Let $\text{TV}^* = \min_\theta \mathbb{E}_{\mathcal{D}} \text{TV}(P, Q_\theta)$ and $\Delta_n = \bar{h}(\beta) \sqrt{(R \ln 2 + \log \frac{2}{\delta})/(2n)}$. With probability $\geq 1 - \delta$,

$$\mathbb{E}_\mu \text{TV}(P, Q_{\hat{\theta}}) \leq \sqrt{\frac{h(\beta) \text{TV}^* + 2\Delta_n + \mathcal{E}_{\text{opt}}}{2\beta(1 - \beta)}} + \text{TV}(\mu, \mathcal{D}).$$

There is no dependence on T and no exposure-bias term. For $\beta = \frac{1}{2}$ the leading term is $\sqrt{2 \ln 2 \text{TV}^*}$.

- (iv) Misspecified, likelihood [proved under Assumptions A1, A2 and A5] . Off-policy soft-label forward KL gives Corollary 2.21. The approximation term obeys $\mathcal{A} \geq T\varepsilon_0$ (Corollary 2.5), and for students with a linear read-out ε_0 is at least the softmax-bottleneck floor (Theorem 2.3).

Proof. (i), (ii) and (iv) collect earlier results; Assumption A5 is needed only for the bottleneck clause of (iv). For (iii), write $\hat{\mathcal{J}}$ for the empirical objective. By Theorem 4.15(vii), $\mathcal{J}(\hat{\theta}) \leq \hat{\mathcal{J}}(\hat{\theta}) + \Delta_n \leq \hat{\mathcal{J}}(\theta_{\text{TV}}) + \mathcal{E}_{\text{opt}} + \Delta_n \leq \mathcal{J}(\theta_{\text{TV}}) + \mathcal{E}_{\text{opt}} + 2\Delta_n$ and $\mathcal{J}(\theta_{\text{TV}}) \leq h(\beta) \text{TV}^*$ (Proposition 1.7(i)). Then apply Theorem 4.15(ii) and Proposition 1.18 with $G = 1$. $\square$

Reading the sandwich. Near-lossless distillation is possible *only if* the student’s usable description length exceeds the teacher’s functional information on the deployment distribution. This condition is necessary, not sufficient: sufficiency needs realisability, which $R \ln 2 > I_\infty$ does not imply. When the teacher is realisable, the sample complexity of behavioural ε -losslessness is $\tilde{O}(R/\varepsilon^2)$ sequences independent of the horizon. Exposure bias is not a fundamental obstacle: it is an artefact of pairing a divergence with the wrong prefix law. When it is impossible, the best achievable behavioural error is TV^* , and PA-JSD is guaranteed to reach $O(\sqrt{\text{TV}^*})$, whereas KL objectives carry no such guarantee.

6.2 Status of the results

status	results
proved, unconditional	TV as worst-case regret (1.4); JSD _β sandwich (1.7); local equivalence (1.8); event bounds (1.10); margins (1.11); chain rules (1.12); hybrid bounds (1.14, 1.15); master decomposition (2.1); compounding theorems (2.14–2.19); curvature (2.11); gradient formulas and unbiased estimators (4.1, 4.8–4.11); temperature limits (4.5); bounded mixture estimator (4.12); agnostic competitiveness (4.13); facts floors and metric entropy (3.2–3.4); one-shot converse (3.7); CCA/CKA facts (5.1); decoding-level thresholds and bounds for greedy decoding, beam search and greedy verification (7.1–7.15)
proved under assumptions	capacity floor (A4); softmax-bottleneck floor (A5); uniform deviation (A1); horizon-free MLE (A3, A4); PA-JSD estimation (A4); end-to-end bounds; last-layer redundancy (A5)
proved within a stylised model	power-law and exponential floors (3.8, 3.9); no capacity gap under optimal allocation (3.10); no identifiability gain and preconditioning (5.3, 5.5); over-constraint and Fisher alignment (5.7, 5.9)
known results cited heuristics	Pinsker/BH; SGD rates; $k/2n$ asymptotics; SQ lower bounds for parities dominant term (2.22); size and data exponents (3.11, 3.12); capacity gap as aliasing (3.13); deep-network representation matching (5.10)
conjecture	distillation gap law (3.14)

6.3 Experiments that test the key assumptions

Several quantities recur below. Sequence-level TV has the unbiased, $[0, 1]$ -bounded estimator $\widehat{\text{TV}} = \frac{1}{n} \sum_i (1 - Q_\theta(y_i | x_i)/P(y_i | x_i))_+$ with $y_i \sim P^{x_i}$. It requires only exact sequence log-probabilities under both models (`check_sequences.py`), and admits Hoeffding intervals. Per-position curves $e_t^{\text{off,KL}}$, $e_t^{\text{on,rKL}}$ and e_t^{TV} are computed exactly over the vocabulary at sampled prefixes.

E1. Horizon, exposure and recoverability. *Hypotheses:* Corollary 1.16, Theorems 2.15, 2.16 and 2.18, and Proposition 4.10. *Protocol:* one teacher; two student sizes; five objectives: off-policy soft forward KL; on-policy reverse KL with the full estimator of Proposition 4.9; the same with stop-gradient; token-level GKD JSD_β; PA-JSD. *Measure:* (a) $\widehat{\text{TV}}(T')$ on prefixes truncated at $T' \in \{16, \dots, 2048\}$, against $\sqrt{\frac{1}{2} \sum_{t \leq T'} e_t^{\text{off,KL}}}$ and $\sum_{t \leq T'} e_t^{\text{off,TV}}$. The ratio of $\widehat{\text{TV}}$ to each bound locates the task in the diffuse or concentrated regime of Example 1.17. (b) Cumulative cost growth. With $c_t = \mathbf{1}\{\text{step } t \text{ violates a checkable constraint}\}$, for example a format, unit-test or arithmetic-step check, fit $J_c(q_\theta; T') - J_c(p; T') \propto T'^\kappa$. The predictions are $\kappa \approx 2$ for TV-type off-policy training on non-recoverable tasks, $\kappa \leq 1.5$ for off-policy forward KL, and $\kappa \approx 1$ for on-policy training when $u = O(1)$. (c) Recoverability. At student-visited prefixes, force each of the top- k tokens and estimate $\hat{u}_t(s) = \max_a \hat{Q}_t^p(s, a) - \min_a \hat{Q}_t^p(s, a)$ from teacher continuations. (d) The prefix-weight process $w_t(s)$ of Proposition 4.10: the fraction of positions where $|w_t - \beta| > 0.1$, which is where token-level GKD departs from sequence-level JSD. *Falsifier:* on-policy gains that do not correlate with small $\hat{u}$, or κ unaffected by the prefix law, would refute the performance-difference account of exposure bias.

E2. Capacity floors and exponents. *Hypotheses:* Theorem 2.3, Propositions 3.8 and 3.9, Heuristics 3.11 and 3.13. *Protocol:* compute the teacher’s centred log-probability matrix on $N \approx 10^5$ held-out contexts, row-weighted by $\Pi^{1/2}$ and, separately, by per-context Fisher square

roots. Estimate its spectrum and a decay exponent $\hat{\alpha}$. Train students across widths d_S , parameter counts N_S and quantisation levels b , at a token budget large enough that $\mathcal{E}_{\text{est}} + \mathcal{E}_{\text{opt}}$ is negligible. Check this by doubling the budget. *Measure*: (a) $\varepsilon_{\text{off}}^{\text{KL}}$ against the bottleneck floor for each d_S (the floor must never be violated, and its tightness is informative); (b) the slope of $\log \varepsilon_{\text{off}}^{\text{KL}}$ against $\log(N_S b_{\text{eff}})$, compared with $-(\hat{\alpha} - 1)$; (c) curvature of that curve as N_S approaches the teacher’s size (the predicted power-law-to-exponential transition); (d) for a fixed student, the ground-truth cross-entropy as a function of teacher size at several token budgets n . The optimal teacher size should grow with n , and the penalty for a too-strong teacher should shrink roughly as k/n . *Falsifier*: capacity gaps that persist undiminished as n grows would contradict Heuristic 3.13 and point to optimisation, not aliasing.

E3. Soft versus hard labels, and representation matching. *Hypotheses*: Proposition 2.8, Heuristic 3.12, Propositions 5.2, 5.5 and 5.9 and Theorem 5.7. *Protocol*: fixed student; supervision by full logits, top- k logits ($k \in \{1, 8, 64\}$), or sampled tokens; token budgets over three orders of magnitude. Representation terms at depths $\{L/4, L/2, 3L/4, L\}$: none, $\mathcal{R}_{\text{LS}}$, $\mathcal{R}_{\text{CKA}}$, or $\mathcal{R}_{\text{F}}$, with K estimated by Hutchinson probes. *Measure*: (a) fitted data exponents b in $\varepsilon_{\text{off}}^{\text{KL}} \approx A + Bn^{-b}$. The prediction is $b_{\text{soft}} > b_{\text{topk}} > b_{\text{hard}}$, with $b_{\text{soft}}/b_{\text{hard}} \approx \hat{\alpha}$. (b) Per-step gradient variance for soft versus hard labels, against the predicted gap $\mathbb{E}_{\nu} \text{tr} \mathcal{I}(s)$. (c) The alignment score $a = \|P_{\text{PCA}} U_F\|_F^2 / d_S$ at each layer, and the over-constraint cost $\Delta \mathcal{A}(\gamma) = \varepsilon_{\text{off}}^{\text{KL}}(\gamma) - \varepsilon_{\text{off}}^{\text{KL}}(0)$. Prediction: $\Delta \mathcal{A}$ is decreasing in a , and ≈ 0 for $\mathcal{R}_{\text{F}}$. (d) Steps to reach a target KL, against the condition-number ratio κ_0/κ_{γ} of Proposition 5.5. (e) Whether last-layer matching adds anything on top of full logits at high τ . It should not. *Falsifier*: benefits of $\mathcal{R}_{\text{LS}}$ under full-logit supervision that are uncorrelated with both conditioning and alignment would indicate a mechanism outside this theory.

7 Decoding-level losslessness

Proposition 1.11 is a pathwise, deterministic sufficient condition. This section answers the quantitative question it leaves open: under the deployment law, how likely is it that the greedy or beam-search outputs of teacher and student differ, given per-token divergences and the teacher’s margins? There are four answers. (a) Every f -divergence has an exact *flip threshold*, which depends only on the teacher’s two largest probabilities (Proposition 7.4). (b) Given per-token errors along the teacher’s greedy paths, the largest possible disagreement probability is the value of a knapsack problem over the distribution of these thresholds. Up to the mass of a single input, this value has a closed dual form (Theorem 7.7). (c) Sequence-level divergences, and hence the per-token errors that training controls, see the merge cost of the two most likely continuations of each greedy prefix. For TV and KL this is exactly the per-token threshold multiplied by the teacher’s probability of its own greedy prefix. That weight can be exponentially small in T , so behavioural and decoding losslessness are logically independent (Theorem 7.9 and Corollary 7.10). (d) Beam trajectories obey the same laws, with the probability gap at the beam’s cut in place of the margin (Theorem 7.14). Beam outputs are bounded by the same quantities, but not always sharply.

Setting. Ties are broken by a fixed total order on $\mathcal{V}$: $\text{gr}(r)$ is the first maximiser of $r \in \Delta(\mathcal{V})$ in that order. Greedy decoding of the teacher emits $y_t = \text{gr}(p_t(\cdot \mid x, y_{<t}))$, and $g_p(x) \in \mathcal{Y}$ is its output; $g_{q_\theta}(x)$ is defined in the same way. The teacher’s greedy path is $s_1^* = x$, $a_t^* = \text{gr}(p_t(\cdot \mid s_t^*))$, $s_{t+1}^* = (s_t^*, a_t^*)$. At s_t^* we write $p_{t,1}^* \geq p_{t,2}^*$ for the two largest teacher probabilities, $\gamma_t = p_{t,1}^* - p_{t,2}^*$ for the margin of Proposition 1.11, $m_t = p_{t,1}^* + p_{t,2}^*$ for the top-two mass, and

$$\pi_t^*(x) := P^x(y_{<t} = a_{<t}^*) = \prod_{j < t} p_{j,1}^*$$

for the teacher's probability of its own greedy prefix. All of these depend on x . The student *flips* at step t if $\text{gr}(q_t(\cdot \mid s_t^*)) \neq a_t^*$. The *decoding risk* is $\mathcal{L}_{\text{dec}}(\theta; \mu) := \mathbb{P}_{x \sim \mu}[g_{q_\theta}(x) \neq g_p(x)]$.

Lemma 7.1 (First flip; zero temperature). *[proved]*

- (i) $g_{q_\theta}(x) \neq g_p(x)$ if and only if the student flips at some step. The two greedy outputs agree up to the first flip and differ there.
- (ii) For $\tau > 0$, let $P^{x,\tau}$ and $Q_\theta^{x,\tau}$ be the sequence laws obtained by replacing every next-token distribution r by $r^{1/\tau} / \sum_a r(a)^{1/\tau}$. If the maximisers along both greedy paths are unique, then $\text{TV}(P^{x,\tau}, Q_\theta^{x,\tau}) \rightarrow \mathbf{1}\{g_{q_\theta}(x) \neq g_p(x)\}$ as $\tau \rightarrow 0$. If this holds for μ -almost every x , the decoding risk is the zero-temperature limit of the behavioural risk.

Proof. (i) By induction on t : until the first flip, the student's greedy path visits the teacher's greedy states. (ii) Each tempered next-token distribution tends to the point mass at its unique maximiser. Hence $P^{x,\tau}(g_p(x)) \rightarrow 1$ and $Q_\theta^{x,\tau}(g_{q_\theta}(x)) \rightarrow 1$, and the TV between two point masses is the indicator that they differ. The last claim follows by dominated convergence. $\square$

7.1 Flip thresholds

Definition 7.2 (Flip threshold; merge cost). For a divergence D on $\Delta(\mathcal{V})$ and $p \in \Delta(\mathcal{V})$ with $a^* = \text{gr}(p)$, the *flip threshold* is

$$\kappa_D(p) := \inf \left\{ D(p \parallel q) : q(a^*) \leq \max_{b \neq a^*} q(b) \right\}.$$

Every flip satisfies the constraint, whatever the tie-breaking rule. For convex f with $f(1) = 0$, let $g_f(u, v) := v f(u/v)$ be its perspective, with the conventions of Definition 1.1. For $P_1 \geq P_2 \geq 0$ with $R := 1 - P_1 - P_2 \geq 0$, the *merge cost* is

$$k_f(P_1, P_2) := \min_{0 \leq c \leq \frac{1}{2}} \left[g_f(P_1, c) + g_f(P_2, c) + g_f(R, 1 - 2c) \right].$$

The merge cost is the least f -divergence at which two atoms of a law, of masses $P_1 \geq P_2$, can be put in the wrong order. The next lemma proves this, and every threshold in this section is an instance of it.

Lemma 7.3 (Merge lemma). *[proved] [numerically checked]* Let P be a law on a countable set, and let S_+ , S_- be disjoint finite sets of atoms with $P^+ := \min_{S_+} P \geq P^- := \max_{S_-} P$. For every f -divergence,

$$\inf \{ D_f(P \parallel Q) : Q(a) \leq Q(b) \text{ for some } a \in S_+, b \in S_- \} = k_f(P^+, P^-).$$

The infimum is attained by the law that gives mass c^* , a minimiser in k_f , to an atom $a^+ \in S_+$ of mass P^+ and to an atom $b^- \in S_-$ of mass P^- , and multiplies P by $(1 - 2c^*)/(1 - P^+ - P^-)$ elsewhere. If $P^+ + P^- = 1$, the bracket in k_f is non-increasing in c , and we take $c^* = \frac{1}{2}$. Suppose moreover that $k_f(P^+, P^-) < \infty$ and that $P^- > 0$ or $f(0) < \infty$. Then for every $\delta > 0$ some Q with $Q(a^+) < Q(b^-)$ has $D_f(P \parallel Q) \leq k_f(P^+, P^-) + \delta$. Without the last condition this can fail: for reverse KL ($f(0) = \infty$) and $P^- = 0$, every Q with $Q(b^-) > 0$ has $\text{KL}(Q \parallel P) = \infty$.

Proof. Let $Q(a) \leq Q(b)$ with $a \in S_+$, $b \in S_-$ and $D_f(P \parallel Q) < \infty$. *Exchange.* If $Q(a^+) > Q(a)$, exchange the Q -masses of a and a^+ . Then, if $Q(b^-) < Q(b)$, exchange the Q -masses of b and b^- . Afterwards $Q(a^+) \leq Q(b^-)$. Neither exchange increases D_f . Each one gives the larger of two Q -masses to the atom with the larger P -mass, and the perspective is submodular: $g_f(u', v') + g_f(u, v) \leq g_f(u', v) + g_f(u, v')$ for $u \leq u'$ and $v \leq v'$. To see this, note that for $v > 0$, $g_f(u', v) - g_f(u, v)$ is $u' - u$ times the slope of the chord of f between u/v and u'/v . Both endpoints decrease as v grows, and chord slopes of a convex function increase with either

endpoint. The case $v = 0$ follows by continuity. *Three cells.* Data processing onto the partition $\{a^+\}, \{b^-\}$, rest gives $D_f(P\|Q) \geq D_f(\bar{P}\|\bar{Q})$, with $\bar{P} = (P^+, P^-, R)$ and $\bar{Q}_1 \leq \bar{Q}_2$. *Equalise.* The map $\bar{Q} \mapsto D_f(\bar{P}\|\bar{Q})$ is convex and vanishes at $\bar{P}$, which violates the constraint if $P^+ > P^-$ (if $P^+ = P^-$, both sides are 0). So the segment from $\bar{Q}$ to $\bar{P}$ meets $\{\bar{Q}_1 = \bar{Q}_2\}$ at a point of no larger divergence, and $D_f(P\|Q) \geq k_f(P^+, P^-)$. The stated law has a likelihood ratio that is constant on each cell, so its divergence equals the three-cell value $k_f(P^+, P^-)$. (If $R = 0$, the derivative of the bracket is $\psi(P^+/c) + \psi(P^-/c) - 2\psi(0) \leq 0$, where $\psi(u) = f(u) - uf'(u)$ is non-increasing, so $c^* = \frac{1}{2}$ is a minimiser.) *Strict order.* The least $D_f(P\|Q)$ subject to $Q(b^-) \geq Q(a^+) + \delta$ is convex and non-decreasing in δ and equals k_f at $\delta = 0$. It is finite for small $\delta > 0$: move mass δ into b^- from another atom that the minimiser charges. This adds a finite cost at b^- because $P^- > 0$ or $f(0) < \infty$. Hence it tends to k_f as $\delta \downarrow 0$. $\square$

Proposition 7.4 (Exact flip thresholds). *[proved] [numerically checked] Let $p_1 \geq p_2$ be the two largest probabilities of p , $\gamma = p_1 - p_2$ and $m = p_1 + p_2$.*

- (i) *For every f -divergence, $\kappa_f(p) = k_f(p_1, p_2)$: only the two largest probabilities matter. The infimum is attained. If $\kappa_f(p) < \infty$, then for every $\delta > 0$ a strict flip (one that flips under any tie rule) costs at most $k_f(p_1, p_2) + \delta$.*
- (ii) *Closed forms and lower bounds:*

$D(p\ q)$	$\kappa_D(p)$	lower bound
TV	$\gamma/2$	–
KL($p\ q$)	$p_1 \log \frac{2p_1}{m} + p_2 \log \frac{2p_2}{m}$	$\gamma^2/(2m)$
KL($q\ p$)	$-\log(1 - (\sqrt{p_1} - \sqrt{p_2})^2)$	$\gamma^2/(2m)$
H^2	$1 - \sqrt{1 - \frac{1}{2}(\sqrt{p_1} - \sqrt{p_2})^2}$	$\gamma^2/(8m)$
$\chi^2(p\ q)$	$(1 - m + \sqrt{m^2 + \gamma^2})^2 - 1$	$2(\sqrt{m^2 + \gamma^2} - m)$
$\text{JSD}_\beta(p\ q)$	$k_{f_\beta}(p_1, p_2)$	$\beta(1 - \beta)\gamma^2/2$
logit oscillation $\max_a \log \frac{q(a)}{p(a)} - \min_a \log \frac{q(a)}{p(a)}$	$\log(p_1/p_2)$	–
centred logits $\ C(\log q - \log p)\ _2$	$\log(p_1/p_2)/\sqrt{2}$	–

- (iii) (Local form.) *If f is twice continuously differentiable near 1 with $f''(1) > 0$, then $\kappa_f(p) = \frac{f''(1)\gamma^2}{2m}(1 + o(1))$ as $\gamma \rightarrow 0$ with m fixed. For forward KL, reverse KL and H^2 , this leading term is also a global lower bound.*

Proof. (i) Apply Lemma 7.3 with $S_+ = \{a^*\}$ and $S_- = \mathcal{V} \setminus \{a^*\}$. Its strict clause applies, because $p_2 = 0$ forces $p_1 = 1$, and then $k_f(1, 0) = \infty$ if $f(0) = \infty$. (ii) For TV the bracket in k_f is $\frac{1}{2}(|p_1 - c| + |p_2 - c| + |m - 2c|)$, minimal at $c = m/2$. For forward KL it is $p_1 \log \frac{p_1}{c} + p_2 \log \frac{p_2}{c} + R \log \frac{R}{1-2c}$, which is stationary at $c = m/2$. For reverse KL it is $c \log \frac{c^2}{p_1 p_2} + (1 - 2c) \log \frac{1-2c}{R}$. Stationarity gives $c = \sqrt{p_1 p_2}/Z$ with $Z = R + 2\sqrt{p_1 p_2} = 1 - (\sqrt{p_1} - \sqrt{p_2})^2$, and the value is $-\log Z$. For H^2 , maximise $\sqrt{2c} \cdot \frac{\sqrt{p_1} + \sqrt{p_2}}{\sqrt{2}} + \sqrt{1 - 2c} \cdot \sqrt{R}$ by Cauchy–Schwarz. For χ^2 , minimise $\frac{2(p_1^2 + p_2^2)}{2c} + \frac{R^2}{1-2c}$ using $\frac{a^2}{x} + \frac{b^2}{y} \geq \frac{(a+b)^2}{x+y}$, and note $2(p_1^2 + p_2^2) = m^2 + \gamma^2$. For the lower bounds: the KL threshold is $m \text{kl}(\frac{p_1}{m} \|\frac{1}{2}) \geq 2m(\frac{\gamma}{2m})^2$ by Pinsker. Next, $-\log(1 - x) \geq x$, $1 - \sqrt{1 - x/2} \geq x/4$, and $(\sqrt{p_1} - \sqrt{p_2})^2 = \gamma^2/(\sqrt{p_1} + \sqrt{p_2})^2 \geq \gamma^2/(2m)$. The χ^2 bound drops the square, and the JSD_β bound combines Proposition 1.7(i) with $\text{TV} \geq \gamma/2$. For logits, let $\delta = \log q - \log p$. A flip to b needs $\delta_b - \delta_{a^*} \geq \log \frac{p(a^*)}{p(b)} \geq \log \frac{p_1}{p_2}$, and $\|C\delta\|_2^2 = \min_c \|\delta - c\mathbf{1}\|^2 \geq \frac{1}{2}(\delta_b - \delta_{a^*})^2$. The perturbation $\delta = \frac{1}{2} \log \frac{p_1}{p_2} (e_b - e_{a^*})$, with b the runner-up, attains both. (iii) The choice $c = m/2$ gives $\frac{m}{2}[f(1 + \frac{\gamma}{m}) + f(1 - \frac{\gamma}{m})] = \frac{f''(1)\gamma^2}{2m}(1 + o(1))$. Conversely, write the bracket as $A + B$. Here $B := 2cf(\frac{m}{2c}) + (1 - 2c)f(\frac{R}{1-2c}) = D_f(\text{Bern}(m)\|\text{Bern}(2c)) \geq 0$, and $A := c[f(u_+) + f(u_-) - 2f(\frac{u_+ + u_-}{2})]$ with $u_\pm = p_{1,2}/c$. At a minimiser $A + B = O(\gamma^2)$, so $B \rightarrow 0$ forces $2c \rightarrow m$. Then $u_\pm \rightarrow 1$ and $A = c(\frac{\gamma}{2c})^2 f''(\xi)$ with $\xi \rightarrow 1$. The global lower bounds are those of (ii). $\square$

Near a tie, TV needs error $\gamma/2$ to flip, but every smooth divergence needs only order γ^2/m . The factor $1/m$ is invisible to Pinsker's inequality, which gives only $\text{KL} \geq \gamma^2/2$.

7.2 Greedy decoding: errors on the greedy path

Definition 7.5 (Knapsack function). For a cost $\omega : \mathcal{X} \rightarrow [0, \infty]$ and a budget $\varepsilon \geq 0$,

$$\Phi_\omega(\varepsilon) := \sup \left\{ \mu(A) : \int_A \omega \, d\mu \leq \varepsilon \right\}, \quad \bar{\Phi}_\omega(\varepsilon) := \inf_{\tau > 0} \left\{ \frac{\varepsilon}{\tau} + \mathbb{E}_\mu \left(1 - \frac{\omega}{\tau} \right)_+ \right\}.$$

$\Phi_\omega(\varepsilon)$ is the largest μ -mass that a budget ε can buy when input x costs $\omega(x)$.

Lemma 7.6 (Knapsack duality). *[proved] [numerically checked] $\bar{\Phi}_\omega(\varepsilon) = \sup \{ \mathbb{E}_\mu w : 0 \leq w \leq 1, \mathbb{E}_\mu [w\omega] \leq \varepsilon \}$. This fractional problem is solved by buying inputs in increasing order of ω . Moreover $\Phi_\omega - \max_x \mu(\{x\}) \leq \Phi_\omega \leq \bar{\Phi}_\omega$ and $\Phi_\omega(\varepsilon) \leq \inf_{\tau > 0} \{ \varepsilon/\tau + \mu[\omega < \tau] \}$.*

Proof. For feasible w and $\tau > 0$, $\mathbb{E}w \leq \mathbb{E}w + \frac{1}{\tau}(\varepsilon - \mathbb{E}[w\omega]) = \frac{\varepsilon}{\tau} + \mathbb{E}[w(1 - \omega/\tau)] \leq \frac{\varepsilon}{\tau} + \mathbb{E}(1 - \omega/\tau)_+$. Now let w buy inputs in increasing order of ω until the budget is spent, with a fractional purchase only at the last level τ^* reached. Then $w = 1$ on $\{\omega < \tau^*\}$, $w = 0$ on $\{\omega > \tau^*\}$ and $\mathbb{E}[w\omega] = \varepsilon$, so both inequalities are equalities at $\tau = \tau^*$. If the budget is never exhausted, they become equalities as $\tau \rightarrow \infty$. On a countable space the fractional purchase can be confined to a single atom, and dropping that atom gives a feasible set. Finally, $(1 - \omega/\tau)_+ \leq \mathbf{1}\{\omega < \tau\}$. $\square$

Theorem 7.7 (Greedy decoding: sharp bounds from errors on the greedy path). *[proved] [numerically checked] Let D be a divergence on $\Delta(\mathcal{V})$. Put $e_t^*(x) := D(p_t \| q_t)(s_t^*)$, $\kappa_t^*(x) := \kappa_D(p_t(\cdot | s_t^*))$ and $\bar{\kappa}(x) := \min_t \kappa_t^*(x)$.*

(i) (Pathwise.) $\mathbf{1}\{g_{q_\theta}(x) \neq g_p(x)\} \leq \max_t \mathbf{1}\{e_t^* \geq \kappa_t^*\} \leq \min\{1, \max_t e_t^*/\kappa_t^*\}$, with $0/0 = \infty/\infty := 1$.

(ii) (Probability.) Let $\varepsilon^* := \mathbb{E}_\mu \sum_t e_t^*$ be the summed per-token error on greedy paths. Then

$$\mathcal{L}_{\text{dec}} \leq \Phi_{\bar{\kappa}}(\mathbb{E}_\mu \max_t e_t^*) \leq \bar{\Phi}_{\bar{\kappa}}(\varepsilon^*) \leq \inf_{\tau > 0} \left\{ \mu[\bar{\kappa} < \tau] + \frac{\varepsilon^*}{\tau} \right\},$$

$$\mathcal{L}_{\text{dec}} \leq \sum_{t=1}^T \bar{\Phi}_{\kappa_t^*}(\mathbb{E}_\mu e_t^*).$$

(iii) (Sharpness.) If D is an f -divergence then, for every teacher, input law and $\varepsilon > 0$,

$$\bar{\Phi}_{\bar{\kappa}}(\varepsilon) - \max_x \mu(\{x\}) \leq \sup_{q_\theta} \left\{ \mathcal{L}_{\text{dec}}(\theta; \mu) : \mathbb{E}_\mu \sum_t e_t^* \leq \varepsilon \right\} \leq \Phi_{\bar{\kappa}}(\varepsilon),$$

with equality on the right wherever $\Phi_{\bar{\kappa}}$ is left-continuous. The same holds with $\max_t$ in place of $\sum_t$. The extremal students differ from the teacher at one greedy state per input.

(iv) (Uniform errors.) If $e_t^*(x) \leq \eta$ for all t and x , then $\mathcal{L}_{\text{dec}} \leq \mu[\bar{\kappa} \leq \eta]$. For f -divergences, some such student reaches $\mu[\bar{\kappa} < \eta]$.

Proof. (i) By Lemma 7.1, a disagreement needs a flip at some s_t^* , and a flip needs $e_t^* \geq \kappa_t^*$ by the definition of κ_D . (ii) Let A be the disagreement set. By (i), $\bar{\kappa} \leq \max_t e_t^*$ on A , so $\int_A \bar{\kappa} \, d\mu \leq \mathbb{E}_\mu \max_t e_t^*$. Then use $\max \leq \sum$ and Lemma 7.6. For the second line, apply the same argument to each flip event and take a union bound. (iii) The right inequality is (ii). For the left, take A with $\int_A \bar{\kappa} \, d\mu < \varepsilon$. For $x \in A$, let $t(x)$ minimise $\kappa_t^*(x)$, and let the student equal the teacher except at the states $s_{t(x)}^*(x)$, $x \in A$. There it uses a strict flip of cost at most $\bar{\kappa}(x) + \delta$ (Proposition 7.4(i)). States of different inputs are different, so this defines a student. It disagrees with the teacher exactly on A , and $\mathbb{E} \sum_t e_t^* = \mathbb{E} \max_t e_t^* \leq \int_A \bar{\kappa} \, d\mu + \delta < \varepsilon$ for small δ . So the supremum is at least $\sup \{ \mu(A) : \int_A \bar{\kappa} < \varepsilon \} = \lim_{\varepsilon' \uparrow \varepsilon} \Phi_{\bar{\kappa}}(\varepsilon')$. This is at least $\bar{\Phi}_{\bar{\kappa}}(\varepsilon) - \max_x \mu(\{x\})$ by Lemma 7.6, because $\bar{\Phi}_{\bar{\kappa}}$ is concave and hence continuous on $(0, \infty)$. (iv) By (i), a disagreement needs $\bar{\kappa} \leq e_t^* \leq \eta$. For the lower bound, flip every input with $\bar{\kappa} < \eta$ as in (iii). $\square$

Remark 7.8 (Reading Theorem 7.7). Greedy decoding is plug-in classification of the teacher’s argmax at each greedy state (Audibert and Tsybakov, 2007). The knapsack function plays the role that the calibration function of a surrogate loss plays in classification (Bartlett et al., 2006). Here it is exact, because the teacher is fixed. The worst student spends its error budget on the inputs whose greedy paths contain the cheapest flips, in increasing order of $\bar{\kappa}$. There are two regimes. If $\bar{\kappa} \geq \kappa_0 > 0$ on the support of μ , then $\mathcal{L}_{\text{dec}} \leq \varepsilon^*/\kappa_0$, and $\mathcal{L}_{\text{dec}} = 0$ as soon as every $e_t^* < \kappa_0$. If instead $\bar{\kappa}$ has mass near 0, a budget ε buys a disagreement probability much larger than ε (Corollary 7.11). The per-step form in (ii) uses the per-position error curves of Section 6.3, evaluated on greedy prefixes.

7.3 Greedy decoding: sequence-level and training errors

Training does not control errors on greedy paths. Off-policy distillation controls per-token errors on teacher-sampled prefixes, and on-policy distillation controls them on student-sampled prefixes. Both are tied to sequence-level divergences by Theorem 1.12 and Proposition 1.14. A teacher sample passes through s_t^* only with probability π_t^* , and this turns out to be the exact exchange rate.

Theorem 7.9 (Path-weighted thresholds). *[proved] [numerically checked] For every f -divergence and every input x ,*

$$\inf_{q_\theta} \left\{ D_f(P^x \| Q_\theta^x) : g_{q_\theta}(x) \neq g_p(x) \right\} = \omega_f(x) := \min_{1 \leq t \leq T} k_f(\pi_t^* p_{t,1}^*, \pi_t^* p_{t,2}^*).$$

In particular, writing ω_{rKL} for the threshold of $\text{KL}(Q_\theta^x \| P^x)$,

$$\begin{aligned} \omega_{\text{TV}} &= \frac{1}{2} \min_t \pi_t^* \gamma_t, & \omega_{\text{KL}} &= \min_t \pi_t^* \kappa_{\text{KL}}(p_t(\cdot | s_t^*)), \\ \omega_{\text{rKL}} &= \min_t -\log \left(1 - \pi_t^* (\sqrt{p_{t,1}^*} - \sqrt{p_{t,2}^*})^2 \right). \end{aligned}$$

Consequently $\mathcal{L}_{\text{dec}}(\theta; \mu) \leq \Phi_{\omega_f}(\mathcal{L}_{D_f}(\theta; \mu)) \leq \bar{\Phi}_{\omega_f}(\mathcal{L}_{D_f}(\theta; \mu))$. The supremum of $\mathcal{L}_{\text{dec}}$ over students with $\mathcal{L}_{D_f} \leq \varepsilon$ obeys the two-sided bound of Theorem 7.7(iii), with ω_f in place of $\bar{\kappa}$. Under $\mu = \mathcal{D}$, Theorem 1.12 gives $\mathcal{L}_{\text{KL}} = T\varepsilon_{\text{off}}^{\text{KL}}$ and $\mathcal{L}_{\text{rKL}} = T\varepsilon_{\text{on}}^{\text{rKL}}$, and Proposition 1.14 gives $\mathcal{L}_{\text{TV}} \leq T \min\{\varepsilon_{\text{off}}^{\text{TV}}, \varepsilon_{\text{on}}^{\text{TV}}\}$. For TV and KL, the extremal students change one next-token distribution per input, and for them all these relations are equalities. So the bounds in terms of per-token training errors are sharp too.

Proof. Fix x and drop it from the notation. *Lower bound.* Let t be the first flip (Lemma 7.1). Up to step t the student follows the teacher’s greedy path, so $Q(s_t^*) = \prod_{j < t} q_j(a_j^* | s_j^*) > 0$. The flip then gives $Q(s_t^*, a_t^*) \leq Q(s_t^*, b)$ for some $b \neq a_t^*$. Apply Lemma 7.3 to the law of $y_{\leq t}$ with $S_+ = \{(s_t^*, a_t^*)\}$ and $S_- = \{(s_t^*, b) : b \neq a_t^*\}$, whose teacher masses are $\pi_t^* p_{t,1}^*$ and at most $\pi_t^* p_{t,2}^*$. Finally, $D_f(P \| Q) \geq D_f(P_{\leq t} \| Q_{\leq t})$ by data processing. *Attainment.* We may assume $\omega_f(x) < \infty$. Choose a minimising t with $p_{t,2}^* > 0$ if $f(0) = \infty$. This is possible. If $p_{t,2}^* = 0$, let $j < t$ be the last step with $p_{j,2}^* > 0$. The steps strictly between j and t are deterministic, so $\pi_j^* p_{j,1}^* = \pi_t^*$. Apply Lemma 7.3 to a law with atoms of masses π_t^* , $P_2 := \pi_j^* p_{j,2}^*$ and 0, first with S_- the null atom and then with S_- both other atoms. The second constraint set is larger, so $k_f(\pi_t^*, P_2) \leq k_f(\pi_t^*, 0)$, and j is also a minimiser. Such a j exists, since otherwise $\pi_t^* = 1$ and $k_f(1, 0) = \infty$. At this t , take the law of Lemma 7.3 for $y_{\leq t}$ and extend it to sequences with the teacher’s conditional law given $y_{\leq t}$. Its likelihood ratio is constant on the three cells, so $D_f(P \| Q) = \omega_f(x)$. With the strict version of Lemma 7.3, its greedy path either leaves the teacher’s before step t or flips at s_t^* . Either way the outputs differ, at cost at most $\omega_f(x) + \delta$. States of different inputs are disjoint, so the per-input constructions combine. The probability statements then follow as in the proof of Theorem 7.7(ii)–(iii). *Special cases.* For TV and KL, the minimiser in k_f is $c = \frac{P_1 + P_2}{2}$ and leaves the rest unchanged. So $k_f(\pi P_1, \pi P_2) = \pi k_f(P_1, P_2)$,

and the extremal law changes only $q_t(\cdot | s_t^*)$. Both prefix laws then put mass π_t^* on s_t^* and agree elsewhere, so the hybrid sums of Proposition 1.14 equal $\text{TV}(P, Q)$. For reverse KL, use the closed form of Proposition 7.4(ii) with (p_1, p_2) replaced by $\pi_t^*(p_{t,1}^*, p_{t,2}^*)$. $\square$

Corollary 7.10 (Behavioural and decoding losslessness are independent). *[proved] [numerically checked]*

- (i) (Confident teachers.) If $\pi_T^*(x) \geq \pi_0$ for μ -almost every x , then $\omega_{\text{TV}} \geq \pi_0 \bar{\kappa}_{\text{TV}}$ and $\omega_{\text{KL}} \geq \pi_0 \bar{\kappa}_{\text{KL}}$. Sequence-level losslessness then controls decoding as in Theorem 7.7, at the price of dividing the budget by π_0 .
- (ii) (Uncertain teachers.) Let the teacher emit i.i.d. tokens from $\{0, 1\}$ with $p(0) = \frac{1+\gamma}{2}$. Change only the student's next-token distribution at the prefix 0^{T-1} , to $(\frac{1}{2} - \delta, \frac{1}{2} + \delta)$. Then the greedy output of every input changes, while $\text{TV}(P^x, Q_\theta^x) = (\frac{\gamma}{2} + \delta)(\frac{1+\gamma}{2})^{T-1}$ and $\text{KL}(P^x \| Q_\theta^x) \rightarrow (\frac{1+\gamma}{2})^{T-1} \kappa_{\text{KL}}$ as $\delta \rightarrow 0$. For $\gamma = 0.2$ and $T = 100$ both are below $2 \cdot 10^{-23}$, yet $\mathcal{L}_{\text{dec}} = 1$.
- (iii) (Converse.) The student that emits $\text{gr}(p_t(\cdot | s))$ with probability one at every state has $\mathcal{L}_{\text{dec}} = 0$. Yet $\text{TV}(P^x, Q_\theta^x) = 1 - P^x(g_p(x))$, and $\text{KL}(P^x \| Q_\theta^x) = \infty$ unless P^x is a point mass.

The path weight π_t^* is thus the exact exchange rate between the two notions. It is close to 1 only when the teacher is confident along its own greedy path.

Proof. (i) $\pi_t^* \geq \pi_T^*$ for $t \leq T$, and $k_f(\pi P_1, \pi P_2) = \pi k_f(P_1, P_2)$ for TV and KL. (ii) The two laws differ only in the next-token law at $s = 0^{T-1}$, which the teacher reaches with probability $(\frac{1+\gamma}{2})^{T-1}$. When two autoregressive laws differ at a single state s , $\text{TV}(P^x, Q_\theta^x) = P^x(s) \text{TV}(p(\cdot | s), q(\cdot | s))$. The KL value follows from Theorem 1.12. (iii) Q_θ^x is the point mass at $g_p(x)$. $\square$

Corollary 7.11 (Rates under a margin condition). *[proved] [numerically checked]* Let $\bar{\gamma}(x) := \min_t \pi_t^*(x) \gamma_t(x)$ be the path-weighted greedy margin, and suppose that $\mu[\bar{\gamma} \leq u] \leq Cu^\alpha$ for all $u > 0$. This is a margin condition of the type introduced by Mammen and Tsybakov (1999) and Tsybakov (2004). Then for every student

$$\mathcal{L}_{\text{dec}} \leq \left(\frac{2(1+\alpha)}{\alpha} \mathcal{L}_{\text{TV}} \right)^{\frac{\alpha}{1+\alpha}} C^{\frac{1}{1+\alpha}} \quad \text{and} \quad \mathcal{L}_{\text{dec}} \leq \left(\frac{2(2+\alpha)}{\alpha} \mathcal{L}_{\text{KL}} \right)^{\frac{\alpha}{2+\alpha}} C^{\frac{2}{2+\alpha}}.$$

Both exponents are attained. Take $T = 1$, $V = 2$, and μ uniform on n inputs with margins $(i/n)^{1/\alpha}$, $i \leq n$, so that $C = 1$. As $n \rightarrow \infty$, the worst-case $\mathcal{L}_{\text{dec}}$ at $\mathcal{L}_{\text{TV}} = \varepsilon \leq \frac{\alpha}{2(1+\alpha)}$ converges to the first bound; for larger ε the bound exceeds 1. At $\mathcal{L}_{\text{KL}} = \varepsilon$, its ratio to the second bound tends to 1 as $n \rightarrow \infty$ and then $\varepsilon \rightarrow 0$. Through Pinsker's inequality, the first bound would give only the exponent $\frac{\alpha}{2(1+\alpha)}$ in $\mathcal{L}_{\text{KL}}$. If instead the per-token TV on greedy paths is at most η everywhere and $\mu[\min_t \gamma_t \leq u] \leq C_* u^\alpha$, then $\mathcal{L}_{\text{dec}} \leq C_*(2\eta)^\alpha$ by Theorem 7.7(iv).

Proof. By Theorem 7.9 and Lemma 7.6, $\mathcal{L}_{\text{dec}} \leq \bar{\Phi}_\omega(\varepsilon)$ with $\varepsilon = \mathcal{L}_D$. Moreover $\mathbb{E}(1 - \omega/\tau)_+ = \frac{1}{\tau} \int_0^\tau \mu[\omega < u] du$. If $\mu[\omega < u] \leq C' u^\alpha$ for all u , then

$$\bar{\Phi}_\omega(\varepsilon) \leq \inf_{\tau > 0} \left\{ \frac{\varepsilon}{\tau} + \frac{C' \tau^\alpha}{1 + a} \right\} = \left(\frac{(1+a)\varepsilon}{a} \right)^{\frac{a}{1+a}} C'^{\frac{1}{1+a}},$$

with the infimum at $\tau^{1+a} = \frac{(1+a)\varepsilon}{aC'}$. For TV, $\omega_{\text{TV}} = \bar{\gamma}/2$, so $(a, C') = (\alpha, 2^\alpha C)$. For KL, $\pi_t^* \kappa_{\text{KL}} \geq \pi_t^* \gamma_t^2 / (2m_t) \geq (\pi_t^* \gamma_t)^2 / 2$ by Proposition 7.4(ii) and $\pi_t^* m_t \leq 1$. So $\omega_{\text{KL}} \geq \bar{\gamma}^2 / 2$ and $(a, C') = (\alpha/2, 2^{\alpha/2} C)$. In the example, the function $\tau \mapsto \frac{\varepsilon}{\tau} + \frac{1}{\tau} \int_0^\tau \mu[\omega < u] du$ has derivative $\tau^{-2} [\int_0^\tau (\mu[\omega < \tau] - \mu[\omega < u]) du - \varepsilon]$, whose bracket is non-decreasing. So the function is quasi-convex, and its stationary point is its minimum. In the TV case the limit law satisfies $\mu[\omega < u] = (2u)^\alpha$ for $u \leq \frac{1}{2}$. The stationary point satisfies $\tau \leq \frac{1}{2}$ exactly when $\varepsilon \leq \frac{\alpha}{2(1+\alpha)}$, and then the display is an equality for the limit law. The discrete law converges to it, and Lemma 7.6 loses at most $1/n$. For $T = 1$ and $V = 2$, $\kappa_{\text{KL}} = \text{kl}(\frac{1+\gamma}{2} \| \frac{1}{2}) = \frac{\gamma^2}{2} (1 + O(\gamma^2))$. $\square$

Remark 7.12 (What the adversary exploits, and how training can avoid it). (a) The worst case puts the whole error budget on small-margin states. Replace the L^1 budget by $\mathbb{E}_\mu \max_t (e_t^*)^r \leq \varepsilon^r$, and assume $\mu[\min_t \gamma_t \leq u] \leq Cu^\alpha$. Then Theorem 7.7, applied to D^r , gives the exponent $\frac{r\alpha}{r+\alpha}$ for TV errors. This interpolates between $\frac{\alpha}{1+\alpha}$ ($r = 1$) and α ($r = \infty$). If the errors are independent of the margins, then $\mu[\text{flip at } t] \leq \mathbb{E}F_t(2e_t^*)$, where $F_t(u) = \mu[\gamma_t \leq u]$. For $\alpha \leq 1$, Jensen’s inequality bounds this by $C(2\mathbb{E}e_t^*)^\alpha$, which is linear in the error when $\alpha = 1$. (b) The factor π_t^* in Theorem 7.9 is the price of training on sampled prefixes. Suppose that $\mu = \mathcal{D}$ and that, given each x , the training prefix law puts mass at least λ on $s_t^*(x)$. Examples are $\nu_t = (1 - \lambda)d_t^p + \lambda \delta_{s_t^*}$ and soft-label distillation on the teacher’s greedy outputs. Then the training loss is at least $\lambda \varepsilon^*$, so Theorem 7.7 applies with $\varepsilon^* \leq F/\lambda$, and no path weight appears.

7.4 Hard labels on decoded outputs

Proposition 7.13 (A margin-free bound for sequence-level distillation). *[proved] [numerically checked] Fix a beam width $k \geq 1$, with ties broken as in Section 7.5, so that $k = 1$ is greedy decoding. Let $\hat{y}(x)$ and $\hat{y}_\theta(x)$ be the teacher’s and the student’s outputs. If $Q_\theta^x(\hat{y}(x)) > \frac{1}{2}$, then $\hat{y}_\theta(x) = \hat{y}(x)$. Hence*

$$\mathbb{P}_\mu[\hat{y}_\theta \neq \hat{y}] \leq \mathbb{E}_\mu \min \left\{ 1, \log_2 \frac{1}{Q_\theta^x(\hat{y}(x))} \right\} \leq \frac{1}{\ln 2} \mathbb{E}_\mu [-\log Q_\theta^x(\hat{y}(x))].$$

For greedy decoding, a disagreement needs $q_t(a_t^ | s_t^*) \leq \frac{1}{2}$ at the first flip. The constant $\frac{1}{2}$ cannot be improved.*

Proof. If $Q(\hat{y}) > \frac{1}{2}$, every prefix $\hat{y}_{\leq t}$ has Q -mass above $\frac{1}{2}$, so it has more mass than any other prefix of the same length. By induction on t , $\hat{y}_{\leq t}$ is a candidate at level t and is the student’s best candidate, so it stays in the student’s beam. At the end it is the student’s most probable hypothesis. Next, $1\{Q \leq \frac{1}{2}\} \leq \min\{1, \log_2(1/Q)\}$. At a greedy flip, $q(a_t^*) \leq q(b) \leq 1 - q(a_t^*)$. Finally, $q_t(\cdot | s_t^*) = (\frac{1}{2} - \delta, \frac{1}{2} + \delta)$ on two tokens is a flip for every $\delta > 0$. $\square$

The right-hand side is the objective of sequence-level knowledge distillation (Kim and Rush, 2016): maximum likelihood on the teacher’s beam-search outputs. Proposition 7.13 shows that this objective is a calibrated surrogate for decoding-level losslessness, with no margin in the constant. It asks the student to be *confident* on the teacher’s decoded outputs, not to be faithful to the teacher’s distribution. These are different targets (Corollary 7.10). A soft-label student that matches a teacher with small margins has $Q(\hat{y}) \approx P(\hat{y})$, which may be far below $\frac{1}{2}$. For such teachers the margin-dependent bounds of Theorems 7.7 and 7.9 are the relevant ones.

7.5 Beam search

Beam search of width k keeps a set $\mathcal{B}_t$ of at most k prefixes of length t , starting from $\mathcal{B}_0 = \{x\}$. The candidates at level t are $\mathcal{C}_t = \{(u, a) : u \in \mathcal{B}_{t-1}, a \in \mathcal{V}\}$. $\mathcal{B}_t$ consists of the k candidates of largest prefix probability under the decoding model. Ties are broken by a fixed order on prefixes that agrees with the order on $\mathcal{V}$ for candidates with a common parent, so that $k = 1$ is greedy decoding. The output is the most probable element of $\mathcal{B}_T$. With the absorbing end-of-sequence convention, every hypothesis has length T , so this is beam search without length normalisation. The *trajectory* is $(\mathcal{B}_1, \dots, \mathcal{B}_T, \text{output})$. For the teacher’s beam search, let $P_t^+ \geq P_t^-$ be the k -th and $(k+1)$ -th largest values of P^x on $\mathcal{C}_t$, where $P^x(u)$ is the teacher’s probability of the prefix u . Let $P_{T+1}^+ \geq P_{T+1}^-$ be the two largest values of P^x on $\mathcal{B}_T$. A level with no $(k+1)$ -th candidate cannot change and is dropped, as is level $T+1$ when $k = 1$. We write $P_{\leq t}^x$ for the law of $y_{\leq t}$, and set $P_{\leq T+1}^x := P^x$.

Theorem 7.14 (Beam search). *[proved] [numerically checked] Fix x and $k \geq 1$.*

- (i) If the student's trajectory first differs from the teacher's at level $t \in \{1, \dots, T+1\}$, then $D_f(P_{\leq t}^x \| Q_{\theta, \leq t}^x) \geq k_f(P_t^+, P_t^-)$ for every f -divergence. In particular $\text{TV}(P_{\leq t}^x, Q_{\theta, \leq t}^x) \geq \frac{1}{2}(P_t^+ - P_t^-)$, and $\sum_{j \leq \min(t, T)} \mathbb{E}_{s \sim d_j^p | x} \text{KL}(p_j \| q_j)(s) = \text{KL}(P_{\leq t}^x \| Q_{\theta, \leq t}^x) \geq k_{\text{KL}}(P_t^+, P_t^-)$.
- (ii) (Beam-local form.) For $D \in \{\text{TV}, \text{KL}\}$, the same lower bounds hold for

$$\sum_{j \leq \min(t, T)} \sum_{u \in \mathcal{B}_{j-1}} P^x(u) D(p_j \| q_j)(u).$$

Only per-token errors at the teacher's beam states enter, weighted by the teacher's probabilities of those states.

- (iii) (Exactness.) Let $\omega_f^{(k)}(x) := \min_t k_f(P_t^+, P_t^-)$. The outputs can differ only if $D_f(P^x \| Q_\theta^x) \geq \omega_f^{(k)}(x)$, so $\mathbb{P}_\mu[\text{beam outputs differ}] \leq \bar{\Phi}_{\omega_f^{(k)}}(\mathcal{L}_{D_f})$. If $f(0) < \infty$, or if every candidate has positive teacher probability, then $\omega_f^{(k)}(x)$ is the infimum of $D_f(P^x \| Q_\theta^x)$ over students whose trajectory differs from the teacher's, and for trajectories the bound is sharp in the sense of Theorem 7.7(iii). All the f -divergences of Proposition 7.4 except reverse KL have $f(0) < \infty$. For reverse KL the condition is needed: with $T = 1$, $p = (0.9, 0.1, 0, 0)$ and $k = 3$, the cut lies between two null candidates, so $\omega_{\text{rKL}}^{(3)} = 0$, yet a finite $\text{KL}(Q_\theta^x \| P^x)$ forces $Q(c) = Q(d) = 0$, and changing the beam then costs at least $\log \frac{1}{0.9}$. For $k = 1$, $(P_t^+, P_t^-) = \pi_t^*(p_{t,1}^*, p_{t,2}^*)$, and (iii) is Theorem 7.9, which needs no condition.
- (iv) (The cut gap cannot be replaced by the output's own gaps.) Let $k = 2$, $T = 2$ and $\mathcal{V} = \{a, b, c\}$. Take first-token probabilities $(0.36, 0.33, 0.31)$ and prefix probabilities $aa : 0.30$, $ab : 0.059$, $ac : 0.001$, $ba : 0.20$, $bb : 0.10$, $bc : 0.03$, $ca : 0.309$ and $cb = cc : 0.0005$. The teacher keeps $\{a, b\}$, then $\{aa, ba\}$, and outputs aa . The cut gaps are 0.02, 0.10 and 0.10, so $\omega_{\text{TV}}^{(2)} = 0.01$. The gaps between the ancestors of aa and their best competitors are $P(a) - P(c) = 0.05$, $P(aa) - P(bb) = 0.20$ and $P(aa) - P(ba) = 0.10$, which would give the threshold 0.025. Yet moving mass $0.01 + \delta$ from the subtree of b to that of c makes the student keep $\{a, c\}$, then $\{ca, aa\}$, and output ca , at $\text{TV} = 0.01 + \delta$. The student "repairs" a search error of the teacher, since $P(ca) = 0.309 > P(aa)$.

Proof. (i) Let $t \leq T$. The beams before level t coincide, so the candidate sets $\mathcal{C}_t$ coincide. If every member of the teacher's beam had larger Q -mass than every non-member, the student would keep the same beam, whatever its tie rule. So $Q(a) \leq Q(b)$ for some member a and non-member b . Apply Lemma 7.3 to the law of $y_{\leq t}$ with $S_+ = \mathcal{B}_t$ and $S_- = \mathcal{C}_t \setminus \mathcal{B}_t$. For $t = T+1$, apply it with S_+ the teacher's output and S_- the rest of $\mathcal{B}_T$. The KL identity is Theorem 1.12 applied to the first t tokens. (ii) Map $y_{\leq t}$ to its path through the teacher's beam tree: to $y_{\leq t}$ itself if $y_{\leq t} \in \mathcal{B}_{t-1}$, and otherwise to $(j, y_{< j})$, where j is the first level with $y_{\leq j} \notin \mathcal{B}_j$. Candidates remain atoms, so (i) holds for the image laws. These are the laws of a sequential process on the beam tree. Its step from $u \in \mathcal{B}_{j-1}$ is $p_j(\cdot | u)$, with the non-beam children merged when $j < t$. The chain rule and the hybrid argument of Theorem 1.12 and Proposition 1.14 hold for such processes, with visiting probabilities $P^x(u)$, and merging children can only decrease KL and TV. (iii) The lower bound follows from (i), because $D_f(P \| Q) \geq D_f(P_{\leq t} \| Q_{\leq t})$ and the outputs can differ only if the trajectories do. For attainment, take the law of Lemma 7.3 at a minimising level and extend it as in the proof of Theorem 7.9. The condition makes the strict clause of Lemma 7.3 available. If the trajectory has not changed before that level, the candidate sets agree there, and the strict version reverses the k -th and $(k+1)$ -th candidates. (iv) Direct computation (`check_decoding.py`). $\square$

7.6 Speculative decoding

Proposition 7.15 (Greedy versus sampled verification). *[proved] [numerically checked] Let the student draft tokens that the teacher verifies.*

- (i) (Speculative sampling.) A token drafted at state s is accepted with probability $\sum_a \min\{p_t(a | s), q_t(a | s)\} = 1 - \text{TV}_t(s)$, and the output has exactly the teacher’s law (Leviathan et al., 2023).
- (ii) (Greedy verification.) Suppose the student drafts greedily and the teacher accepts the longest drafted prefix that agrees with its own greedy continuation. This is the verification rule of blockwise parallel decoding (Stern et al., 2018), and the zero-temperature case of (i). Every verified prefix lies on the teacher’s greedy path. A round started at s_t^* with L drafted tokens accepts $N = \min\{j \geq 0 : \text{the student flips at } s_{t+j}^*\} \wedge L$ of them. So $\mathbb{E}_\mu N = \sum_{j=1}^L (1 - \mathbb{P}_\mu[\text{a flip at one of } s_t^*, \dots, s_{t+j-1}^*])$, and Theorems 7.7 and 7.9, applied to these windows, bound the expected acceptance length from below.
- (iii) (Pointwise comparison.) At a state s with margin $\gamma(s) > 0$, the greedy rejection indicator is at most $\min\{1, 2\text{TV}_t(s)/\gamma(s)\}$. This is $2/\gamma(s)$ times the rejection probability of speculative sampling, and the constant 2 cannot be improved.

Proof. (i) The acceptance probability is $\sum_a q(a) \min\{1, p(a)/q(a)\}$. For exactness of the output law, see Leviathan et al. (2023). (ii) The teacher emits its greedy token at each position, whether it accepts the draft or corrects it. So each round starts at a teacher greedy state, and by Lemma 7.1 the accepted run ends at the first flip. (iii) This is the TV row of Proposition 7.4(ii). $\square$

Sampled verification rejects in proportion to the per-token TV, whatever the margins. Greedy verification is a threshold rule. It never rejects while $\text{TV}_t < \gamma_t/2$ along the greedy path. But when errors concentrate on small-margin states, its worst-case rejection rate is of order $\varepsilon^{\alpha/(1+\alpha)}$ (Corollary 7.11). This is a square root when the margins have a bounded density at 0, and it exceeds the linear rate of speculative sampling for small ε .

Remark 7.16 (Measuring the bounds). All the quantities are cheap to compute. One greedy (or beam) decode of the teacher per input gives the states s_t^* , the margins γ_t , the top-two masses m_t and the weights π_t^* , or the cut probabilities $P_t^\pm$. One student forward pass on the same prefixes gives the e_t^* . The observed $\mathcal{L}_{\text{dec}}$ can then be compared with $\bar{\Phi}_{\bar{\kappa}}(\varepsilon^*)$, with its per-step version, and with the path-weighted bound of Theorem 7.9 at the measured $T\varepsilon_{\text{off}}^{\text{KL}}$. The ratio of the observed $\mathcal{L}_{\text{dec}}$ to the worst case shows how strongly the student’s errors concentrate on small-margin states.

8 Open mathematical problems

These are ordered by how much their solution would advance near-lossless distillation.

P1. The functional information of trained teachers. Characterise, or give computable bounds on, $I_\infty(\pi, \mathcal{D})$ from Theorem 3.1 for realistic priors, for example the distribution of networks produced by a training procedure. Is there a data-dependent estimator of the *functional* description length of a single trained teacher, such as the minimal R at which some R -bit student achieves ε -losslessness, with matching upper and lower bounds? This is the quantity that decides whether lossless distillation is possible at all.

P2. Horizon-free agnostic guarantees with tractable estimators. PA-JSD is horizon-free at the population level (Theorem 6.1(iii)), but its gradient estimator has sequence-level score-function variance. Is there an objective that is simultaneously (a) an exact sequence-level divergence, (b) agnostically TV-competitive with a constant (not square-root) factor, and (c) has a gradient estimator whose variance is bounded uniformly in T ? For deterministic experts, Foster et al. (2024) suggest that log-loss behaviour cloning is already horizon-free; the stochastic, misspecified case is open.

- P3. The optimal agnostic exponent.** Proposition 4.13 gives $\text{TV}(\hat{\theta}) = O(\sqrt{\text{TV}^*})$. Is the square root necessary for *any* smooth, tractable surrogate? Equivalently, what is the best ϕ with $\text{TV}(\arg \min \mathcal{D}) \leq \phi(\text{TV}^*)$ over divergences $\mathcal{D}$ that admit unbiased per-token gradients? A lower bound of the form $\phi(t) \gtrsim \sqrt{t}$ for all f -divergences with $f''(1) > 0$ would be a clean impossibility result.
- P4. A first-principles distillation scaling law.** Prove Conjecture 3.14, or a corrected form, in a model rich enough to exhibit both the softmax bottleneck and depth. Examples are deep linear networks with a softmax read-out, or random-feature models with a power-law spectrum. The model should include the soft-versus-hard data exponents (Heuristic 3.12) and the finite-sample capacity gap (Heuristic 3.13). Determine whether the transition to exponential decay (Proposition 3.9) survives realistic spectra.
- P5. Recoverability of language models.** Give conditions on the teacher under which the recoverability constant u of Theorem 2.16 is $O(1)$, or bounded in expectation rather than in the worst case. Relate u to measurable properties of the teacher, such as its value function under task rewards or the mixing of its induced Markov chain on prefixes. A distribution-dependent u would turn the $O(uT\varepsilon_{\text{on}})$ bound into a usable prediction.
- P6. Fixed points of stop-gradient on-policy distillation.** Characterise the fixed points of the DAGger-type map in Proposition 4.9. When do they coincide with minimisers of a sequence-level divergence? Is there a Lyapunov function? Under what conditions does the iteration converge for overparameterised students?
- P7. Representation matching in deep networks.** Extend Theorem 5.7 and Proposition 5.9 beyond the linear-Gaussian model. Is there a Fisher-aligned target, depending on s through the Jacobian, whose constraint provably has zero approximation cost for a nonlinear teacher–student pair? Can one quantify sample-complexity and computational gains of intermediate supervision for transformer students, beyond parity-type separations?
- P8. Optimal input design for distillation.** The teacher can be queried anywhere. Which input distribution $\mathcal{D}$, possibly adaptive, minimises the deployment risk under μ for a fixed query budget, given the covariate-shift terms of Proposition 1.18? This is an active-learning or optimal-design problem with a known labelling function. Minimax rates for it are unknown in the sequence setting.
- P9. Decoding-level losslessness beyond the worst case.** Section 7 settles the worst case. It gives exact flip thresholds (Proposition 7.4), the knapsack characterisation of the largest disagreement probability (Theorem 7.7), the path-weighted exchange rate with sequence-level and training errors (Theorem 7.9), and cut-gap thresholds for beam trajectories (Theorem 7.14). Three questions remain. (a) Trained students are not adversarial. Characterise the joint law of per-token errors and margins that distillation produces. This would place real students between the worst-case exponent $\frac{\alpha}{1+\alpha}$ and the margin-blind exponent α (Remark 7.12). (b) For beam width $k \geq 2$, find the exact output-level threshold. For TV and KL it lies between $\omega_f^{(k)}$ and the final-level merge cost $k_f(P_{T+1}^+, P_{T+1}^-)$, because moving mass between the two best final hypotheses always changes the output; Theorem 7.14(iv) shows that the lower end can be the right one. (c) Truncated sampling (top- k , nucleus, min- p) interpolates between the $1 - \text{TV}$ acceptance of speculative sampling (Leviathan et al., 2023) and the greedy threshold. Derive the corresponding sharp bounds.

A Deferred proofs

A.1 Proof of Theorem 2.7 (horizon-free MLE bound)

Write $D_H^2(P, Q) = \sum_y (\sqrt{P(y)} - \sqrt{Q(y)})^2 = 2H^2(P, Q)$. For any P, Q on $\mathcal{Y}$,

$$\mathbb{E}_{y \sim P} \sqrt{Q(y)/P(y)} = \sum_y \sqrt{P(y)Q(y)} = 1 - \frac{1}{2}D_H^2(P, Q) \leq \exp(-\frac{1}{2}D_H^2(P, Q)). \quad (6)$$

Fix $\theta \in \Theta_S$ and let $Z_i(\theta) = \frac{1}{2} \log(Q_\theta^{x_i}(y^{(i)})/P^{x_i}(y^{(i)}))$. The pairs $(x_i, y^{(i)})$ are i.i.d., so by (6) and Jensen,

$$\mathbb{E} \exp\left(\sum_i Z_i(\theta)\right) = \left(\mathbb{E}_{x \sim \mathcal{D}} [1 - \frac{1}{2}D_H^2(P^x, Q_\theta^x)]\right)^n \leq \exp\left(-\frac{n}{2}\mathbb{E}_x D_H^2(P^x, Q_\theta^x)\right).$$

Markov's inequality gives, for each θ , with probability $\geq 1 - \delta/|\Theta_S|$,

$$\sum_i Z_i(\theta) \leq -\frac{n}{2}\mathbb{E}_x D_H^2(P^x, Q_\theta^x) + \log(|\Theta_S|/\delta).$$

A union bound makes this hold for all θ simultaneously. By realisability and the definition of the MLE, $\sum_i Z_i(\hat{\theta}) \geq \sum_i Z_i(\theta^0) = 0$. Hence $\mathbb{E}_x D_H^2(P^x, Q_{\hat{\theta}}^x) \leq 2 \log(|\Theta_S|/\delta)/n$, that is, $\mathbb{E}_x H^2 \leq \log(|\Theta_S|/\delta)/n$. Finally, $\text{TV} \leq H\sqrt{2 - H^2} \leq \sqrt{2}H$ (Proposition 1.5), and Jensen gives $\mathbb{E}_x \text{TV} \leq \sqrt{2}\mathbb{E}_x H^2$. The horizon T never enters: y is treated as a single symbol from $\mathcal{Y}$, and only $\log|\Theta_S|$ appears (Zhang, 2006; Foster et al., 2024). $\square$

A.2 Proof of the performance-difference identity (Theorem 2.16)

Let $V_t^p(s) = \mathbb{E}_{a \sim p_t(\cdot|s)} Q_t^p(s, a)$, with $V_{T+1}^p \equiv 0$. Then $Q_t^p(s, a) = c_t(s, a) + \mathbb{E}[V_{t+1}^p(s_{t+1}) | s_t = s, y_t = a]$, and the transition $s_{t+1} = (s, a)$ is the same under both policies. Along a student trajectory,

$$\sum_{t=1}^T c_t(s_t, y_t) - V_1^p(s_1) = \sum_{t=1}^T [c_t(s_t, y_t) + V_{t+1}^p(s_{t+1}) - V_t^p(s_t)].$$

Take expectations under q_θ . The t -th summand has conditional expectation $\mathbb{E}_{a \sim q_t(\cdot|s_t)} Q_t^p(s_t, a) - V_t^p(s_t)$ given s_t . Moreover $\mathbb{E} V_1^p(s_1) = J_c(p)$, since $s_1 = x \sim \mathcal{D}$ under both policies. This yields the identity. For the bound, note that for f with $\max f - \min f \leq u$, $|\mathbb{E}_q f - \mathbb{E}_p f| \leq u \text{TV}(p, q)$. $\square$

A.3 Reverse water-filling asymptotics (Propositions 3.8 and 3.9)

For a Gaussian vector with independent coordinates of variances λ_i and squared-error distortion, the distortion-rate function is $D(R) = \sum_i \min(w, \lambda_i)$, with w fixed by $\sum_i \frac{1}{2} \log_+(\lambda_i/w) = R$ nats (Cover and Thomas, 2006, Thm. 10.3.3). Take $\lambda_i = ci^{-\alpha}$, $\alpha > 1$, and let $k = \max\{i : \lambda_i > w\}$, so that $w \in [\lambda_{k+1}, \lambda_k)$. Then $R = \frac{\alpha}{2} \sum_{i \leq k} \log \frac{k}{i} + \frac{k}{2} \log \frac{\lambda_k}{w}$. The last term lies in $(0, \frac{k}{2}\alpha \log(1 + \frac{1}{k})) \subseteq (0, \frac{\alpha}{2}]$. By Stirling, $\sum_{i \leq k} \log \frac{k}{i} = k - \frac{1}{2} \log(2\pi k) + o(1)$. Hence $k = \frac{2R}{\alpha}(1 + o(1))$. The distortion is $k w + \sum_{i > k} ci^{-\alpha} = ck^{1-\alpha}(1 + \frac{1}{\alpha-1})(1 + o(1))$. Substituting k , and halving for the $\frac{1}{2}\|\cdot\|^2$ form of the local KL, gives the constant in Proposition 3.8. For a finite spectrum of length k_T , once $w < \lambda_{k_T}$ all components are active. Then $R = \frac{1}{2} \sum_{i \leq k_T} \log(\lambda_i/w)$ gives $w = (\prod_i \lambda_i)^{1/k_T} e^{-2R/k_T}$ and $D = k_T w$, which is Proposition 3.9 after the change of units $R \mapsto R \ln 2$ and halving. $\square$

A.4 Numerical verification

The scripts in `verification/` check the results marked [numerically checked], by random search, exact enumeration or optimisation. For multi-part results, the parts covered are those listed below. `check_divergences.py` covers Propositions 1.5, 1.7, 1.8, 1.10, 2.11, 4.1, 4.3, 4.5 and 4.6,

the counterexample of Proposition 4.17, and Lemma 2.2. This includes both comparison directions and the JSD_β sandwich of Proposition 1.8, all three event bounds, and parts (ii)–(iv) of Proposition 2.11. `check_sequences.py` enumerates a $V = 3$, $T = 4$ autoregressive pair and verifies Theorems 1.12, 2.15 and 2.16 and Propositions 1.14, 2.19, 4.8, 4.9 and 4.12. It also checks the TV estimator, the bias of the stop-gradient estimator, the unbiasedness of the pathwise/score split in Definition 4.14, and the bias that results from adding the two recipes. `check_capacity.py` verifies Propositions 3.8 and 3.9, Theorem 2.3, and Corollary 3.2, the $R = 0$ tightness of Corollary 3.3, and Theorem 3.1 on 300 random instances by exact computation of $I(\mathbf{T}; Y)$. `check_representations.py` verifies Propositions 5.1 and 5.9 and Theorem 5.7 in a random instance of the linear-Gaussian model. `check_decoding.py` verifies Lemmas 7.3 and 7.6, Propositions 7.4, 7.13 and 7.15, Theorems 7.7, 7.9 and 7.14, and Corollaries 7.10 and 7.11. It checks the flip thresholds against direct constrained optimisation over the simplex, the knapsack characterisation against brute force over subsets, and the sequence-level and beam-search statements by exact enumeration, including the attaining students and the example of Theorem 7.14(iv). It also uses teachers with zero probabilities and deterministic steps, and reproduces the reverse-KL examples that make the conditions in Lemma 7.3 and Theorem 7.14(iii) necessary. All checks pass.

References

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, 2024.
- Hirotsugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.
- Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. arXiv:2404.05405, 2024.
- Shun-ichi Amari. α -divergence is unique, belonging to both f -divergence and Bregman divergence classes. *IEEE Transactions on Information Theory*, 55(11):4925–4931, 2009.
- Kushal Arora, Layla El Asri, Hareesh Bahuleyan, and Jackie Chi Kit Cheung. Why exposure bias matters: An imitation learning perspective of error accumulation in language generation. In *Findings of ACL*, 2022.
- Jean-Yves Audibert and Alexandre B. Tsybakov. Fast learning rates for plug-in classifiers. *The Annals of Statistics*, 35(2):608–633, 2007.
- Jimmy Ba and Rich Caruana. Do deep nets really need to be deep? In *Advances in Neural Information Processing Systems*, 2014.
- Peter L. Bartlett and Shahar Mendelson. Rademacher and Gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002.
- Peter L. Bartlett, Michael I. Jordan, and Jon D. McAuliffe. Convexity, classification, and risk bounds. *Journal of the American Statistical Association*, 101(473):138–156, 2006.
- Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*, 2015.

- Lucas Beyer, Xiaohua Zhai, Amélie Royer, Larisa Markeeva, Rohan Anil, and Alexander Kolesnikov. Knowledge distillation: A good teacher is patient and consistent. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.
- Avrim Blum, Merrick Furst, Jeffrey Jackson, Michael Kearns, Yishay Mansour, and Steven Rudich. Weakly learning DNF and characterizing statistical query learning using Fourier analysis. In *ACM Symposium on Theory of Computing*, 1994.
- Dankmar Böhning. Multinomial logistic regression algorithm. *Annals of the Institute of Statistical Mathematics*, 44(1):197–200, 1992.
- Léon Bottou and Olivier Bousquet. The tradeoffs of large scale learning. In *Advances in Neural Information Processing Systems*, 2008.
- Jean Bretagnolle and Catherine Huber. Estimation des densités: risque minimax. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, 47:119–137, 1979.
- Cristian Buciluă, Rich Caruana, and Alexandru Niculescu-Mizil. Model compression. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2006.
- Dan Busbridge, Amitis Shidani, Floris Weers, Jason Ramapuram, Etai Littwin, and Russ Webb. Distillation scaling laws. In *International Conference on Machine Learning*, 2025.
- Jang Hyun Cho and Bharath Hariharan. On the efficacy of knowledge distillation. In *IEEE/CVF International Conference on Computer Vision*, 2019.
- Thomas A. Courtade and Tsachy Weissman. Multiterminal source coding under logarithmic loss. *IEEE Transactions on Information Theory*, 60(1):740–761, 2014.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.
- Hugo Cui, Bruno Loureiro, Florent Krzakala, and Lenka Zdeborová. Generalization error rates in kernel regression: The crossover from the noiseless to noisy regime. In *Advances in Neural Information Processing Systems*, 2021.
- Tri Dao, Govinda M. Kamath, Vasilis Syrgkanis, and Lester Mackey. Knowledge distillation as semiparametric inference. In *International Conference on Learning Representations*, 2021.
- Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.
- Dylan J. Foster, Sham M. Kakade, Jian Qian, and Alexander Rakhlin. The statistical complexity of interactive decision making. arXiv:2112.13487, 2021.
- Dylan J. Foster, Adam Block, and Dipendra Misra. Is behavior cloning all you need? Understanding horizon in imitation learning. In *Advances in Neural Information Processing Systems*, 2024.
- Tommaso Furlanello, Zachary C. Lipton, Michael Tschannen, Laurent Itti, and Anima Anandkumar. Born-again neural networks. In *International Conference on Machine Learning*, 2018.
- Saeed Ghadimi and Guanghui Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.

- Seyed Kamyar Seyed Ghasemipour, Richard Zemel, and Shixiang Gu. A divergence minimization perspective on imitation learning methods. In *Conference on Robot Learning*, 2019.
- Nathan Godey, Éric de la Clergerie, and Benoît Sagot. Why do small language models underperform? Studying language model saturation via the softmax bottleneck. arXiv:2404.07647, 2024.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. In *International Conference on Learning Representations*, 2024.
- Hrayr Harutyunyan, Ankit Singh Rawat, Aditya Krishna Menon, Seungyeon Kim, and Sanjiv Kumar. Supervision complexity and its role in knowledge distillation. In *International Conference on Learning Representations*, 2023.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. arXiv:1503.02531, 2015.
- Jordan Hoffmann et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, 2022.
- Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. TinyBERT: Distilling BERT for natural language understanding. In *Findings of EMNLP*, 2020.
- Sham Kakade and John Langford. Approximately optimal approximate reinforcement learning. In *International Conference on Machine Learning*, 2002.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. arXiv:2001.08361, 2020.
- Liyiming Ke, Sanjiban Choudhury, Matt Barnes, Wen Sun, Gilwoo Lee, and Siddhartha Srinivasa. Imitation learning as f-divergence minimization. In *Workshop on the Algorithmic Foundations of Robotics*, 2020.
- Michael Kearns. Efficient noise-tolerant learning from statistical queries. *Journal of the ACM*, 45(6):983–1006, 1998.
- Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of EMNLP*, 2016.
- Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. DistiLLM: Towards streamlined distillation for large language models. In *International Conference on Machine Learning*, 2024.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, 2019.
- Tanishq Kumar et al. Scaling laws for precision. arXiv:2411.04330, 2024.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, 2023.
- Alexander Lin, Jeremy Wohlwend, Howard Chen, and Tao Lei. Autoregressive knowledge distillation through imitation learning. In *Proceedings of EMNLP*, 2020.
- Jianhua Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.

- David Lopez-Paz, Léon Bottou, Bernhard Schölkopf, and Vladimir Vapnik. Unifying distillation and privileged information. In *International Conference on Learning Representations*, 2016.
- Enno Mammen and Alexandre B. Tsybakov. Smooth discrimination analysis. *The Annals of Statistics*, 27(6):1808–1829, 1999.
- Aditya Krishna Menon, Ankit Singh Rawat, Sashank J. Reddi, Seungyeon Kim, and Sanjiv Kumar. A statistical perspective on distillation. In *International Conference on Machine Learning*, 2021.
- Tom Minka. Divergence measures and message passing. Technical Report MSR-TR-2005-173, Microsoft Research, 2005.
- Ilya Mironov. Rényi differential privacy. In *IEEE Computer Security Foundations Symposium*, 2017.
- Seyed Iman Mirzadeh, Mehrdad Farajtabar, Ang Li, Nir Levine, Akihiro Matsukawa, and Hassan Ghasemzadeh. Improved knowledge distillation via teacher assistant. In *AAAI Conference on Artificial Intelligence*, 2020.
- Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When does label smoothing help? In *Advances in Neural Information Processing Systems*, 2019.
- Mary Phuong and Christoph Lampert. Towards understanding knowledge distillation. In *International Conference on Machine Learning*, 2019.
- Mark S. Pinsker. *Information and Information Stability of Random Variables and Processes*. Holden-Day, 1964.
- Yury Polyanskiy and Yihong Wu. *Information Theory: From Coding to Learning*. Cambridge University Press, 2025.
- Marc’Aurelio Ranzato, Sumit Chopra, Michael Auli, and Wojciech Zaremba. Sequence level training with recurrent neural networks. In *International Conference on Learning Representations*, 2016.
- Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. FitNets: Hints for thin deep nets. In *International Conference on Learning Representations*, 2015.
- Stéphane Ross and J. Andrew Bagnell. Efficient reductions for imitation learning. In *International Conference on Artificial Intelligence and Statistics*, 2010.
- Stéphane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *International Conference on Artificial Intelligence and Statistics*, 2011.
- Igal Sason and Sergio Verdú. f-divergence inequalities. *IEEE Transactions on Information Theory*, 62(11):5973–6006, 2016.
- Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE National Convention Record*, 7:142–163, 1959.
- Samuel Stanton, Pavel Izmailov, Polina Kirichenko, Alexander A. Alemi, and Andrew Gordon Wilson. Does knowledge distillation really work? In *Advances in Neural Information Processing Systems*, 2021.

- Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise parallel decoding for deep autoregressive models. In *Advances in Neural Information Processing Systems*, 2018.
- Zhiqing Sun, Hongkun Yu, Xiaodan Song, Renjie Liu, Yiming Yang, and Denny Zhou. Mobile-BERT: A compact task-agnostic BERT for resource-limited devices. In *Proceedings of ACL*, 2020.
- Jiaxi Tang, Rakesh Shivanna, Zhe Zhao, Dong Lin, Anima Singh, Ed H. Chi, and Sagar Jain. Understanding and improving knowledge distillation. arXiv:2002.03532, 2020.
- Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive representation distillation. In *International Conference on Learning Representations*, 2020.
- Alexandre B. Tsybakov. Optimal aggregation of classifiers in statistical learning. *The Annals of Statistics*, 32(1):135–166, 2004.
- Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- Tim van Erven and Peter Harremoës. Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014.
- Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, 2020.
- Yuqiao Wen, Zichao Li, Wenyu Du, and Lili Mou. f-divergence minimization for sequence-level knowledge distillation. In *Proceedings of ACL*, 2023.
- Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982.
- Noam Wies, Yoav Levine, and Amnon Shashua. Sub-task decomposition enables learning in sequence to sequence tasks. In *International Conference on Learning Representations*, 2023.
- Wing Hung Wong and Xiaotong Shen. Probability inequalities for likelihood ratios and convergence rates of sieve MLEs. *Annals of Statistics*, 23(2):339–362, 1995.
- Xiaohan Xu et al. A survey on knowledge distillation of large language models. arXiv:2402.13116, 2024.
- Zhilin Yang, Zihang Dai, Ruslan Salakhutdinov, and William W. Cohen. Breaking the softmax bottleneck: A high-rank RNN language model. In *International Conference on Learning Representations*, 2018.
- Sergey Zagoruyko and Nikos Komodakis. Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer. In *International Conference on Learning Representations*, 2017.
- Chen Zhang et al. Towards the law of capacity gap in distilling language models. arXiv:2311.07052, 2023.
- Tong Zhang. From ϵ -entropy to KL-entropy: Analysis of minimum information complexity density estimation. *Annals of Statistics*, 34(5):2180–2210, 2006.